

KASHTRE

CLINICAL MODULE

SYSTEMS REQUIREMENTS DOCUMENT v6.1

CONSOLIDATED DRAFT | PHASES 1 TO 10

COMPLETE CLINICAL FUNCTIONAL REQUIREMENTS BASELINE

Module Key: CLINICAL_ORCHESTRATOR

Production Baseline: Laravel 10.48.x / PHP 8.3 (Composer requirement ^8.1) / MySQL 8+ / Vue 3

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This consolidated draft brings Phases 1 to 10 into one controlled Clinical Module specification. Phase 1 establishes the governing rules that every later Clinical workflow must obey. It does not authorize production implementation until the complete v6.1 SRD, companion Laravel EDD, traceability and acceptance testing are approved.

This consolidated Systems Requirements Document defines the complete functional requirements for the KashTre Clinical Module across the patient-care lifecycle, from patient and encounter context, clinical documentation, diagnosis and care planning through ordering, medication administration, clinical observations, diagnostic-result review, transitions of care and production assurance. The Clinical Module serves as the clinical orchestration layer of the KashTre platform and operates through controlled integration with the Main Module, Inventory, Pharmacy, LIMS, Imaging, Theatre, Surgery and other authorized services. Particular emphasis is placed on patient safety, contextual authorization, clinical accountability, point-of-care usability, immutable provenance and the accurate translation of clinical activities into downstream operational events without requiring users to duplicate data entry or make inventory, coverage or financial decisions outside their clinical responsibilities.

Point-of-care governing principle: clinical users record or confirm what physically occurred during patient care. Clinical supplies the patient, encounter and clinical-event context; Inventory determines and records the physical stock consequence; and the Main Module determines any applicable Approved Pool, coverage or financial consequence. A clinical user shall not be required to duplicate information already captured during a medication administration, procedure or other clinical activity.

Document Control

Field	Phase 1 Rule
Purpose	Define the Clinical Module foundation, ownership boundaries, authorization model and shared-service contracts.
Authority	Applies to every later Clinical v6.1 phase and supersedes conflicting local-unit, blanket-role and unrestricted-access assumptions.
Baseline	Clinical SRD v6.0 and addendum, reviewed and corrected rather than copied blindly.
Companion contracts	Main Module Shared Unit Engine Contract; Shared AI Services Gateway SRD; Main Module identity, permissions and client-space services.
Implementation status	Functional draft only. Laravel engineering design follows after functional approval.
Change control	Changes to a Phase 1 governing rule require recorded impact analysis across all later phases.

Revision Intent

- Retain valid v6.0 concepts including Laravel-native architecture, dynamic care assignment, CDEs, task projections, major transitions and shared AI integration.
- Correct the obsolete assumption that Clinical owns a writable unit master or conversion registry.
- Replace role-name checks with atomic permissions, configurable permission bundles and contextual authorization.
- Require active client-space assignment for ordinary client-space work.
- Define safe exceptions for consultation, cross-cover, remote services, governance and emergency access.

1. Purpose, Scope and Outcomes

CLN-FND-001 The Clinical Module shall orchestrate clinical documentation, decisions, observations, orders, medication administration, results follow-up, care plans, clinical tasks and major clinical transitions across outpatient, inpatient and post-discharge workflows.

CLN-FND-002 Phase 1 shall govern every Clinical function, API, page, background task, integration and report, whether implemented now or in a later release.

CLN-FND-003 The Clinical Module shall not duplicate authoritative Main Module registries for users, official titles, facilities, client spaces, permissions or canonical units.

CLN-FND-004 Every later SRD feature shall define its permission, scope, client-space requirement, patient relationship, record-state conditions, exceptions, audit events and safe-failure behavior.

CLN-FND-005 Routine bedside care shall remain responsive and task-driven; configurable process engines shall govern major transitions and other explicitly configured high-risk workflows, not every ordinary clinical action.

Governing outcome: no developer should decide access, ownership or integration behavior by inference.

2. Definitions and Concept Separation

Concept	Definition	Must Not Be Confused With
Tenant	Top-level isolated customer or governance domain.	Facility or client space
Entity	Configured legal or operational organization within a tenant.	Department
Facility	Physical or virtual service-delivery site.	Tenant
Client space	Operational work destination such as ward, clinic, room, theatre or virtual clinic.	Organizational node
Patient space	The patient's effective location or service context within an encounter.	User assignment

Organizational assignment	The user's HR reporting and administrative home.	Clinical client-space activation
Client-space assignment	Effective authorization context permitting work in a client space.	Professional privilege
Official title	Main Module employment title such as Registered Nurse or Medical Officer.	Atomic permission
Permission	A stable code allowing one system action in principle.	Scope or patient relationship
Permission bundle	Configurable group of atomic permissions.	Hardcoded role
Clinical privilege	Additional competence or legal authority for a high-risk action.	Ordinary title bundle
Care relationship	Active link between user/team/role and patient/encounter.	Facility membership
Break glass	Audited, time-limited emergency access outside ordinary context.	Administrator access
Base unit	Canonical Clinical normalization unit selected from Main Module Unit Engine.	Preferred display unit

3. Architectural and Source-of-Truth Boundaries

3.1 Main Module Authority

CLN-OWN-001 The Main Module shall remain authoritative for user identity, authentication status, official titles, designations, cadres, qualifications, employment state, facilities, client spaces, roles, permissions, permission bundles and permission assignments.

CLN-OWN-002 The Main Module shall expose stable public identifiers and supported service contracts. Clinical shall not treat copied labels as authoritative identity.

CLN-OWN-003 The Main Module shall own the canonical Shared Unit Management, Composition and Conversion Engine.

CLN-OWN-004 The Main Module shall remain authoritative for inventory master items and the identifiers required by Clinical ordering and medication translation.

3.2 Clinical Module Authority

CLN-OWN-010 Clinical shall own patient-care relationships, clinical content, clinical record states, clinical workflow authority, CDE policy, clinical unit-usage policy, clinical tasks, warnings, overrides and clinical audit context.

CLN-OWN-011 Clinical shall publish the catalogue of atomic clinical permission codes required by its workflows; the Main Module shall register and assign those codes.

CLN-OWN-012 Clinical shall make the final contextual authorization decision for Clinical resources after consuming Main Module identity and entitlement data.

3.2.1 Observation-First Clinical Documentation

Observation-first governing principle: applicable forms and clinical context define what observations are required and how they are rendered; the system converts those requirements into due work; clinical users record each observation once; and the system automatically renders every applicable form, flowsheet, chart, score, report and summary from the authoritative observations.

CLN-OWN-013 Clinical shall maintain atomic Observations as the authoritative patient-specific clinical facts used by observation worklists, forms, flowsheets, charts, scores, decision support, reports and exports.

CLN-OWN-014 A Clinical Form Definition shall ordinarily define observation requirements and rendering rules and shall not become a competing source of observations already stored in the Clinical record.

CLN-OWN-015 A clinical user shall record each observation once. Every authorized consuming view shall reuse that Observation where patient, encounter, CDE, occurrence time, status, quality and context are applicable.

CLN-OWN-016 Clinical shall derive observation work from applicable client-space defaults, active forms, orders, protocols, care plans, medication-monitoring requirements and authorized patient-specific instructions.

CLN-OWN-017 The Clinical Form Engine shall be a requirements, composition and rendering capability and shall not operate as a parallel observation database.

3.3 HR Module Relationship

CLN-OWN-020 Clinical shall consume effective employment, organizational, credential and client-space assignment information supplied by the authoritative platform and HR capabilities.

CLN-OWN-021 An organizational assignment shall not by itself authorize access to a patient chart or activate work in a client space.

3.4 Shared AI Gateway

CLN-OWN-030 Clinical shall consume AI exclusively through the Shared AI Gateway and shall not embed independent provider credentials or direct provider integrations.

CLN-OWN-031 AI output shall remain draft, suggested or advisory until an authenticated authorized user reviews and accepts it.

3.5 LIMS, Imaging, Inventory, Theatre and Surgery

CLN-OWN-040 Clinical shall create and track clinical intent and follow-up but shall not silently assume ownership of another module's fulfilment, result authorization, stock truth, image archive or theatre administration.

CLN-OWN-041 Every cross-module object shall preserve source module, source public ID, version, status, timestamps and correlation ID.

3.6 Point-of-Care Consumption Ownership

CLN-OWN-042 Clinical shall own the patient, encounter, clinical activity and source-event context used to initiate a patient-linked Consumption Session but shall not become the authoritative source of physical stock balances.

CLN-OWN-043 Inventory shall remain authoritative for item resolution, packaging, batch or lot, expiry, stock location, physical quantity, stock depletion, wastage classification and inventory reconciliation.

CLN-OWN-044 The Main Module shall remain authoritative for prior-payment verification, Approved Pool or entitlement evaluation, pricing, coverage, responsible-account determination and financial posting.

CLN-OWN-045 The clinical user shall record or confirm physical usage and shall not be required to determine whether an item is prepaid, covered, included, billable or financially attributable.

4. Tenancy, Facility and Client-Space Context

CLN-CTX-001 Every Clinical request and persisted Clinical record shall be tenant-scoped.

CLN-CTX-002 Facility and client-space context shall be explicit where the workflow is location-dependent.

CLN-CTX-003 A user shall not select a client space that the authoritative assignment service does not return as currently effective, except through an approved exceptional pathway.

CLN-CTX-004 The patient's effective client space shall be derived from the active encounter and patient-movement record, not from a stale page selection.

CLN-CTX-005 Clinical shall reject cross-tenant identifiers even when an internal numeric identifier happens to exist.

CLN-CTX-006 A facility-level permission shall not automatically become tenant-wide.

CLN-CTX-007 Client-space changes shall be effective-dated and shall not retrospectively rewrite who was authorized at the time of a historical action.

4.1 Assignment Types

Assignment Type	Ordinary Use	Activation Basis
PERMANENT	Routine home work location	Effective-dated approved assignment
ROSTER_DRIVEN	Scheduled work in one or more spaces	Published roster and active shift window
TEMPORARY	Time-limited transfer or support	Approved start and end
ROTATIONAL	Planned rotation	Rotation schedule
RELIEF	Configured relief pool deployment	Accepted deployment
CROSS_COVER	On-call or temporary cover	Cover assignment and time window

REMOTE_SERVICE	Authorized virtual clinical work	Remote-service assignment and patient link
EMERGENCY	Crisis deployment	Emergency assignment or break glass

CLN-CTX-010 Each assignment shall record user, tenant, facility, client space, assignment type, permitted operational function, effective start, effective end, approving authority, source reference, status and audit metadata.

CLN-CTX-011 Assignment expiry shall remove ordinary authority without deleting historical evidence.

5. Clinical Authorization Model

5.1 Governing Formula

Effective Clinical Authority = Active authenticated user + atomic permission + qualifying scope + active client-space assignment where required + patient/encounter relationship where required + applicable privilege/credential + permitted record state + consent/confidentiality clearance - explicit restrictions.

CLN-AUTH-001 The system shall deny by default.

CLN-AUTH-002 Authentication shall not be treated as authorization.

CLN-AUTH-003 Possession of a permission shall be necessary but not sufficient where contextual checks apply.

CLN-AUTH-004 Authorization shall be enforced server-side for web, mobile, API, queue and integration processing.

CLN-AUTH-005 Hiding a button shall not substitute for server authorization.

CLN-AUTH-006 If permission or context services are unavailable and no approved resilient cache can prove authority, the system shall fail closed for write and restricted-read actions.

CLN-AUTH-007 Every authorization decision shall use stable identifiers rather than display labels.

CLN-AUTH-008 Authorization policy versions shall be auditable and historically resolvable.

5.2 Decision Order

Order	Decision
1	Authenticate actor and client application.
2	Resolve tenant, entity and facility.
3	Resolve requested atomic permission.
4	Confirm Main Module entitlement and assignment effectiveness.
5	Resolve patient, encounter and effective patient space.
6	Confirm client-space scope or a permitted exception.
7	Confirm care relationship where required.
8	Confirm credentials and additional privileges.
9	Confirm resource state and workflow authority.
10	Apply consent, confidentiality and security-label restrictions.
11	Apply explicit denials and separation-of-duty rules.
12	If denied, determine whether a defined break-glass pathway is eligible.
13	Return allow or structured deny reason.
14	Audit the decision when required.

6. Atomic Permissions, Bundles and Official Titles

6.1 Atomic Permission Rules

CLN-PERM-001 Each permission shall grant one clearly named action against one resource family.

CLN-PERM-002 Permissions shall use stable namespaced codes beginning with clinical.

CLN-PERM-003 Read, create, update, sign, approve, cancel, correct, override, export and administer shall be separate where the risk differs.

CLN-PERM-004 Clinical shall publish permission metadata including description, risk tier, default scope, credential requirement, break-glass eligibility and audit level.

CLN-PERM-005 Retired permission codes shall remain resolvable for historical audit but shall not be newly assigned.

6.2 Permission Bundles

CLN-PERM-010 A permission bundle shall be a configurable reusable collection of atomic permissions.

CLN-PERM-011 A bundle shall be versioned, effective-dated, approved and auditable.

CLN-PERM-012 Changing a bundle shall not silently rewrite the historical authority used for prior actions.

CLN-PERM-013 A bundle shall not contain an unrestricted wildcard permission.

6.3 Official Title Mapping

CLN-PERM-020 An official title may be mapped to one or more baseline Clinical permission bundles.

CLN-PERM-021 Title-to-bundle mappings shall be configurable by authorized administrators and shall not be hardcoded in Laravel source code.

CLN-PERM-022 A title-derived bundle shall describe ordinary professional activity but shall not provide unrestricted patient access.

CLN-PERM-023 A title change, suspension or employment termination shall trigger recalculation of effective Clinical authority.

CLN-PERM-024 Higher-seniority wording in a title shall not be interpreted algorithmically as broader authority.

6.4 Illustrative Baseline Bundles

The following are suggested pre-seeded templates. They are not immutable legal statements and require institutional approval.

Template	Illustrative Ordinary Actions	Excluded Without Additional Authority
NURSING_GENERAL	View assigned charts; record observations; nursing assessments and notes; care-plan actions; administer permitted medication; handover.	Prescribe; authorize discharge; independently alter medical orders; high-risk overrides.
MEDICAL_OFFICER_GENERAL	View assigned charts; history and examination; progress notes; diagnoses; routine orders; result review; discharge preparation.	Restricted prescribing; specialist procedures; final actions reserved to configured senior authority.
CONSULTANT_GENERAL	Consultant documentation; configured order and approval actions; care-plan oversight; configured discharge authority.	Actions outside specialty, facility, relationship or credential scope.
ALLIED_HEALTH_GENERAL	View assigned chart sections; assessments; profession-specific plans and notes; task completion.	Medication prescribing and unrelated clinical sections.
WARD_CLERK_LIMITED	Operational lists, movement preparation and configured administrative fields.	Clinical note content, observation entry, orders or medication administration.
CLINICAL_ADMIN_CONFIGURATION	Manage approved Clinical settings and dictionaries.	Routine chart access unless separately entitled.

6.5 Initial Permission Families

- clinical.patient.*
- clinical.encounter.*
- clinical.note.*
- clinical.problem.*
- clinical.diagnosis.*
- clinical.observation.*
- clinical.order.*
- clinical.medication.*
- clinical.result.*

- clinical.careplan.*
- clinical.nursing.*
- clinical.transition.*
- clinical.task.*
- clinical.settings.*
- clinical.report.*
- clinical.audit.*
- clinical.break_glass.*

CLN-PERM-030 Later phases shall expand every family into an action-level catalogue and shall not rely on the wildcard forms shown for grouping.

7. Scope Model

Scope	Meaning
SELF	Only the user's own non-patient records where applicable.
ASSIGNED_PATIENTS	Patients explicitly assigned to the user.
CARE_TEAM	Patients linked to an active care team containing the user.
CLIENT_SPACE	Patients in one or more assigned client spaces.
DEPARTMENT	Configured departmental oversight, not automatic chart-wide write access.
FACILITY	Configured facility oversight.
ENTITY	Configured entity oversight.
TENANT	Exceptional tenant-wide authority.
CROSS_ENTITY_GOVERNANCE	Read or oversight functions explicitly approved across entities.
EMERGENCY_ONLY	Authority usable only through emergency controls.

CLN-SCOPE-001 The effective scope shall be the narrowest intersection of permission assignment, client-space assignment, patient relationship and resource restrictions.

CLN-SCOPE-002 A broader organizational role shall not automatically override a narrower clinical scope.

CLN-SCOPE-003 Write authority shall be narrower than or equal to read authority unless an explicitly designed workflow requires otherwise.

CLN-SCOPE-004 Exports and bulk access shall require separate permissions and stronger scope evaluation.

8. Patient and Encounter Relationships

CLN-REL-001 Clinical shall support individual, role-based, team-based and hybrid care assignments.

CLN-REL-002 A client-space assignment may establish ordinary eligibility to work in a location, while a patient relationship may further narrow which patients the user may access.

CLN-REL-003 Care assignments shall record patient, encounter, relationship type, actor or team, responsibility, effective time, source, status and handover information.

CLN-REL-004 The system shall support primary clinician, primary nurse, assigned nurse, covering clinician, consultant, allied-health provider, care-team member and accepted referral relationships.

CLN-REL-005 Relationship expiry or handover shall not delete historical actions.

CLN-REL-006 Opening a chart shall not itself create a care relationship.

CLN-REL-007 A user shall not self-assign to a patient unless an authorized workflow allows it and records the reason.

8.1 Ordinary Ward Example

Condition	Required
User state	Active and authenticated
Baseline authority	clinical.observation.create
Title source	Registered Nurse mapped to approved baseline bundle
User location	Active assignment to Children's Ward

Patient context	Active encounter in Children's Ward
Relationship	Assigned nurse, ward care team, or policy-approved ward coverage
Record state	Encounter open and observation entry permitted
Outcome	Observation entry allowed and audited

9. Credentials, Privileges and Restrictions

CLN-PRIV-001 High-risk authority shall be represented separately from ordinary title-derived permissions.

CLN-PRIV-002 A permission may require an active credential, competency, completed training, professional licence, specialty privilege or local authorization.

CLN-PRIV-003 Credentials and privileges shall be effective-dated and shall support suspension and expiry.

CLN-PRIV-004 Clinical shall verify the required privilege at action time, not only at login.

CLN-PRIV-005 Explicit restrictions shall override grants unless a formally configured emergency rule applies.

- Controlled-medicine prescribing
- Chemotherapy verification or administration
- Blood-product authorization
- Independent procedure performance
- Sedation
- High-alert medication override
- Death validation
- Specialist result authorization
- Clinical record correction approval

10. Delegation, Acting Roles and Cross-Cover

CLN-DEL-001 Delegation shall be explicit, time-limited, scoped and auditable.

CLN-DEL-002 A delegator shall not delegate authority they do not possess.

CLN-DEL-003 Delegated authority shall not outlive the delegator's own authority or the stated expiry time.

CLN-DEL-004 Delegation shall identify delegator, delegate, permissions or bundle, scope, patient/client-space constraints, start, end, reason and approval where required.

CLN-DEL-005 Acting appointments shall consume authoritative Main Module or HR records and shall not be inferred from user activity.

CLN-DEL-006 Cross-cover shall identify the covered service or client space and active time window.

CLN-DEL-007 The UI shall make temporary authority visible to the user without implying permanent role change.

11. Break-Glass Emergency Access

CLN-BTG-001 Break glass shall be available only for configured emergency categories and eligible permission families.

CLN-BTG-002 The system shall capture reason code, free-text justification where required, patient or scope, duration, user, device, facility, client space and correlation ID.

CLN-BTG-003 Reauthentication shall be required before activation unless a documented emergency-mode policy states otherwise.

CLN-BTG-004 Emergency access shall be time-limited and shall expire automatically.

CLN-BTG-005 Emergency access shall grant the minimum configured actions required for the emergency.

CLN-BTG-006 Every resource viewed or modified under break glass shall be linked to the emergency-access event.

CLN-BTG-007 Configured reviewers shall receive notification and an after-action report.

CLN-BTG-008 Break glass shall not permit silent deletion, audit suppression or retrospective alteration of the reason.

CLN-BTG-009 Use of break glass shall not create a permanent care relationship or client-space assignment.

11.1 Suggested Pre-Seeded Reasons

- Emergency resuscitation or crash call
- On-call cross-cover when ordinary assignment is unavailable
- Urgent specialist consultation
- Cross-department surge or crisis response
- Primary responsible clinician unavailable
- Preoperative or postoperative stabilization

12. Consent, Confidentiality and Sensitive Records

CLN-CONS-001 Clinical shall support consent and confidentiality directives that narrow access, disclosure or permitted purpose.

CLN-CONS-002 The system shall support sensitivity labels at chart, encounter, document, section or data-element level where configured.

CLN-CONS-003 Ordinary title, permission and client-space assignment shall not automatically override a sensitive-record restriction.

CLN-CONS-004 The interface shall avoid revealing restricted content through counts, snippets, alerts, search results, notifications or exports.

CLN-CONS-005 Access to restricted content shall require the configured additional permission, relationship, consent basis or emergency pathway.

CLN-CONS-006 Consent changes shall be effective-dated and auditable without rewriting prior lawful access decisions.

CLN-CONS-007 The SRD shall not assume one jurisdiction's consent law; deployers shall configure rules under approved organizational and legal policy.

13. Shared Unit Engine Clinical Contract

CLN-UOM-001 Clinical shall not create a writable `clinical_uom_master` or `clinical_uom_conversions_master` as competing sources of truth.

CLN-UOM-002 Clinical shall reference canonical Main Module unit public IDs for CDEs, medication fields, observations, charts and imported results.

CLN-UOM-003 Clinical may define effective-dated domain policies specifying base unit, permitted input units, permitted display units, preferred unit, precision, reference-range unit and physiological-limit unit for each Clinical concept.

CLN-UOM-004 Every quantitative entry shall preserve original value and original unit, normalized value and base unit, conversion rule ID, conversion-rule version and rounding policy where conversion occurs.

CLN-UOM-005 Molar-to-mass conversions shall require analyte context; product and package conversions shall require item or formulation context.

CLN-UOM-006 If a required conversion cannot be verified, Clinical shall block commit or stage the inbound result as an exception. It shall not silently preserve the number under a different unit.

CLN-UOM-007 Reference ranges, critical limits and physiological limits shall be evaluated against the verified normalized value, while the user interface may display a permitted alternative unit.

CLN-UOM-008 Clinical shall provide high-visibility input-unit indicators and explicit unit switching at the point of capture.

CLN-UOM-009 Historical observations shall remain reproducible after a unit or rule is deprecated.

13.1 Glucose Policy Example

Policy Field	Example
Clinical concept	Blood glucose
Base unit	mmol/L canonical Main Module unit ID
Permitted input	mmol/L; mg/dL
Permitted display	mmol/L; mg/dL
Required context	Glucose analyte ID
Persist	Original value/unit plus normalized value/base unit and rule provenance

14. Shared Time Engine and Clinical Temporal Provenance

14.1 Purpose and Governing Outcome

This section defines how the Clinical Module shall acquire, interpret, preserve, display, exchange and audit dates and times across tenants, entities, facilities, client spaces, users, devices and external systems.

The Clinical Module operates as a formal consumer of the Main Module Shared Time Engine. The Main Module shall own authoritative timezone configuration and shared time-resolution services. Clinical shall own the immutable temporal provenance of Clinical records and events.

The shared platform architecture requires historical events to be persisted as unambiguous UTC timestamps, uses IANA timezone identifiers instead of hardcoded offsets, separates user presentation preferences from business time and prohibits dependence on application-server or database-server local time. [\[KashTre_SR...Time_Zones | PDF\]](#)

Governing outcome: An authorized reviewer shall be able to determine:

1. When a clinical or physical event actually occurred.
2. When the event was received by KashTre.
3. When the record was authored or entered.
4. When the record was durably committed.
5. When it was verified, signed, corrected or reconciled.
6. Which timezone and UTC offset applied to the event.
7. How that timezone was resolved.
8. Whether the time was exact, estimated, adjusted, retrospective, device-reported or otherwise uncertain.
9. Whether the event arrived late or out of sequence.
10. Whether the displayed time has been converted for the current viewer.

No developer shall infer these distinctions from a generic created_at or updated_at field.

14.2 Companion Architecture and Authority

The requirements in this section shall be read together with:

- [KashTre_SRD_Multi_Tenant_Time_Zones.pdf](#)
- [KashTre_EDD_Multi_Tenant_Time_Zones.pdf](#)
- The Main Module identity, tenant, entity, facility and client-space contracts
- The Clinical Module Shared Unit Engine contract
- Clinical Phase 2 patient, encounter and workspace requirements
- Clinical Phase 6 medication administration requirements
- Clinical Phase 7 observation and device-integration requirements
- Clinical Phase 9 transition and continuity requirements
- Clinical Phase 10 interoperability and production-assurance requirements

The shared timezone EDD establishes a centralized TimeEngineService, injectable clock abstraction, tenant and branch timezone resolution, UTC and local-time storage separation, offline drift estimation and time-confidence metadata.

[\[KashTre_ED...Time_Zones | PDF\]](#)

CLN-TIME-001 The Clinical Module shall be registered as a formal consumer of the Main Module Shared Time Engine.

CLN-TIME-002 The Main Module Shared Time Engine shall remain authoritative for:

- Trusted UTC acquisition
- Valid IANA timezone identifiers
- Tenant default timezone
- Entity or branch timezone

- Facility timezone
- Client-space timezone where supported
- User presentation-timezone preferences
- Timezone inheritance and resolution
- Shared timezone conversion rules
- Timezone configuration versions and effective periods
- Shared infrastructure clock policy
- Timezone data updates and validation

CLN-TIME-003 Clinical shall not create or maintain a competing writable timezone, daylight-saving, UTC-offset or timezone-conversion master.

CLN-TIME-004 Clinical shall remain authoritative for the temporal meaning and provenance of Clinical records, including occurrence, effective, authored, recorded, received, verified, signed, corrected and reconciled times.

CLN-TIME-005 Where Clinical applies a stricter timestamp, device-clock, ambiguity or confidence rule for patient safety, the stricter Clinical rule shall apply without changing Main Module ownership of shared timezone configuration.

14.3 Definitions and Concept Separation

Concept	Definition	Must Not Be Confused With
UTC instant	Unambiguous point on the global timeline stored in Coordinated Universal Time	Local wall-clock time
Local datetime	Calendar date and clock time in a specified timezone	UTC instant without timezone context
IANA timezone	Region-based timezone identifier such as Africa/Freetown	Fixed offset or abbreviation
UTC offset	Difference between local time and UTC at a particular instant	Permanent timezone identity
Occurrence time	Time the clinical, operational or physical event actually happened	Time the record was entered
Effective time	Clinically meaningful time or period to which a record applies	Database creation time
Scheduled time	Time at which an activity is intended to occur	Actual occurrence time
Authored time	Time an author composed or submitted content	Event occurrence time
Recorded time	Time KashTre durably committed the record	User-entered occurrence time
Received time	Time KashTre received an inbound device or external-system message	Device measurement time
Verified time	Time an eligible user verified a record	Authored or recorded time
Signed time	Time an eligible user finalized or authenticated a record	Occurrence time
Corrected time	Time a formal correction was committed	Original event time
Source time	Raw date and time supplied by a user, device or external system	Trusted server time
Device measurement time	Time reported by a device for the actual measurement	Device transmission or server receipt time
Temporal snapshot	Immutable timezone, offset, local-time and resolution context stored with an event	Current timezone configuration
Presentation timezone	Timezone used to display an event to a user	Timezone in which the event occurred
Time precision	Level actually known, such as date, minute, second or millisecond	Assumed precision
Time confidence	Structured assessment of the reliability of a reported or estimated time	Clinical validity of the measured value
Late entry	Record entered materially after the event occurred	Correction of an erroneous record

Retrospective entry	Entry in which the user states an earlier event time	Contemporaneous recording
Estimated time	Time derived through a controlled estimation method	Exact observed time
Ambiguous local time	Local datetime that maps to more than one UTC instant	Duplicate clinical event
Nonexistent local time	Local datetime skipped during a timezone transition	Invalid calendar date

IANA timezone identifiers provide regional timezone context, while an explicit offset remains part of a precise dateTime or instant representation. [\[fhir.hl7.org\]](https://fhir.hl7.org/)

14.4 Core Temporal Principles

CLN-TIME-010 Every persisted Clinical record shall have a server-generated creation timestamp.

CLN-TIME-011 Every mutable Clinical record shall have a server-generated last-update timestamp.

CLN-TIME-012 Every clinically meaningful event shall have an occurrence or effective time in addition to its recorded time.

CLN-TIME-013 Clinical shall not use one generic timestamp to represent multiple clinically distinct times.

CLN-TIME-014 Where occurrence, effective, authored, recorded, received, verified, signed, corrected and processed times differ, they shall be preserved separately.

CLN-TIME-015 Historical events shall be persistently ordered using unambiguous UTC instants and deterministic tie-breaking information.

CLN-TIME-016 A display conversion shall not modify the persisted UTC instant or historical temporal snapshot.

CLN-TIME-017 A timezone configuration change shall apply prospectively according to its approved effective period and shall not rewrite historical temporal facts.

CLN-TIME-018 Clinical shall preserve the precision actually known and shall not fabricate missing hours, minutes, seconds, fractions or timezone context.

CLN-TIME-019 Date-only, approximate, uncertain and exact date-time values shall remain structurally distinguishable.

14.5 Trusted Clock and Infrastructure Independence

The shared platform SRD requires synchronized infrastructure clocks, injectable clock abstraction and complete independence from server locality. [\[KashTre SR...Time Zones | PDF\]](#)

CLN-TIME-020 Clinical application services shall obtain authoritative current UTC time exclusively from the approved shared injectable clock abstraction.

CLN-TIME-021 Production infrastructure nodes shall use an approved synchronized clock source with monitored drift and operational alerting.

CLN-TIME-022 Clinical business logic shall not call or rely directly on:

- Browser local time
- Mobile-device local time
- Application-server local timezone
- Database-server local timezone
- Queue-worker local timezone
- Container or hosting-region local timezone
- Bedside-device local time as authoritative server time

CLN-TIME-023 Browser and mobile-device clocks may supply source-event metadata but shall not generate authoritative recorded, received, verified, signed or audit timestamps.

CLN-TIME-024 Database defaults may support durable timestamp creation only when they use the approved UTC clock source and conform to the Clinical temporal contract.

CLN-TIME-025 A change in web-server, application-server, queue-worker, container, database or hosting-region timezone shall not change Clinical outcomes.

CLN-TIME-026 Clock health, last synchronization time and detected infrastructure drift shall be observable through authorized operational monitoring.

CLN-TIME-027 A node whose clock exceeds the approved infrastructure-drift threshold shall be removed from time-sensitive processing or placed under an approved degraded-mode policy.

14.6 Timezone Resolution

14.6.1 Clinical Event Resolution Track

The effective timezone for a Clinical event shall be resolved in the following order:

1. Explicit event timezone, where the workflow permits an authorized explicit timezone.
2. Patient-care facility timezone.
3. Client-space timezone, where explicitly configured and applicable.
4. Entity or branch timezone.
5. Tenant default timezone.
6. Platform UTC fallback through a controlled exception.

CLN-TIME-030 Clinical shall resolve event timezone from the patient's effective care-delivery context rather than merely from the logged-in user's current location.

CLN-TIME-031 Facility and client-space timezone resolution shall use stable public identifiers and effective-dated Main Module configuration.

CLN-TIME-032 An explicit event timezone shall require a workflow that supports it and shall preserve the selecting actor, reason and source.

CLN-TIME-033 Clinical shall not accept an arbitrary client-supplied timezone identifier without server-side validation.

CLN-TIME-034 The platform UTC fallback shall not be silently interpreted as the patient-care facility timezone.

CLN-TIME-035 Use of the UTC fallback shall be auditable and shall enter a configuration or data-quality review queue where the missing timezone may affect clinical interpretation.

14.6.2 Presentation Resolution Track

The user presentation timezone may be resolved in the following order:

1. User preferred timezone.
2. Active viewing-context timezone.
3. Facility timezone.
4. Entity or branch timezone.
5. Tenant default timezone.
6. Platform UTC.

CLN-TIME-036 User timezone preference shall control presentation only.

CLN-TIME-037 Changing a user presentation preference shall not alter:

- Stored UTC instants
- Originating local datetime
- Originating timezone
- Applicable UTC offset
- Event sequence
- Clinical date
- Due-state calculation already determined for a completed event
- Audit history
- Financial or Inventory processing

CLN-TIME-038 Where a user views information in a timezone different from the originating timezone, the interface shall identify the displayed timezone when ambiguity could affect clinical interpretation.

CLN-TIME-039 The originating facility time shall remain available from the event detail or provenance view.

14.7 Clinical Temporal Snapshot

CLN-TIME-040 Every clinically significant historical event shall preserve a temporal snapshot sufficient to reconstruct its original interpretation.

The snapshot shall contain, where applicable:

- Authoritative UTC instant
- Original or resolved local datetime
- IANA timezone identifier
- Applicable UTC offset in minutes
- Timezone-resolution source type
- Timezone-resolution source public ID
- Timezone policy or configuration version
- Time source
- Time precision
- Time confidence
- Estimated-time indicator
- Adjusted-time indicator
- Late-entry indicator
- Retrospective-entry indicator
- Device clock status
- Correlation and provenance identifiers

CLN-TIME-041 The local datetime and UTC offset stored in the temporal snapshot shall represent the rules that applied to that event at that instant.

CLN-TIME-042 Clinical shall not reconstruct historical local time solely from the facility's current timezone configuration.

CLN-TIME-043 Retirement, renaming or replacement of a timezone configuration shall not make historical event snapshots unresolvable.

CLN-TIME-044 Temporal snapshots shall be immutable after event finalization, except through a formal correction process that preserves the original snapshot.

CLN-TIME-045 The EDD shall define a reusable temporal-value implementation pattern so that individual engineering teams do not create inconsistent timestamp structures.

14.8 Actual and Historical Events

CLN-TIME-050 A completed or historical Clinical event shall be persisted as an unambiguous UTC instant.

CLN-TIME-051 Historical event records shall retain the local datetime, timezone and offset applicable when the event occurred or was interpreted.

CLN-TIME-052 Later changes to timezone rules shall not move a historical event to a different UTC instant.

CLN-TIME-053 The Clinical timeline shall ordinarily position an event according to its clinically meaningful occurrence or effective time.

CLN-TIME-054 The timeline shall preserve and expose recorded, received, verified and corrected times where those differ from occurrence time.

CLN-TIME-055 Events sharing the same occurrence instant shall use deterministic ordering based on event priority, sequence, source, recorded time and stable public identifier as defined in the EDD.

CLN-TIME-056 Deterministic same-time ordering shall not imply clinical causation.

CLN-TIME-057 An event received late shall appear according to its validated occurrence time while visibly retaining its receipt and recording times.

14.9 Future and Recurring Clinical Schedules

The shared platform architecture distinguishes historical events from future schedules because future UTC conversion may change when timezone rules change. [\[KashTre SR...Time Zones | PDF\]](#)

CLN-TIME-060 A future activity intended to occur at a specific local civil time shall preserve:

- Intended local date and time
- IANA timezone
- Recurrence rule, where applicable
- Applicable scheduling policy
- Ambiguity and gap policy
- Source configuration version
- Next evaluated UTC execution time
- Last calculation time

CLN-TIME-061 Future local-time schedules shall be converted using the supported timezone rules applicable when the occurrence is due.

CLN-TIME-062 A recurring activity defined as occurring at a local clock time shall not be implemented solely as a fixed number of elapsed seconds.

CLN-TIME-063 Schedule recalculation following a timezone-rule change shall affect future occurrences only and shall not alter completed events.

CLN-TIME-064 Recalculation shall preserve the schedule version, previous execution time, new execution time, reason and affected future occurrences.

CLN-TIME-065 Clinical schedules governed by this section include:

- Medication administration opportunities
- Observation schedules
- Monitoring plans
- Infusion checks
- PRN reassessments
- Care-plan reviews
- Clinical tasks
- Follow-up activities
- Appointment-linked clinical work
- Transition and discharge follow-up
- Recurring procedure or treatment activities

CLN-TIME-066 The system shall prevent duplicate due events when timezone configuration or daylight-saving rules change.

CLN-TIME-067 A timezone update shall not silently make an already completed event overdue, early, late or omitted.

14.10 Event-Specific Timestamp Families

14.10.1 Medication Administration

A Medication Administration Event may require:

- Scheduled time
- Permitted administration window
- Preparation time
- Independent-check time
- Actual start time
- Actual completion time
- Witness time
- Recorded time
- Signed time

- Reassessment due time
- Reassessment completion time
- Correction time

CLN-TIME-070 The actual administration time shall not be replaced by the time at which the nurse completes documentation.

CLN-TIME-071 Medication events recorded retrospectively shall preserve both actual administration time and recorded time.

CLN-TIME-072 Preparation, double-check, administration, witnessed waste, infusion-rate change, pause, resume, bag change and reassessment shall preserve separate occurrence times where they represent separate events.

CLN-TIME-073 Medication schedules shall ordinarily use the patient-care facility timezone unless the authorized order or service context explicitly specifies another timezone.

14.10.2 Consumption Sessions and Events

A consumption workflow may require:

- Physical-use occurrence time
- Consumption Session creation time
- User confirmation time
- Deferral time
- Inventory receipt time
- Inventory posting time
- Main Module evaluation time
- Reconciliation completion time
- Correction or reversal time

CLN-TIME-074 Physical-use time shall remain separate from Consumption Session confirmation and downstream processing times.

CLN-TIME-075 Medication product lines generated from an administration shall inherit the applicable administration occurrence time and temporal snapshot.

CLN-TIME-076 Incidental supplies shall ordinarily inherit the linked clinical-activity time, but an authorized user may record a materially different physical-use time.

CLN-TIME-077 Inventory posting time and Main Module financial-evaluation time shall not be represented as the time an item was physically used.

CLN-TIME-078 Deferred consumption shall preserve physical-use, deferral, confirmation and reconciliation times separately.

14.10.3 Observations and Devices

A device-originated observation may require:

- Device measurement time
- Device transmission time
- Interface receipt time
- Clinical service receipt time
- User validation time
- Durable-recording time
- Result-release time
- Correction time

CLN-TIME-079 A device measurement shall ordinarily appear clinically according to its validated measurement time, not merely its transmission or receipt time.

CLN-TIME-080 Device-reported time shall remain distinguishable from trusted server timestamps.

14.10.4 Orders, Results and Transitions

CLN-TIME-081 Orders shall distinguish authored, signed, activated, scheduled, held, resumed, discontinued, cancelled and completed times where applicable.

CLN-TIME-082 Results shall distinguish specimen or study time, result-generation time, source-release time, Clinical receipt time, review time, acknowledgement time and correction time where applicable.

CLN-TIME-083 Patient movement shall distinguish requested, approved, departed, arrived, accepted and completed times.

CLN-TIME-084 Encounter closure shall distinguish clinical finish time, closure-check completion time and final closure time.

14.11 Device and External-System Time

CLN-TIME-090 Each integrated clinical device shall have a stable device identity and an approved mapping to tenant, facility, client space where required and time-resolution context.

CLN-TIME-091 Clinical shall preserve the raw source timestamp received from the device or external system.

CLN-TIME-092 The raw timestamp shall not be overwritten when Clinical calculates a corrected, normalized or estimated UTC instant.

CLN-TIME-093 Device-originated events shall preserve, where available:

- Raw device datetime
- Device timezone or offset
- Device sequence number
- Device transmission time
- Server receipt time
- Estimated clock drift
- Normalized UTC instant
- Time confidence
- Clock-validation status
- Adjustment method
- Integration and message identifiers

CLN-TIME-094 A device timestamp without an offset may be interpreted using the approved device-to-location and timezone mapping.

CLN-TIME-095 Use of a device mapping shall preserve the mapping ID, version and effective period.

CLN-TIME-096 If device identity, location, timezone or clock status cannot be established, the event shall be staged, warned or quarantined according to risk.

CLN-TIME-097 Clinical shall define event-risk-specific drift tolerances.

CLN-TIME-098 The shared financial or low-value transaction drift threshold shall not automatically apply to medication administration, critical observations, blood products, deterioration events or other high-risk Clinical events.

CLN-TIME-099 Automatic adjustment shall not conceal the fact that the device clock was inaccurate or uncertain.

14.12 Offline Clinical Activity

CLN-TIME-100 Offline Clinical records shall preserve:

- Local event ID
- Device ID
- User ID
- Patient and encounter
- Raw device event time
- Device submission time
- Device sequence number
- Timezone context
- Synchronization time
- Server receipt time
- Server-recorded time
- Estimated or validated occurrence time
- Time-confidence classification

CLN-TIME-101 Offline client clocks shall not produce authoritative server-recorded timestamps.

CLN-TIME-102 Synchronization shall preserve the original offline event time and shall not silently replace it with synchronization time.

CLN-TIME-103 Clinical shall evaluate duplicate, stale, future, out-of-order and materially drifted offline events before finalization.

CLN-TIME-104 An estimated occurrence time shall be clearly identified as estimated.

CLN-TIME-105 Where estimation is not clinically safe, the record shall enter reconciliation rather than being automatically finalized.

CLN-TIME-106 Offline synchronization retries shall be idempotent and shall not create duplicate clinical, consumption or audit events.

14.13 Late, Retrospective and Corrected Records

CLN-TIME-110 A late or retrospective record shall preserve the user-stated occurrence time and actual server-recorded time.

CLN-TIME-111 A late entry shall not be made to appear as though it was documented contemporaneously.

CLN-TIME-112 The interface shall identify late or retrospective documentation where clinically, legally or operationally material.

CLN-TIME-113 Backdating shall require an authorized workflow, atomic permission, controlled reason and audit record.

CLN-TIME-114 A user shall not directly edit system-generated recorded, received, verified, signed or audit timestamps.

CLN-TIME-115 A correction shall preserve:

- Original occurrence time
- Original recorded time
- Original signed or verified time
- Original temporal snapshot
- Corrected values
- Correction time
- Correcting actor
- Reason
- Replacement or supersession linkage

CLN-TIME-116 Correcting clinical content shall not automatically change the event occurrence time unless the time itself is the subject of the authorized correction.

CLN-TIME-117 A correction to event time shall preserve both the original and corrected times and shall initiate downstream reconciliation where sequencing, medication, inventory, billing or reporting may be affected.

14.14 Daylight-Saving Gaps and Ambiguous Local Time

The shared platform documents describe repeated local hours, skipped local times, offset selection and adjustment for legacy hardware. Clinical shall apply more conservative rules to completed patient-care events. [\[KashTre_SR...Time_Zones | PDF\]](#), [\[KashTre_ED...Time_Zones | PDF\]](#)

CLN-TIME-120 A local datetime mapping to two UTC instants shall be treated as ambiguous until an approved rule or sufficient source evidence resolves the applicable offset.

CLN-TIME-121 A local datetime that did not exist because of a timezone transition shall not be silently treated as an exact completed clinical-event time.

CLN-TIME-122 Clinical shall not automatically use lenient timezone mapping for completed medication, critical-observation, blood-product or other high-risk events.

CLN-TIME-123 Sequence numbers, adjacent trusted events, device provenance and source metadata may support resolution but shall not replace required human review where clinical risk remains.

CLN-TIME-124 Future schedules may use an approved policy such as:

- Earlier valid offset
- Later valid offset
- Move forward to the next valid time
- Skip occurrence

- Require user confirmation
- Route for clinical review

CLN-TIME-125 The selected schedule policy and any adjustment shall be recorded.

CLN-TIME-126 The raw local input shall be preserved when an ambiguity or gap is adjusted.

CLN-TIME-127 Unresolved temporal ambiguity shall not be concealed by assigning server receipt time as the event occurrence time.

14.15 Display and User Interface

CLN-TIME-130 Clinical interfaces shall use a consistent date and time format appropriate to the tenant and user configuration.

CLN-TIME-131 Critical workflows shall avoid ambiguous numeric-only dates where day and month order could be misunderstood.

CLN-TIME-132 The displayed timezone shall be visible where the viewer timezone differs from the event timezone or where timezone interpretation affects clinical action.

CLN-TIME-133 Event details shall make the originating local time and UTC offset available.

CLN-TIME-134 The interface shall distinguish:

- Scheduled
- Occurred
- Recorded
- Received
- Verified
- Signed
- Corrected
- Estimated
- Late
- Retrospective

CLN-TIME-135 Relative expressions such as “five minutes ago” may supplement but shall not replace an exact date and time in high-risk or audit-sensitive workflows.

CLN-TIME-136 Due, overdue, early and late labels shall be calculated from server-evaluated temporal rules, not browser-local time.

CLN-TIME-137 A user changing presentation timezone shall not cause completed events to move between encounters, reports or clinical dates unless the report explicitly states that grouping uses the selected display timezone.

CLN-TIME-138 Printed and exported records shall identify the timezone used for displayed timestamps.

14.16 Cross-Module Events and APIs

CLN-TIME-140 Every cross-module Clinical event shall include an unambiguous offset-aware occurrence timestamp or UTC instant.

CLN-TIME-141 Where local interpretation is required, the payload shall include:

- IANA timezone identifier
- Applicable UTC offset
- Local datetime
- Time source
- Time precision
- Time confidence where applicable
- Source-event ID
- Correlation ID
- Idempotency key

CLN-TIME-142 API contracts shall not transmit an unqualified local datetime for a completed event unless the field is explicitly identified as raw or unresolved source input.

CLN-TIME-143 Main Module, Inventory, LIMS, Imaging, Theatre, Surgery and external-system processing times shall not overwrite Clinical occurrence time.

CLN-TIME-144 Cross-module retries shall preserve the original event time and idempotency identifier.

CLN-TIME-145 A consuming module may add its own receipt, processing and completion timestamps but shall not relabel them as the originating clinical-event time.

CLN-TIME-146 Contract versions shall define timestamp format, precision, timezone requirements, nullable conditions and error behaviour.

14.17 Shared Time Engine Availability and Clinical Cache

CLN-TIME-150 Clinical may maintain a read-through resilience cache of Main Module timezone-resolution data.

CLN-TIME-151 The cache shall not become an independently editable source of timezone truth.

CLN-TIME-152 Cached entries shall preserve:

- Tenant, entity, facility or client-space public ID
- Resolved IANA timezone
- Resolution source
- Main Module configuration version
- Effective start and end
- Cache retrieval time
- Cache expiry
- Last verified time
- Fresh or stale status

CLN-TIME-153 Use of cached timezone data shall follow an approved risk-based outage policy.

CLN-TIME-154 Where a valid approved cache proves the effective timezone, Clinical may continue according to the configured safe-failure policy.

CLN-TIME-155 Where timezone cannot be resolved safely, Clinical shall not guess or silently use server-local time.

CLN-TIME-156 An action that cannot be safely completed without timezone resolution shall be blocked or staged for reconciliation.

CLN-TIME-157 An event recorded using cached resolution shall retain the cache version and last-verification time in its provenance where required.

CLN-TIME-158 Time Engine recovery shall reconcile staged events without duplicating or silently changing their original temporal facts.

14.18 Audit and Monitoring

CLN-TIME-160 Temporal audit evidence shall record, where applicable:

- Actor
- Device or service identity
- Patient and encounter
- Facility and client space
- Raw source time
- Normalized UTC time
- Resolved local time
- Timezone and offset
- Resolution source
- Configuration version
- Drift estimate
- Time confidence

- Adjustment
- Override
- Late-entry reason
- Correction
- Outcome

CLN-TIME-161 Timezone configuration changes affecting Clinical shall be versioned, effective-dated and auditable.

CLN-TIME-162 Operational monitoring shall identify:

- Unsynchronized application nodes
- Excessive clock drift
- Invalid timezone identifiers
- Missing facility timezone assignments
- Stale Clinical timezone-cache entries
- Device clock failures
- Ambiguous timestamps
- Nonexistent local timestamps
- Future-dated completed events
- Excessive late entries
- Failed schedule recalculations
- Cross-module temporal inconsistencies

CLN-TIME-163 The system shall generate alerts for time-related conditions that could affect medication safety, observation interpretation, clinical sequencing or legal-record integrity.

CLN-TIME-164 Audit ordering shall use authoritative UTC timestamps and deterministic sequence information.

CLN-TIME-165 Audit displays may provide local-time conversion but shall preserve and expose UTC ordering.

14.19 Configuration

14.19.1 Main Module Configuration

The Main Module shall configure and govern:

- Tenant default timezone
- Entity or branch timezone
- Facility timezone
- Client-space timezone where supported
- User presentation-timezone preference
- Timezone-resolution hierarchy
- Timezone configuration versions
- IANA timezone validation
- Shared clock and synchronization policy
- Platform fallback
- Shared service cache-control metadata

14.19.2 Clinical Configuration

Clinical may configure:

- Event-risk classes
- Device clock-drift tolerances
- Event-specific warning, staging and quarantine rules
- Allowed retrospective-entry windows
- Late-entry reason requirements
- Time-correction approval rules
- Future-schedule ambiguity and gap policies

- Display requirements by workflow
- Time-related escalation rules
- Allowed cache use during Shared Time Engine outage

CLN-TIME-170 Clinical configuration shall not permit administrators to create arbitrary timezone identifiers or offsets.

CLN-TIME-171 Clinical event-risk and drift policies shall be versioned, reviewed, approved and effective-dated.

CLN-TIME-172 A configuration change shall not rewrite historical records.

CLN-TIME-173 High-risk defaults shall fail safely when no approved Clinical temporal policy exists.

14.20 Acceptance Scenarios

ID	Scenario	Expected Result
PH1-TIME-AT-001	A blood-pressure machine measures at 10:04, transmits at 10:05 and the reading is validated at 10:07.	Measurement, transmission, receipt, validation and recorded times remain distinct.
PH1-TIME-AT-002	A nurse administers medicine at 10:05 and completes documentation at 10:08.	The administration appears at 10:05 with 10:08 retained as recorded time.
PH1-TIME-AT-003	Confirm Consumption completes after medication administration.	Physical-use, confirmation, Inventory posting and Main evaluation times remain separate.
PH1-TIME-AT-004	A remote clinician views an event using another timezone.	The display converts appropriately while originating facility time and stored UTC remain unchanged.
PH1-TIME-AT-005	A facility changes its configured timezone.	Historical records retain their original timezone, offset and local interpretation.
PH1-TIME-AT-006	A bedside device clock is materially ahead.	The raw time is preserved and the event is warned, staged or quarantined according to risk.
PH1-TIME-AT-007	A device sends local time without an offset.	The approved device-location mapping resolves timezone and records provenance.
PH1-TIME-AT-008	A high-risk event occurs during a repeated local hour.	The system does not silently choose an offset when ambiguity remains clinically material.
PH1-TIME-AT-009	A local time does not exist during a forward clock transition.	Raw input is preserved and the event follows the approved review or scheduling policy.
PH1-TIME-AT-010	A recurring medication is scheduled for the same local time each day.	Future opportunities remain aligned to intended local civil time.
PH1-TIME-AT-011	The Shared Time Engine becomes unavailable.	Clinical uses only an approved current cache or safely stages or blocks the action without guessing.
PH1-TIME-AT-012	An offline event synchronizes twice.	Idempotency prevents duplicate records while preserving the original source time.
PH1-TIME-AT-013	A user enters a retrospective observation.	Both the stated occurrence time and actual recorded time are preserved.
PH1-TIME-AT-014	An authorized correction changes an event time.	Original and corrected times, reason, actor and downstream reconciliation remain traceable.
PH1-TIME-AT-015	Two events have the same UTC occurrence instant.	Deterministic ordering occurs without implying causation.
PH1-TIME-AT-016	The database server timezone changes.	Clinical scheduling, recording and display outcomes remain unchanged.
PH1-TIME-AT-017	A user changes the profile display timezone.	Visual rendering changes, but historical records, clinical dates and audit ordering do not.
PH1-TIME-AT-018	An Inventory response arrives after its linked Clinical event.	Inventory processing time is retained without replacing Clinical occurrence time.
PH1-TIME-AT-019	An external result arrives one day after the specimen event.	The timeline uses the applicable clinical event time and visibly retains source-release and receipt times.

PH1-TIME-AT-020	A timezone configuration update alters future schedule conversion.	Future events are recalculated with versioned provenance; completed events are unchanged.
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14.21 Mandatory Negative Tests

The following tests shall be included in the Phase 1 test baseline:

- Browser clock supplied as authoritative recorded time
- Application-server local time used for clinical scheduling
- Database-server timezone change altering results
- Unqualified local datetime accepted as a completed event
- Hardcoded fixed offset used instead of an IANA timezone
- User presentation preference changing historical event meaning
- Current facility timezone used to reinterpret historical records
- Device timestamp overwriting server receipt time
- Server receipt time falsely labelled as device measurement time
- Raw device time lost after normalization
- Material device drift accepted without policy evaluation
- Financial drift threshold applied automatically to high-risk Clinical events
- Ambiguous repeated-hour timestamp silently resolved
- Nonexistent local clinical event time silently advanced
- Retrospective entry presented as contemporaneous
- Signed timestamp directly edited
- Correction deleting the original timestamp
- Offline synchronization producing duplicate events
- Retry assigning a new occurrence time
- Future schedule stored only as a fixed UTC value where local-time intent must be preserved
- Timezone-service outage causing silent server-local fallback
- Stale cache used beyond approved validity
- Cross-module processor replacing Clinical occurrence time
- Export omitting timezone where interpretation is ambiguous
- Same-time events ordered nondeterministically
- Timezone configuration change rewriting completed medication administrations

14.22 Reporting Requirements

Authorized users shall be able to report:

- Clinical records with missing occurrence times
- Records where occurrence and recorded times differ beyond configured thresholds
- Late and retrospective entries
- Time corrections and overrides
- Device clock drift
- Low-confidence or estimated event times
- Ambiguous or nonexistent local times
- Events recorded using UTC fallback
- Events recorded using cached timezone resolution
- Stale timezone mappings
- Future-dated completed events
- Out-of-order external or device events
- Schedule changes caused by timezone updates
- Cross-module timestamp discrepancies

- Facilities or client spaces without valid timezone configuration
- Infrastructure clock-health exceptions

CLN-TIME-RPT-001 Temporal reports shall state the timezone used for display and grouping.

CLN-TIME-RPT-002 A presentation-timezone change shall not alter the underlying report population unless the report explicitly defines viewer-timezone grouping.

CLN-TIME-RPT-003 Audit and medico-legal exports shall include sufficient temporal provenance to reconstruct the originating event time.

14.23 Phase 1 Traceability

Area	Requirement Prefix	Primary Evidence
Ownership	CLN-TIME-001 to CLN-TIME-005	Shared-service architecture review
Core temporal rules	CLN-TIME-010 to CLN-TIME-019	Record-contract inspection
Trusted clock	CLN-TIME-020 to CLN-TIME-027	Infrastructure and clock testing
Resolution	CLN-TIME-030 to CLN-TIME-039	Multi-tenant timezone tests
Temporal snapshot	CLN-TIME-040 to CLN-TIME-045	Historical reproducibility tests
Historical events	CLN-TIME-050 to CLN-TIME-057	Timeline and ordering tests
Future schedules	CLN-TIME-060 to CLN-TIME-067	Scheduling and transition tests
Event families	CLN-TIME-070 to CLN-TIME-084	Medication, consumption, observation and transition tests
Devices	CLN-TIME-090 to CLN-TIME-099	Device simulator and drift tests
Offline	CLN-TIME-100 to CLN-TIME-106	Offline synchronization tests
Late and corrected entries	CLN-TIME-110 to CLN-TIME-117	Provenance and correction tests
DST ambiguity	CLN-TIME-120 to CLN-TIME-127	Timezone boundary tests
User interface	CLN-TIME-130 to CLN-TIME-138	Web and mobile UI tests
Integration	CLN-TIME-140 to CLN-TIME-146	Contract and interoperability tests
Resilience	CLN-TIME-150 to CLN-TIME-158	Outage and cache tests
Audit	CLN-TIME-160 to CLN-TIME-165	Audit and monitoring tests
Configuration	CLN-TIME-170 to CLN-TIME-173	Configuration governance tests
Reporting	CLN-TIME-RPT	Reporting verification

14.24 Registered Phase 1 Temporal Gaps

Gap ID	Resolution Required	Release Blocking
CLN-P1-TIME-GAP-001	Confirm Clinical registration as a formal Shared Time Engine consumer.	Yes
CLN-P1-TIME-GAP-002	Confirm authoritative facility and client-space timezone fields and APIs.	Yes
CLN-P1-TIME-GAP-003	Approve the Clinical timezone-resolution precedence.	Yes
CLN-P1-TIME-GAP-004	Define the shared timestamp serialization and precision contract.	Yes
CLN-P1-TIME-GAP-005	Approve the reusable Clinical temporal snapshot structure.	Yes
CLN-P1-TIME-GAP-006	Define risk-based device clock-drift tolerances.	Yes

CLN-P1-TIME-GAP-007	Define ambiguous and nonexistent local-time policies for future schedules.	Yes
CLN-P1-TIME-GAP-008	Define rules for high-risk completed events with uncertain device time.	Yes
CLN-P1-TIME-GAP-009	Approve offline Clinical time-confidence handling.	Yes
CLN-P1-TIME-GAP-010	Approve late-entry and time-correction permissions and reasons.	Yes
CLN-P1-TIME-GAP-011	Define Shared Time Engine cache validity and outage behaviour.	Yes
CLN-P1-TIME-GAP-012	Confirm NTP/PTP monitoring, alert ownership and unhealthy-node handling.	Yes
CLN-P1-TIME-GAP-013	Reconcile Clinical terminology of facility, entity and client space with shared “branch” terminology.	Yes
CLN-P1-TIME-GAP-014	Define printed, exported and interoperability timezone-display requirements.	Yes
CLN-P1-TIME-GAP-015	Add temporal requirements to the final requirements traceability matrix and companion EDD.	Yes

14.25 Completion Gate

The temporal component of Phase 1 shall be considered complete only when:

- Clinical is formally registered as a Shared Time Engine consumer.
- Main Module and Clinical ownership boundaries are approved.
- IANA timezone identifiers are mandatory across applicable contracts.
- No Clinical business rule depends on server-local time.
- Historical-event and future-schedule models are approved.
- The Clinical temporal snapshot is defined.
- Medication and Consumption Session timestamps are mapped.
- Observation and device-time provenance is defined.
- Late-entry and correction rules are approved.
- Device drift and time-confidence policies are approved.
- DST ambiguity and gap policies are approved.
- Offline synchronization behaviour is approved.
- Time Engine cache and outage behaviour are approved.
- API timestamp formats and precision are approved.
- Acceptance and negative tests are assigned.
- Every registered temporal gap has an accountable owner.
- The companion Clinical EDD contains an implementation and migration design.

Phase rule: No later Clinical phase may weaken the distinction between event time, source time, received time, recorded time, verification time, signing time, correction time and downstream processing time without formal impact assessment and approval.

Appendix: Minimum Temporal Record

Group	Minimum content
Event identity	Event public ID, event type, source module and source-event ID
Subject context	Tenant, entity, facility, client space, patient and encounter where applicable
Occurrence	UTC occurrence or effective instant, or period start and end
Local interpretation	Local datetime, IANA timezone and UTC offset

Resolution	Timezone source type, source public ID and configuration version
Source	User, device, external system or trusted server
Processing	Received, recorded and processed times where applicable
Attestation	Authored, verified, signed and witnessed times where applicable
Correction	Original time, corrected time, correction-recorded time and reason
Confidence	Precision, confidence, estimated indicator and adjustment indicator
Device	Device ID, raw time, sequence, drift and clock status where applicable
Provenance	Actor, correlation ID, idempotency key, version and audit reference

This complete section provides the Phase 1 governing contract. Phase 6 and Phase 7 should only add event-specific rules for medication, consumption, observations and devices, while Phase 10 should define the integration and production-assurance obligations.

15. Shared AI Gateway Contract

CLN-AI-001 Clinical shall send AI requests only through registered Shared AI Gateway services.

CLN-AI-002 Clinical shall send the minimum necessary data for the authorized purpose.

CLN-AI-003 An AI request shall include tenant, requesting module, requesting user, service code, correlation ID and approved context metadata.

CLN-AI-004 AI-generated notes, extracted observations, diagnoses, order suggestions, summaries and protocol suggestions shall remain drafts or suggestions.

CLN-AI-005 AI shall not approve, sign, publish, prescribe, place an order, release a result, activate a protocol or execute a major transition.

CLN-AI-006 The accepting user shall independently possess the permission, scope, client-space relationship and privilege required for the resulting action.

CLN-AI-007 Clinical shall preserve the AI transaction reference, model/provider provenance supplied by the Gateway, prompt-template version where supplied, user edits and acceptance or rejection outcome.

CLN-AI-008 If the AI Gateway is unavailable, ordinary authenticated manual workflows shall remain available where safe.

16. Shared Service Availability and Safe Failure

Dependency	Safe-Failure Rule
Identity/authentication	No protected Clinical action without authenticated identity.
Permission service	Deny restricted action unless approved cached entitlement remains valid and policy permits cache use.
Client-space assignment	Deny ordinary client-space write if current assignment cannot be established.
Unit Engine	Do not commit a value requiring unavailable conversion; allow already verified canonical display where safe.
AI Gateway	Continue manual workflows; do not fabricate AI output.
Inventory	Do not promise stock truth; stage or display unavailable status.
LIMS/Imaging	Preserve order and follow-up; do not invent result state.
Audit service	High-risk write shall fail if its required audit record cannot be durably recorded.

CLN-RES-001 Every shared dependency shall define timeout, retry, idempotency, circuit-breaker and reconciliation requirements in the EDD.

CLN-RES-002 Clinical shall distinguish unavailable, unauthorized, not found, stale and invalid responses.

CLN-RES-003 A retry shall not duplicate a signed note, order, administration, transition or audit event.

16. Audit, Provenance and Historical Reproducibility

CLN-AUD-001 Clinical shall maintain immutable audit evidence for access, create, update, sign, approve, cancel, correct, override, export, break-glass and configuration actions.

CLN-AUD-002 Audit evidence shall record actor, impersonation or delegation context, tenant, facility, client space, patient, encounter, resource, action, outcome, timestamp, reason, correlation ID, device/client, permission used and policy version where applicable.

CLN-AUD-003 Denied high-risk attempts shall be auditable without leaking protected data to the denied user.

CLN-AUD-004 Clinical provenance shall distinguish author, recorder, verifier, signer, approver and system process.

CLN-AUD-005 Corrections shall preserve the original content and link the replacement, reason and authorizing action.

CLN-AUD-006 Audit and provenance shall remain queryable after users, titles, units, policies or client-space assignments change.

17. Configuration Interfaces

17.1 Main Module Configuration Dependencies

- Official titles and employment state
- Atomic permission registry
- Permission bundles and versions
- Title-to-bundle mappings
- User and role assignments
- Facility and client-space records
- User client-space assignments
- Credentials and privileges
- Delegations and restrictions
- Shared Unit Engine

17.2 Clinical Configuration

- Clinical permission metadata and required context
- Care relationship types
- Clinical privilege requirements
- Break-glass categories and review rules
- Sensitive-record labels and restrictions
- CDE unit-use policies
- Clinical audit severity
- Safe-cache and outage policies
- AI service enablement by workflow
- Cross-module status mappings

CLN-CFG-001 Every configuration page shall require a dedicated management permission and shall not be exposed by ordinary clinical title bundles.

CLN-CFG-002 Configuration changes shall support draft, review, approval, activation, deprecation and historical resolution where the change can affect clinical safety or access.

CLN-CFG-003 No configuration shall be activated when required references are missing or invalid.

18. Operational Interfaces

CLN-UI-001 Clinical navigation shall be derived from effective permissions but server authorization shall remain authoritative.

CLN-UI-002 The user shall see the active facility, client space, assignment type and temporary authority context.

CLN-UI-003 Patient lists shall be filtered by effective assignment and relationship rather than showing unrestricted facility-wide lists by default.

CLN-UI-004 A denial shall provide a usable reason category such as missing permission, expired assignment, wrong client space, missing relationship, record locked or additional privilege required, without exposing restricted content.

CLN-UI-005 The interface shall clearly identify break-glass mode and its expiry.

CLN-UI-006 Mobile and web interfaces shall enforce the same authorization outcomes.

19. Notifications and Events

CLN-EVT-001 Clinical shall consume user, title, assignment, permission, client-space, credential and unit-policy changes through approved versioned interfaces or events.

CLN-EVT-002 Consumers shall use event IDs and idempotent processing.

CLN-EVT-003 Security-sensitive changes including account suspension, assignment expiry, privilege suspension and break-glass activation shall trigger prompt cache invalidation.

CLN-EVT-004 Notifications shall not include restricted clinical detail unless the recipient is authorized to receive it.

- user.status.changed
- user.official_title.changed
- permission.assignment.changed
- client_space.assignment.changed
- credential.status.changed
- unit.version.activated
- unit.retired
- clinical.break_glass.activated
- clinical.break_glass.expired
- clinical.access.review.required

20. Reporting and Oversight

CLN-RPT-001 Authorized administrators shall be able to report effective permissions by user, title, bundle, facility and client space.

CLN-RPT-002 The system shall report users with permissions but no active operational assignment, and users with assignments but missing required permissions or credentials.

CLN-RPT-003 The system shall report expiring assignments, credentials, delegations and privileges.

CLN-RPT-004 Break-glass reports shall show reason, scope, duration, accessed resources, actions and review outcome.

CLN-RPT-005 Reports and exports shall themselves be permission- and scope-controlled and audited.

21. Acceptance Scenarios

ID	Scenario	Expected Result
PH1-AT-001	Registered Nurse with observation permission, active ward assignment and qualifying patient relationship records temperature.	Allowed; action and context audited.
PH1-AT-002	Same nurse attempts observation entry for a patient in an unassigned ICU.	Denied with wrong-client-space reason.
PH1-AT-003	Medical Officer has title bundle but prescribing privilege is expired.	Prescribing action denied; ordinary permitted documentation remains available.
PH1-AT-004	User receives approved cross-cover assignment for a night shift.	Configured authority is active only for approved scope and time.
PH1-AT-005	Unassigned clinician invokes break glass during resuscitation.	Time-limited minimum access after required controls; reviewer notified.
PH1-AT-006	Main Module permission service is unavailable.	Restricted write fails closed unless an approved current cached entitlement rule applies.
PH1-AT-007	Glucose entered in mg/dL where base is	Clinical verifies permitted unit and Main

	mmol/L.	Unit Engine conversion; stores source, normalized value and provenance.
PH1-AT-008	Unit Engine cannot verify conversion.	Save blocked or inbound result staged; no silent conversion.
PH1-AT-009	AI suggests an order for a user lacking order-sign permission.	Suggestion may be displayed as policy permits, but cannot be committed or signed.
PH1-AT-010	Sensitive chart appears in search for ordinary user.	No restricted snippet or content is exposed; access governed by additional controls.
PH1-AT-011	Official title changes.	Effective baseline bundle recalculated and stale authorization invalidated.
PH1-AT-012	Assignment expires while user remains logged in.	Ordinary client-space authority is removed promptly; historical work remains intact.

22. Mandatory Negative Tests

- Cross-tenant public ID substitution
- Direct API call when UI button is hidden
- User with permission but no client-space assignment
- User with client-space assignment but no permission
- User with correct title but expired credential
- Access using inactive encounter
- Access after transfer to another client space
- Self-assignment without authority
- Delegation beyond delegator authority
- Delegation after expiry
- Break-glass without reason
- Break-glass after expiry
- Sensitive data leakage in search, alerts or notifications
- Unit label substitution without canonical ID
- Molar-to-mass conversion without analyte context
- AI response committed without human authorization
- Queue worker acting without recorded service identity and tenant
- Replay of a previously accepted high-risk request

23. Phase 1 Traceability

Area	Requirement Prefix	Primary Evidence
Foundation	CLN-FND	Architecture review
Ownership	CLN-OWN	Source-of-truth contract tests
Context	CLN-CTX	Tenant and client-space tests
Authorization	CLN-AUTH	Policy and negative tests
Permissions	CLN-PERM	Catalogue and bundle tests
Scope	CLN-SCOPE	Object-level access tests
Relationships	CLN-REL	Care-assignment tests
Privileges	CLN-PRIV	Credential-expiry tests
Delegation	CLN-DEL	Time and scope tests
Break glass	CLN-BTG	Emergency and after-action tests
Consent	CLN-CONS	Sensitive-data tests
Units	CLN-UOM	Conversion contract tests
AI	CLN-AI	Human-approval tests
Resilience	CLN-RES	Outage and idempotency tests
Audit	CLN-AUD	Immutable evidence tests

24. Registered Phase 1 Gaps

Gap ID	Resolution Required	Release Blocking
CLN-P1-GAP-001	Confirm current Main Module permission schema and APIs.	Yes
CLN-P1-GAP-002	Confirm authoritative source and exact model for user client-space assignments.	Yes
CLN-P1-GAP-003	Approve initial atomic permission catalogue and official-title bundle templates.	Yes
CLN-P1-GAP-004	Approve credential and privilege registry ownership.	Yes
CLN-P1-GAP-005	Approve care-relationship types and relationship creation authority.	Yes
CLN-P1-GAP-006	Approve sensitivity-label and consent policy for applicable jurisdictions.	Yes
CLN-P1-GAP-007	Approve break-glass categories, time limits and reviewers.	Yes
CLN-P1-GAP-008	Approve resilient cache rules for permission and assignment outages.	Yes
CLN-P1-GAP-009	Reconcile all clinical_uom_* references to the Main Module Unit Engine.	Yes
CLN-P1-GAP-010	Define service identities and scopes for background jobs and integrations.	Yes

25. Phase 1 Completion Gate

- Main Module, Clinical, HR and integration ownership boundaries approved.
- Atomic permission naming rules approved.
- Title-to-bundle model approved as configurable, not hardcoded.
- Client-space assignment established as mandatory for ordinary location-bound work.
- Care relationship and scope intersection approved.
- Credential, delegation, restriction and break-glass models approved.
- Shared Unit Engine contract accepted and local writable unit masters marked for reconciliation.
- Shared AI Gateway human-approval rule accepted.
- Audit and safe-failure requirements accepted.
- Every registered Phase 1 gap assigned an owner and acceptance test.

Phase 2 shall not redefine or weaken Phase 1 authorization, source-of-truth or shared-service rules without a formally approved change and impact assessment.

Appendix A. Initial Clinical Permission Metadata Template

Field	Required Content
Permission code	Stable clinical.* action code
Display name	Human-readable action
Resource family	Patient, note, observation, order, medication, result, transition, settings and so on
Action	View, create, update, sign, approve, cancel, correct, override, export or administer
Risk tier	Routine, elevated, high or critical
Default scope	Narrowest recommended scope
Client-space required	Yes, no or conditional
Relationship required	Relationship types or conditional rule
Credential required	Credential or privilege code

Break-glass eligible	Yes or no with category
Audit level	Standard, enhanced or critical
Separation of duty	Conflicting permissions or creator-approver restriction
Status and version	Lifecycle and effective dates

Appendix B. Sources and Standards Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module v6.1 working draft.
- KashTre Main Module Shared Unit Engine Contract v1.0 Draft.
- KashTre Shared AI Services Gateway SRD v3.1.
- KashTre HR Module SRD v6.25.
- HL7 EHR System Functional Model guidance on emergency access, consent, audit and access control.
- OWASP authorization guidance on deny by default, least privilege, server-side enforcement and testing.

Appendix C. Review Record

Role	Name	Decision	Date	Comments
Clinical Product Owner				
Main Module Architect				
Clinical Architect				
HR Architect				
Security Lead				
Database Lead				
API Lead				
Quality Assurance				
Clinical Governance				
Data Protection/Legal				

PHASE 2

PATIENT, ENCOUNTER AND CLINICAL WORKSPACE FOUNDATION

Phase 2 defines how users identify the correct patient, select and manage the correct encounter, resolve patient location and responsibility, and work safely within patient lists, ward views and the longitudinal chart.

Document Control and Phase Authority

Field	Rule
Purpose	Define patient identity use, encounter context, patient-space context and Clinical workspace behavior.
Dependency	All Phase 1 ownership, authorization, client-space, care-relationship, unit, AI, audit and safe-failure rules remain mandatory.
Functional boundary	This phase defines workspace foundations. Detailed notes, diagnoses, orders, medication, observations and results are specified in later phases.
Baseline	Clinical SRD v6.0 and addendum, corrected through Phase 1.
Implementation	Functional requirements only. Laravel schemas, policies, APIs and Vue components belong in the companion EDD.
Release state	Not approved for production until consolidated SRD, EDD and acceptance gates are approved.

Phase 2 outcomes

- Prevent wrong-patient and wrong-encounter work.
- Give each user a contextually filtered work surface.
- Preserve continuous patient history without merging distinct encounters.
- Represent patient movement and location without confusing client spaces with organizational ownership.
- Make alerts, care responsibility, chart state and confidentiality visible without disclosing restricted content.
- Specify concurrency, closure, reopening, downtime and reconciliation behavior.

1. Governing Principles

CLN-P2-GOV-001 The system shall require an explicit patient and encounter context before committing encounter-bound clinical information.

CLN-P2-GOV-002 The system shall distinguish patient identity, patient record, encounter, episode, visit, appointment, client space, patient space, bed and care relationship.

CLN-P2-GOV-003 The patient banner shall remain visible throughout patient-context work and shall not be replaced by a generic page title.

CLN-P2-GOV-004 The system shall use at least two configured patient identifiers before high-risk patient-specific actions and shall not use room or bed number as a patient identifier.

CLN-P2-GOV-005 An active encounter shall not imply unrestricted access; all Phase 1 authorization checks remain required.

CLN-P2-GOV-006 A longitudinal chart shall aggregate authorized history while preserving source encounter, author, time, state and provenance for every item.

CLN-P2-GOV-007 Patient movement shall change location context prospectively and shall not rewrite the location attached to completed historical actions.

CLN-P2-GOV-008 The user interface shall fail safely when identity, encounter, assignment or movement information is stale, contradictory or unavailable.

Mandatory safety rule: no clinical action may be saved merely because a browser tab already contains a patient record. The server shall revalidate patient, encounter, authority and record state at commit time.

2. Conceptual Model and Definitions

Concept	Definition	Key Rule
Person	A human represented by authoritative identity services.	One person may have one or more patient identifiers under governed matching rules.
Patient	The care-recipient record used for healthcare activities.	Patient identity is not an encounter.

Patient identifier	Stable identifier with assigning authority and type.	Display labels and names are not identifiers.
Encounter	A bounded interaction for healthcare service or health assessment.	Must carry status, class/type, patient and service context.
Episode of care	A configured grouping of related encounters.	Does not erase encounter boundaries.
Appointment	Planned service occurrence.	Does not prove that care occurred.
Client space	Operational service location or virtual service context.	Owned by Main Module; used for access and work routing.
Patient space	Patient's effective location within an encounter.	Derived from movements, not manually assumed.
Bed/slot	A resource within a patient space.	Never a patient identifier.
Care relationship	Effective responsibility link between user/team/role and patient/encounter.	Required where Phase 1 policy specifies.
Chart	Authorized longitudinal projection of clinical information.	Not a single mutable document.
Workspace	Role- and context-filtered UI for current work.	Must not enlarge authorization scope.

3. Source-of-Truth Responsibilities

3.1 Main Module

CLN-P2-OWN-001 The Main Module shall remain authoritative for patient demographic identity where configured, enterprise patient identifiers, users, facilities, client spaces and location registries.

CLN-P2-OWN-002 Clinical shall consume stable public identifiers and assigning-authority metadata and shall not manufacture substitute enterprise identifiers.

CLN-P2-OWN-003 Client-space names, hierarchy and active status shall be consumed from the Main Module.

3.2 Clinical Module

CLN-P2-OWN-010 Clinical shall own clinical encounter state, patient clinical workspace projections, care relationships, clinical chart state and clinical movement consequences within its boundary.

CLN-P2-OWN-011 Clinical shall preserve links to authoritative patient, facility, client-space and user IDs without duplicating those masters as editable registries.

3.3 Other modules

CLN-P2-OWN-020 Appointments, billing clearance, bed administration, laboratory fulfilment, imaging fulfilment and inventory availability shall remain owned by their authoritative modules while Clinical consumes verified status and references.

CLN-P2-OWN-021 A remote module status shall not be rewritten locally as though Clinical were the source of truth.

4. Patient Search, Discovery and Selection

4.1 Authorized search

CLN-P2-PTS-001 Patient search shall require clinical.patient.search and an allowed tenant, facility or other configured scope.

CLN-P2-PTS-002 Searchable fields may include approved patient identifier, name, date of birth, telephone, appointment/visit reference and other configured demographic attributes.

CLN-P2-PTS-003 Search results shall be minimum-necessary and shall not reveal diagnoses, sensitive labels, notes or full addresses unless separately authorized.

CLN-P2-PTS-004 The system shall distinguish exact identifier matches from broader demographic matches.

CLN-P2-PTS-005 Users shall not search across tenants. Cross-entity search shall require a specific governance permission and purpose.

CLN-P2-PTS-006 Search shall handle aliases, spelling variants and previous names only through authoritative identity data and shall not auto-merge records.

4.2 Result presentation

Displayed Field	Default Rule
Patient identifier	Show approved masked or full value according to policy.
Name	Show authoritative display name and configured name-use indicator.
Date of birth/age	Show enough information for differentiation.
Sex/other demographic fields	Show only configured identity data needed for safe differentiation.
Current encounter	Show authorized status, class and service location.
Photograph	Optional, policy-controlled and not a substitute for identifiers.
Sensitive status	Show a neutral restriction indicator without exposing the restricted reason.
Deceased indicator	Prominent where authoritative and permitted.

CLN-P2-PTS-010 Selecting a result shall open a confirmation step or patient banner before a patient-specific action begins.

CLN-P2-PTS-011 The system shall warn when several records have materially similar demographics.

CLN-P2-PTS-012 The user shall not be allowed to merge, split or correct enterprise patient identity from the Clinical workspace unless an explicit authoritative workflow and permission exist.

5. Positive Patient Identification

CLN-P2-ID-001 Before medication administration, specimen collection, blood-product activity, image/procedure initiation, invasive procedure, transfer acceptance, discharge completion and other configured high-risk actions, the user shall confirm at least two approved identifiers.

CLN-P2-ID-002 Approved identifiers shall be configured and may include patient name, date of birth, enterprise patient number or another identity attribute. Bed, room and ward shall not count as patient identifiers.

CLN-P2-ID-003 Barcode, wristband, biometric or device-assisted identification may support verification but shall not bypass mismatch handling.

CLN-P2-ID-004 If scanned identity does not match the active chart, the system shall block the action, preserve both references in a security-safe event and require resolution.

CLN-P2-ID-005 When a patient cannot reliably provide identity, the system shall support configured emergency, newborn, unconscious and unknown-patient processes with temporary identifiers and later reconciliation.

CLN-P2-ID-006 Temporary identifiers shall be visually distinct and shall not be reused.

CLN-P2-ID-007 Identity confirmation shall occur again when the patient context changes, not merely once per login session.

5.1 Wrong-patient prevention controls

- Persistent patient banner
- Two-identifier confirmation for configured actions
- Visual differentiation of similar names
- Recent-patient list with identifiers and locations
- Tab title containing safe configured patient context
- Mismatch blocking
- Confirmation after long inactivity
- Server revalidation at commit
- Audit of patient-context switches
- No silent carry-over of draft data between patients

6. Encounter Creation and Activation

6.1 Encounter classes

Class	Examples	Required Context
OUTPATIENT	General or specialty clinic visit	Service, facility, date/time and responsible context
EMERGENCY	Emergency intake and care	Emergency service, acuity context and arrival time
INPATIENT	Admission and stay	Admitting service, facility and location/movement context
DAY_CASE	Same-day planned care	Procedure/service and expected period
VIRTUAL	Telehealth or remote care	Virtual client space and remote-service authority
HOME/COMMUNITY	Authorized field or home service	Service area and assigned team
OBSERVATION	Time-limited monitored encounter	Observation service and review plan

CLN-P2-ENC-001 Encounter classes, types and status values shall be configurable through controlled dictionaries and mapped to interoperability codes where approved.

CLN-P2-ENC-002 Encounter creation shall require clinical.encounter.create and appropriate service/facility scope.

CLN-P2-ENC-003 Each encounter shall record patient, unique public ID, identifier, class, type, status, service, facility, planned and actual period, source/referral, reason, responsible clinician/team where known, initial client space where applicable and audit metadata.

CLN-P2-ENC-004 Appointment conversion shall preserve the appointment reference but shall not treat a scheduled appointment as completed clinical attendance.

CLN-P2-ENC-005 The system shall check for duplicate or overlapping encounters according to configurable rules and present existing encounters before creating another.

CLN-P2-ENC-006 An authorized user may create an emergency encounter with minimum safe data where policy permits; missing information shall enter a visible completion queue.

7. Encounter Status Lifecycle

Status	Meaning	Permitted Direction Examples
PLANNED	Expected future encounter	ARRIVED, CANCELLED
ARRIVED	Patient has presented	TRIAGE/IN_PROGRESS, CANCELLED, ENTERED_IN_ERROR
TRIAGE	Triage activity active	IN_PROGRESS, ON_HOLD
IN_PROGRESS	Clinical care active	ON_HOLD, TRANSFER_PENDING, FINISHED
ON_HOLD	Temporarily paused	IN_PROGRESS, FINISHED
TRANSFER_PENDING	Movement awaiting completion	IN_PROGRESS, TRANSFERRED
FINISHED	Clinical activity ended, closure checks pending	CLOSED, REOPENED_BY_EXCEPTION
CLOSED	Final operational closure	REOPENED only by approved workflow
CANCELLED	Planned/arrived encounter cancelled	No clinical completion; correction rules apply
ENTERED_IN_ERROR	Encounter created for wrong subject/context	Immutable evidence plus corrective link

CLN-P2-ENC-010 Status transitions shall be explicit, permission-controlled and audited.

CLN-P2-ENC-011 A status change shall validate mandatory prerequisites and outstanding safety items defined by later phases.

CLN-P2-ENC-012 Cancellation and entered-in-error shall be distinct. Cancellation describes a legitimate encounter that did not proceed; entered-in-error corrects an invalid record.

CLN-P2-ENC-013 Closing an encounter shall not delete pending results, follow-up obligations or historical tasks.

CLN-P2-ENC-014 The system shall reject illegal transitions and return a structured reason.

8. Encounter Selection and Context Lock

CLN-P2-CTX-001 When a patient has more than one encounter, the system shall require the user to select or confirm the applicable encounter before encounter-bound entry.

CLN-P2-CTX-002 The active encounter context shall display encounter ID, class/type, status, service, facility, current patient space and period.

CLN-P2-CTX-003 Draft information shall remain bound to the patient and encounter on which it was created.

CLN-P2-CTX-004 Switching patient or encounter shall warn about unsaved work and shall never copy draft values silently.

CLN-P2-CTX-005 Deep links and notifications shall be re-authorized and shall confirm current patient and encounter context before display.

CLN-P2-CTX-006 Browser back/forward navigation shall not restore an unauthorized or stale patient context without revalidation.

CLN-P2-CTX-007 A closed encounter may be viewed according to permission but may not receive ordinary new entries unless the specific entry type supports a late-entry or addendum workflow.

9. Patient Banner and Safety Header

Banner Element	Requirement
Identity	Authoritative display name plus at least one prominent identifier and configured second identifier.
Age context	Date of birth and calculated age; neonatal/pediatric precision where configured.
Encounter	Class/type, status, visit identifier and start date/time.
Location	Facility, client space, room/bed or virtual context, with effective timestamp.
Responsibility	Primary/attending clinician, assigned nurse/team where available.
Safety alerts	Allergy, critical warning, isolation, fall/pressure/VTE or other configured indicators without replacing detailed review.
Confidentiality	Restriction indicator and emergency-access state.
Special state	Deceased, unknown identity, temporary identifier, duplicate-review pending or encounter closed.
Actions	Context-safe access to overview, timeline and permitted workflows.

CLN-P2-BNR-001 The banner shall remain visible or immediately accessible on every patient-context page, modal and mobile workflow.

CLN-P2-BNR-002 Banner content shall refresh when encounter, movement, alert or care-responsibility events change.

CLN-P2-BNR-003 Colour alone shall not convey safety-critical meaning.

CLN-P2-BNR-004 The banner shall not expose restricted details to an unauthorized user.

CLN-P2-BNR-005 Printed and exported clinical documents shall include approved patient and encounter identifiers on every page where required.

10. Patient Location, Movement and Patient Space

10.1 Location hierarchy

CLN-P2-LOC-001 Clinical shall consume Main Module location and client-space identifiers and shall retain the distinction between facility, client space, room, bed/slot and virtual location.

CLN-P2-LOC-002 Each movement shall record source, destination, requested time, effective time, status, actor, reason, encounter, transport/clinical readiness references where applicable and correlation ID.

CLN-P2-LOC-003 A requested or reserved destination shall not become the patient's effective location until the configured acceptance/completion event occurs.

CLN-P2-LOC-004 The current patient space shall be derived from the latest valid completed movement, with detection of overlaps and gaps.

CLN-P2-LOC-005 Clinical actions shall store the effective patient location at action time when location is clinically relevant.

10.2 Movement states

State	Meaning
REQUESTED	Movement requested but not accepted.
ACCEPTED	Receiving destination agrees; patient not yet moved.
IN_TRANSIT	Movement underway where used.
COMPLETED	Destination becomes effective patient space.
REJECTED	Destination declined with reason.
CANCELLED	Valid request withdrawn before completion.
ENTERED_IN_ERROR	Invalid movement retained with correction linkage.

CLN-P2-LOC-010 Movement completion shall trigger recalculation of patient lists, client-space access, care assignments, task routing and alerts without rewriting prior history.

CLN-P2-LOC-011 A user assigned only to the source space shall lose ordinary new-write authority after completed transfer, subject to configured handover and late-entry rules.

CLN-P2-LOC-012 A receiving user shall not gain ordinary authority merely because a transfer was requested; configured assignment and relationship rules still apply.

11. Patient Lists and Worklist Derivation

List	Inclusion Basis
My Patients	Active individual responsibility or accepted referral.
My Current Work	Assigned tasks, due observations, medication windows and follow-up within authorized scope.
My Team	Active care-team relationships.
My Client Space	Current patients in spaces where the user has active operational assignment and permission.
Covering Patients	Effective cross-cover or on-call assignment.
Consultations	Accepted consultation/referral relationship.
Recent Patients	Recently accessed patients within retention and authorization rules.
Exceptions	Authorized unresolved identity, movement, result, task or chart-state problems.

CLN-P2-LST-001 Lists shall be projections derived from authoritative current state and shall not become separate manual sources of truth.

CLN-P2-LST-002 List membership shall never override authorization at chart-open or action-commit time.

CLN-P2-LST-003 Filters shall include approved patient, encounter, location, care team, acuity, task, status and date attributes.

CLN-P2-LST-004 Sensitive patients shall not be revealed through unauthorized list inclusion, counts or filter facets.

CLN-P2-LST-005 The UI shall display when information is delayed or last refreshed.

CLN-P2-LST-006 Bulk selection shall be restricted to specifically permitted low-risk actions and shall never enable bulk signature or indiscriminate chart export.

12. Ward and Client-Space Workspace

12.1 Census header

CLN-P2-WRD-001 An authorized client-space workspace shall show current census counts derived from effective patient-space records and location status.

CLN-P2-WRD-002 Counts shall distinguish occupied, reserved, available, unavailable/maintenance and configured overflow capacity where those concepts are supported.

CLN-P2-WRD-003 Census discrepancies shall be visible through an exception queue and shall not be silently repaired.

12.2 Bed and slot view

CLN-P2-WRD-010 A bed/slot card may show minimum-necessary patient identity, encounter, acuity and responsibility information only to authorized users.

CLN-P2-WRD-011 Clicking an occupied bed shall not bypass patient and encounter authorization.

CLN-P2-WRD-012 An available bed action shall initiate the authoritative allocation or movement workflow rather than directly changing patient location.

CLN-P2-WRD-013 Overflow spaces shall use controlled temporary location records with unique identifiers, effective periods, inherited policies defined explicitly, and retirement workflow.

CLN-P2-WRD-014 A retired overflow space shall remain historically resolvable.

13. Care Team and Responsibility Panel

CLN-P2-CARE-001 The workspace shall show current primary clinician, primary nurse, assigned team, covering users, accepted consultants and allied-health relationships where authorized.

CLN-P2-CARE-002 Each relationship shall display role/responsibility, effective period, source and handover state.

CLN-P2-CARE-003 Users shall not edit care relationships unless they hold the exact create, transfer, accept or end permission required by the workflow.

CLN-P2-CARE-004 Replacing a primary responsible person shall preserve the prior relationship and record handover acceptance where required.

CLN-P2-CARE-005 An unavailable responsible clinician shall trigger configured cover or escalation workflows; the system shall not silently assign a substitute.

14. Clinical Chart Workspace

14.1 Default sections

- Overview and safety summary
- Active encounter
- Longitudinal timeline
- Problems and diagnoses
- Allergies and intolerances
- Medications and medication administration
- Orders and requests
- Results
- Observations and scores
- Notes and documentation
- Care plans and tasks
- Procedures
- Referrals and consultations
- Patient movements and transitions
- Documents and media
- Audit/provenance view for authorized users

CLN-P2-CHT-001 Sections shall be permission-filtered and shall not expose section counts or previews when the section itself is restricted.

CLN-P2-CHT-002 The chart shall clearly distinguish active-encounter information from historical information.

CLN-P2-CHT-003 The chart shall not aggregate conflicting facts into a single unqualified value. Source, time, status and encounter shall remain available.

CLN-P2-CHT-004 Unsigned drafts shall be visually and technically distinct from signed clinical records.

CLN-P2-CHT-005 Corrected, amended and entered-in-error content shall remain linked and visibly state its current legal/clinical status.

CLN-P2-CHT-006 Chart navigation shall preserve patient and encounter context while enforcing authorization on every data request.

14.2 Forms and Clinical Charts

CLN-P2-CHT-007 The patient workspace shall provide a Forms and Clinical Charts section containing active, available, completed, discontinued, superseded and historical forms applicable to the patient and encounter.

CLN-P2-CHT-008 A rendered form shall retrieve applicable authoritative Observations and other approved records rather than require duplicate clinical data entry.

CLN-P2-CHT-009 An authorized user shall be able to select Add Form and activate an eligible approved Form Definition for the patient or encounter.

CLN-P2-CHT-010 Adding a form shall activate its presentation and applicable observation requirements but shall not modify defaults for other patients or the governing client-space configuration.

CLN-P2-CHT-011 A patient-added form shall remain historically associated with the patient and encounter after completion, discontinuation, supersession or entry in error.

15. Longitudinal Timeline

CLN-P2-TML-001 The timeline shall present authorized events by clinically meaningful time while preserving recorded time, effective time and source time where they differ.

CLN-P2-TML-002 Events shall include type, status, author/source, encounter, facility/client space and provenance link.

CLN-P2-TML-003 Filters shall include encounter, event type, service, author/team, date range, status and location.

CLN-P2-TML-004 The timeline shall distinguish preliminary, final, corrected, cancelled and entered-in-error states.

CLN-P2-TML-005 Late-arriving data shall appear according to clinical event time with a visible received/recorded timestamp and shall not impersonate contemporaneous entry.

CLN-P2-TML-006 Same-time ordering shall use deterministic tie-breaking and shall not imply causation.

CLN-P2-TML-007 The timeline shall support source-object navigation only when the user is authorized for that source object.

16. Alerts, Allergies and Critical Context

CLN-P2-ALT-001 The patient workspace shall show active safety alerts appropriate to the user's permissions and workflow context.

CLN-P2-ALT-002 Alerts shall identify severity, source, status, effective period, verification state and required action where applicable.

CLN-P2-ALT-003 Acknowledgement shall not be treated as resolution unless the alert workflow explicitly defines that transition.

CLN-P2-ALT-004 Alert fatigue controls shall not suppress critical alerts without approved policy and audit.

CLN-P2-ALT-005 Allergy and intolerance summaries shall preserve substance, reaction, severity, verification and source information when available.

CLN-P2-ALT-006 The workspace shall show stale, unverified or conflicting safety information as such rather than selecting one silently.

17. Concurrent Access, Drafts and Conflict Handling

CLN-P2-CON-001 The system shall permit appropriate multidisciplinary concurrent viewing and documentation while preventing unsafe lost updates.

CLN-P2-CON-002 Editable resources shall use version or concurrency tokens and reject updates based on stale versions where overwrite would lose information.

CLN-P2-CON-003 A user shall be shown a clear conflict summary and shall not be asked to choose between anonymous raw payloads.

CLN-P2-CON-004 Drafts shall record owner, patient, encounter, type, last saved time, version and device/session context.

CLN-P2-CON-005 Auto-save shall not sign, finalize or publish content.

CLN-P2-CON-006 Draft recovery shall require renewed authorization and shall never recover a draft into another patient or encounter.

CLN-P2-CON-007 The presence of another viewer shall not disclose that viewer's identity unless policy permits.

18. Encounter Closure, Reopening and Late Information

CLN-P2-CLS-001 Encounter closure shall require `clinical.encounter.close` and completion of configured closure checks.

CLN-P2-CLS-002 Closure checks shall identify outstanding orders, results, medication reconciliation, follow-up, documentation, movement and task obligations defined in later phases.

CLN-P2-CLS-003 The system shall distinguish finished from closed so that operational completion checks can occur before final closure.

CLN-P2-CLS-004 Closed encounters shall be read-only except for separately authorized late-entry, addendum, correction, result-arrival and follow-up workflows.

CLN-P2-CLS-005 Reopening shall require a dedicated permission, reason, effective scope and audit; it shall not erase the original closure event.

CLN-P2-CLS-006 A new episode of care shall not be recorded as reopening merely to avoid creating the correct new encounter.

CLN-P2-CLS-007 Post-discharge results and obligations shall remain linked to the original encounter while producing new follow-up tasks where required.

19. Wrong-Patient, Duplicate and Identity-Correction Handling

CLN-P2-COR-001 Clinical shall provide a report-wrong-patient or identity-concern action without allowing ordinary users to merge records directly.

CLN-P2-COR-002 Suspected duplicate, overlay or wrong-patient conditions shall place affected high-risk actions into a controlled review state where safe and configured.

CLN-P2-COR-003 Identity correction shall preserve old and new identifiers, authoritative decision, actor, reason, effective time and impacted-object reconciliation status.

CLN-P2-COR-004 Moving a clinical record from one patient to another shall require an exceptional governed workflow, dual review where configured and complete provenance.

CLN-P2-COR-005 The system shall produce an impact list covering notes, orders, results, medications, specimens, images, billing references and audit events before correction completion.

CLN-P2-COR-006 Downstream modules shall receive idempotent correction events and reconciliation status shall be monitored.

20. Downtime, Offline and Recovery

CLN-P2-DWN-001 Downtime identity and encounter workflows shall use approved local policy and shall preserve temporary identifiers, time, facility, client space and responsible user/device.

CLN-P2-DWN-002 Offline patient lists shall contain only the minimum data required for approved care scope and shall expire according to policy.

CLN-P2-DWN-003 An offline action shall not assume that patient location, encounter status or authority remained unchanged; synchronization shall revalidate and reconcile.

CLN-P2-DWN-004 Conflicts shall enter a review queue rather than silently overwriting online changes.

CLN-P2-DWN-005 The system shall distinguish data captured offline from data committed online and preserve both timestamps.

CLN-P2-DWN-006 Temporary unknown-patient and encounter records shall be reconciled through an approved identity workflow after service restoration.

21. Mobile and Remote Workspace

CLN-P2-MOB-001 Mobile patient lists and chart views shall enforce the same Phase 1 permissions, client-space assignment and care-relationship rules as web interfaces.

CLN-P2-MOB-002 The mobile interface shall keep patient identity and active encounter context visible during action entry.

CLN-P2-MOB-003 Device notifications shall minimize protected content and require authenticated opening before showing patient details.

CLN-P2-MOB-004 Remote-service users shall require an active `REMOTE_SERVICE` assignment or other approved context and shall not inherit facility-wide access from remote login alone.

CLN-P2-MOB-005 Screenshots, local caching, clipboard and background previews shall follow configured mobile security policy.

22. Permissions Catalogue for Phase 2

Permission	Purpose	Typical Extra Context
clinical.patient.search	Search authorized patient registry	Tenant/facility scope
clinical.patient.chart.view	Open authorized chart	Client space or care relationship
clinical.patient.chart.view_sensitive	View restricted chart content	Sensitivity clearance
clinical.patient.identity.concern_report	Report possible wrong identity	Patient context
clinical.encounter.create	Create encounter	Facility/service scope
clinical.encounter.view	View encounter	Authorized patient scope
clinical.encounter.update	Update permitted encounter metadata	Record state
clinical.encounter.status_change	Perform ordinary status transition	Transition rule
clinical.encounter.close	Close encounter	Closure checks
clinical.encounter.reopen	Reopen closed encounter	Reason and elevated authority
clinical.movement.request	Request movement	Source scope
clinical.movement.accept	Accept into destination	Destination scope
clinical.movement.complete	Complete movement	Handover and location checks
clinical.care_assignment.manage	Manage responsibility links	Scope and separation of duty
clinical.client_space.workspace.view	View operational workspace	Active client-space assignment
clinical.timeline.view	View authorized longitudinal events	Patient scope
clinical.audit.context.view	View detailed provenance	Elevated read scope

CLN-P2-PRM-001 These permissions shall be registered in the Main Module and mapped to configurable bundles; titles shall not be checked directly in Laravel business logic.

23. Audit and Event Catalogue

23.1 Mandatory audit events

- Patient search
- Patient chart open
- Patient/encounter switch
- Sensitive chart access or denial
- Encounter create and status transition
- Movement request, accept, reject, cancel and complete
- Care relationship create, accept, transfer and end
- Wrong-patient warning and mismatch
- Identity concern report
- Encounter closure and reopening
- Draft recovery conflict
- Bulk list action
- Offline synchronization and reconciliation

23.2 Platform events

- patient.identity.updated
- patient.identity.merge_pending
- patient.identity.reconciled
- encounter.created
- encounter.status.changed
- encounter.closed
- encounter.reopened
- patient.movement.requested
- patient.movement.completed
- care.relationship.changed
- patient.workspace.invalidated
- patient.alert.changed

- patient.confidentiality.changed

CLN-P2-EVT-001 Events shall use stable public IDs, version, tenant, occurred-at time, correlation ID and idempotency key.

CLN-P2-EVT-002 Event consumers shall not infer a movement is complete from a request event.

24. Reports and Operational Monitoring

- Active encounters by class, status, facility and client space
- Patients without resolved current space
- Overlapping or missing movements
- Encounters lacking responsible clinician/team where required
- Temporary identifiers awaiting reconciliation
- Suspected duplicates and identity overlays
- Closed encounters with unresolved follow-up
- Stale patient-list projections
- Unauthorized or denied chart access trends
- Break-glass chart access
- Client-space census discrepancies
- Drafts older than configured threshold

CLN-P2-RPT-001 Report visibility and exports shall remain permission- and scope-controlled.

CLN-P2-RPT-002 Operational counts shall be reproducible from source records and shall disclose refresh time and filters.

25. Acceptance Scenarios

ID	Scenario	Expected Result
P2-AT-001	Nurse searches exact patient ID within assigned ward.	Minimum safe result shown; chart open reauthorized.
P2-AT-002	Two patients have similar names.	Results show differentiating identifiers and warning; no automatic selection.
P2-AT-003	User scans wristband for a different patient than active chart.	Action blocked and mismatch recorded.
P2-AT-004	Patient has outpatient and inpatient encounters.	User must confirm the applicable encounter before entry.
P2-AT-005	Transfer is requested but not completed.	Current patient space remains source; destination shown as pending.
P2-AT-006	Transfer completes to ICU.	Patient lists and ordinary client-space access recalculate prospectively.
P2-AT-007	Ward nurse opens patient after assignment expiry.	Ordinary write denied despite stale browser page.
P2-AT-008	Two users edit the same mutable draft.	Stale update rejected or reconciled without silent overwrite.
P2-AT-009	Encounter is closed with pending final result.	Result and follow-up remain linked; ordinary new entry blocked.
P2-AT-010	Authorized user adds late information.	Separate late-entry/addendum workflow preserves closure and provenance.
P2-AT-011	Unknown emergency patient is registered.	Unique temporary identity and encounter created with visible reconciliation status.
P2-AT-012	Sensitive patient appears in unauthorized search.	No restricted detail or revealing facet is exposed.
P2-AT-013	AI-generated summary is requested from timeline.	Only authorized source data used; output remains draft and requires human review.
P2-AT-014	Offline entry synchronizes after patient moved.	Server revalidates context and routes conflict to reconciliation.
P2-AT-015	Overflow bed is retired.	Historical location remains resolvable and current capacity updates.

26. Mandatory Negative Tests

- Cross-tenant patient ID substitution
- Patient search without permission
- Chart open from copied deep link after assignment expiry
- Commit to wrong encounter through modified request payload
- Bed number used as sole patient identifier
- Movement request treated as completed location
- Unauthorized patient revealed in list count or search facet
- Closed encounter edited through direct API
- State version overwrites newer data
- Draft recovered into another patient
- Sensitive details exposed in notification preview
- Duplicate encounter created despite exact active encounter match
- Temporary identifier reused
- Offline synchronization silently overwrites movement
- Care relationship self-created without permission
- Bulk export enabled by ordinary chart-view permission

27. Phase 2 Traceability

Area	Prefix	Evidence
Governance	CLN-P2-GOV	Architecture and workflow review
Ownership	CLN-P2-OWN	Contract tests
Patient search	CLN-P2-PTS	Search privacy tests
Identification	CLN-P2-ID	Wrong-patient tests
Encounter	CLN-P2-ENC	Lifecycle tests
Context	CLN-P2-CTX	Patient/encounter binding tests
Banner	CLN-P2-BNR	UI and accessibility tests
Location	CLN-P2-LOC	Movement tests
Lists	CLN-P2-LST	Projection and access tests
Ward	CLN-P2-WRD	Census reconciliation tests
Care	CLN-P2-CARE	Relationship tests
Chart	CLN-P2-CHT	Authorization and state tests
Timeline	CLN-P2-TML	Ordering/provenance tests
Alerts	CLN-P2-ALT	Safety display tests
Concurrency	CLN-P2-CON	Lost-update tests
Closure	CLN-P2-CLS	Closure/reopen tests
Correction	CLN-P2-COR	Identity reconciliation tests
Downtime	CLN-P2-DWN	Offline recovery tests
Mobile	CLN-P2-MOB	Parity tests

28. Registered Phase 2 Gaps

Gap ID	Resolution Required	Blocking
CLN-P2-GAP-001	Confirm authoritative patient identity master and matching/merge service.	Yes
CLN-P2-GAP-002	Approve encounter classes, types and status lifecycle for each care setting.	Yes
CLN-P2-GAP-003	Confirm whether bed administration is Main, Clinical or a dedicated service while preserving Phase 2 boundaries.	Yes
CLN-P2-GAP-004	Approve two-identifier rules for each high-risk workflow.	Yes
CLN-P2-GAP-005	Approve patient banner fields and sensitive-data masking.	Yes

CLN-P2-GAP-006	Approve current patient-space derivation and movement completion authority.	Yes
CLN-P2-GAP-007	Approve care-relationship creation, acceptance, handover and expiry rules.	Yes
CLN-P2-GAP-008	Approve unknown-patient and newborn temporary identity workflows.	Yes
CLN-P2-GAP-009	Approve offline patient-list retention and synchronization policy.	Yes
CLN-P2-GAP-010	Approve chart closure prerequisites and reopening authorities.	Yes
CLN-P2-GAP-011	Approve identity correction, merge and wrong-patient clinical-record reconciliation workflow.	Yes
CLN-P2-GAP-012	Confirm FHIR version/profile approach for Patient, Encounter and Location exchange.	Before interoperability release

29. Phase 2 Completion Gate

- Phase 1 rules incorporated without contradiction.
- Patient identity source and identifier rules approved.
- Two-identifier patient-safety rules approved.
- Encounter schema concepts, classes and statuses approved.
- Patient-space and movement semantics approved.
- Patient banner and chart context approved.
- Patient lists and client-space workspace derivation approved.
- Care responsibility and handover model approved.
- Concurrency, drafts and closure rules approved.
- Identity correction and downtime reconciliation approved.
- Permissions, events, acceptance tests and gaps assigned.

Phase 3 shall use the patient, encounter, location, chart-state and authorization foundations defined here. It shall not create independent patient-context rules for individual note types.

Appendix A. Minimum Encounter Functional Record

Group	Fields
Identity	Public ID, encounter identifier, patient public ID, tenant, entity
Classification	Class, type, service, priority where used
Status	Current status, status history and version
Time	Planned period, actual period, arrival, finish and closure timestamps
Location	Facility, current patient space and movement references
Responsibility	Responsible clinician, nurse, care team and relationship references
Origin	Appointment, referral, source or prior encounter references
Clinical context	Reason, initial acuity or admission context where applicable
Closure	Disposition, closure checks, closed by and reopened references
Audit	Created/updated actor, correlation ID and provenance

Appendix B. Phase 2 Review Checklist

Review Role	Key Review Focus
Clinical Product Owner	Workflow completeness and usability
Patient Administration	Identity, registration and encounter rules

Nursing	Ward lists, handover, patient-space and banner
Medical Leadership	Responsibility, encounter state and chart access
Main Module Architect	Identity, client-space and event contracts
Security/Data Protection	Minimum necessary, masking and access control
Interoperability	Patient, Encounter and Location mappings
Quality Assurance	Acceptance, negative and concurrency tests
Operations	Downtime, reconciliation and monitoring

Appendix C. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phase 1 Draft.
- HL7 FHIR Patient, Encounter and Location materials.
- WHO Patient Identification safety solution.
- ONC SAFER Patient Identification guidance.

PHASE 3

CLINICAL DOCUMENTATION AND LEGAL RECORD CONTROLS

Phase 3 defines creation, review, signature, co-signature, correction, amendment, late entry, copy-forward, voice and AI-assisted documentation while preserving patient safety and the legal clinical record.

Document Control and Phase Authority

Field	Rule
Purpose	Define the complete functional lifecycle and governance of Clinical documentation.
Dependencies	All Phase 1 authorization and Phase 2 patient, encounter, workspace and record-state rules apply.
Technology	The companion implementation shall be Laravel-native. This SRD defines behavior, not partial framework code.
Boundary	Problems, diagnoses, orders, medication, observations and results are referenced but detailed in later phases.
Authority	This phase supersedes conflicting assumptions that a signed note may be overwritten or AI output may become final automatically.
Release	Controlled draft, not approved for production.

Phase 3 outcomes

- Create consistent note-type and template governance.
- Separate draft, completed, signed, co-signed, corrected and entered-in-error states.
- Preserve authorship, attestation and version history.
- Control copy-forward and imported content.
- Keep voice and AI assistance advisory and reviewable.
- Support multidisciplinary documentation without unsafe overwrites.
- Define permissions, audit, reports, acceptance tests and release gaps.

1. Governing Principles

CLN-P3-GOV-001 Clinical documentation shall be patient-bound, encounter-bound where applicable, attributable, time-stamped, versioned and auditable.

CLN-P3-GOV-002 A signed or otherwise finalized record entry shall not be destructively overwritten.

CLN-P3-GOV-003 The system shall distinguish author, recorder, transcriber, verifier, signer, co-signer, approver and source system.

CLN-P3-GOV-004 The person signing a record shall be shown the complete content and context being attested.

CLN-P3-GOV-005 Auto-save, speech recognition, templates, copy-forward and AI shall not constitute signature or clinical attestation.

CLN-P3-GOV-006 Documentation interfaces shall remain dynamic and template-driven without permitting administrators to bypass safety, provenance or authorization requirements.

CLN-P3-GOV-007 Every note type shall declare its patient context, encounter rules, permitted authors, required sections, signature rules, amendment rules, retention and interoperability mapping.

Legal-record rule: final content remains readable and traceable. Subsequent corrections or additions create linked record states rather than erasing history.

2. Documentation Concepts

Concept	Definition	Control
Record entry	A discrete clinical statement or structured entry.	Versioned and attributable.
Note	A documentation record composed of sections and entries.	Bound to note type and context.
Document	A coherent package prepared for persistence, exchange or attestation.	Frozen version when issued.
Draft	Editable, incomplete content not yet attested.	Not treated as final clinical truth.

Completion	Author indicates content is ready for required attestation.	May still require signature/co-signature.
Signature	Authenticated attestation by an authorized person.	Locks signed version.
Co-signature	Additional attestation required by policy.	Does not conceal original author.
Addendum	New supplementary content linked to a final entry.	Does not modify original text.
Correction	Replacement or clarification of erroneous content.	Original remains accessible.
Late entry	New documentation recorded after the clinical event.	Shows event and entry times.
Entered in error	Final record declared invalid for clinical use.	Retained with reason and replacement link.
Copy-forward	Reuse of prior content with provenance.	Must be reviewed and identifiable.
Imported content	Content received from device, module or external source.	Source and validation status preserved.

3. Documentation Types and Registry

3.1 Pre-seeded documentation families

- History and physical examination
- Progress note
- Nursing note
- Consultation note
- Allied-health note
- Triage note
- Procedure note
- Operative and anaesthetic boundary documents
- Transfer summary
- Discharge summary
- Referral note or letter
- Care-plan review note
- Handover note
- Telephone or virtual consultation note
- Patient/family communication note
- Administrative clinical clarification
- Unstructured attachment with governed metadata

CLN-P3-TYP-001 Documentation types shall be stored in a controlled, versioned registry and shall not be hardcoded solely in Laravel controllers or Vue pages.

CLN-P3-TYP-002 Each type shall define stable code, name, purpose, clinical category, allowed encounter classes, permitted authors, required permission, required sections, signature mode, co-sign rule, sensitivity, retention class, correction policy and active period.

CLN-P3-TYP-003 A retired type shall remain resolvable for historical records but shall not be selectable for new entries.

CLN-P3-TYP-004 Local note types may extend the registry through governed configuration but shall not redefine canonical codes silently.

CLN-P3-TYP-005 The interface shall show only note types allowed by the user's effective permissions, title bundle, client-space assignment, relationship, encounter class and professional scope.

4. Templates and Structured Sections

CLN-P3-TPL-001 Templates shall be versioned and effective-dated.

CLN-P3-TPL-002 A template shall define sections, order, required status, data type, terminology binding, help text, conditional visibility, validation, repeatability and signature inclusion.

CLN-P3-TPL-003 Clinical Data Elements referenced by a template shall use stable CDE identifiers and shall not be duplicated as unrelated free-text fields.

CLN-P3-TPL-004 Template changes shall not alter previously signed notes or their rendered meaning.

CLN-P3-TPL-005 A note shall preserve the template ID and exact version used at creation and signature.

CLN-P3-TPL-006 Administrators shall preview and validate templates before activation.

CLN-P3-TPL-007 A template may provide optional suggested text, but it shall not pre-assert unverified clinical facts.

CLN-P3-TPL-008 Required sections may be satisfied by structured data, authored narrative or approved reference links only where explicitly configured.

4.1 Template lifecycle

State	Permitted Use
DRAFT	Configuration work only
IN_REVIEW	Clinical and governance review
APPROVED	Approved but not yet effective
ACTIVE	Available for new documentation
DEPRECATED	Existing use ends on configured date
RETIRED	Historical resolution only
REJECTED	Not available

5. Authoring Workflow

Step	Required System Behavior
1. Select action	Show permitted note types for active patient and encounter.
2. Confirm context	Display patient banner, encounter, location and authoring role.
3. Create draft	Bind patient, encounter, note type, template version, author and timestamps.
4. Enter content	Validate required and conditional fields while allowing safe draft save.
5. Review imported/reused content	Display provenance and require confirmation.
6. Run validation	Show blocking errors separately from warnings.
7. Complete	Freeze a review candidate and resolve required attestations.
8. Sign	Reauthenticate where configured and attest to exact version.
9. Co-sign	Route to authorized co-signer if required.
10. Publish to chart	Display final status and issue downstream events.

CLN-P3-AUT-001 Draft creation shall require an atomic permission for the note family and Phase 1 contextual authorization.

CLN-P3-AUT-002 The server shall revalidate patient, encounter, client space, relationship, privilege and record state at save, complete and sign operations.

CLN-P3-AUT-003 The system shall warn before leaving unsaved content and shall never transfer content into another patient or encounter.

CLN-P3-AUT-004 A user may save an incomplete draft unless the action itself is unsafe or the configured workflow prohibits it.

CLN-P3-AUT-005 A final validation failure shall block completion or signature and shall identify the field or rule requiring action.

6. Required Documentation Metadata

Group	Required Fields
Identity	Public ID, tenant, patient, encounter where applicable, note type
Context	Facility, client space, service, care relationship, specialty
Authorship	Author, recorder/transcriber if different, professional title and acting/delegated context
Time	Clinical event time, authored time, recorded time, signed time and time-zone context

Template	Template ID and version; section definitions used
State	Draft/completion/signature/co-signature/correction status and version
Provenance	Source objects, copied/imported fragments, AI transaction reference
Confidentiality	Sensitivity labels and disclosure restrictions
Audit	Correlation ID, client/device, create/update actors and reasons
Interoperability	Canonical type/code and source/target mappings where approved

CLN-P3-MET-001 Clinical event time and record-entry time shall be stored separately when they differ.

CLN-P3-MET-002 The author's title and authority context at signature time shall be historically preserved.

CLN-P3-MET-003 Changing a user's later title shall not rewrite authorship metadata on prior notes.

7. Draft Management and Auto-Save

CLN-P3-DRF-001 Drafts shall be visible only to the author and other explicitly authorized users according to note policy.

CLN-P3-DRF-002 Auto-save shall create recoverable draft versions without creating chart-final content.

CLN-P3-DRF-003 Draft recovery shall display patient, encounter, note type and last-save information before opening.

CLN-P3-DRF-004 Stale drafts shall enter a configurable review or expiry workflow and shall not be silently deleted.

CLN-P3-DRF-005 The system shall prevent simultaneous sessions from silently overwriting the same draft.

CLN-P3-DRF-006 Abandoned drafts and their retention shall follow configured institutional policy.

CLN-P3-DRF-007 Other users shall not cite an unsigned draft as final clinical evidence unless an emergency policy explicitly permits viewing and labels it clearly.

8. Completion, Signature and Attestation

8.1 Signature modes

Mode	Meaning
AUTHOR_SIGNATURE	Author authenticates professional content.
RECORDER_PLUS_AUTHOR	Recorder enters content; responsible author verifies and signs.
CO_SIGNATURE	Additional authorized person attests according to policy.
SUPERVISOR_ATTESTATION	Supervisor attests where training/scope rules require.
ORGANIZATIONAL_ATTESTATION	Organization authenticates an issued document under policy.
PATIENT_ACKNOWLEDGEMENT	Records acknowledgement, not professional clinical authorship.

CLN-P3-SIG-001 Signature shall apply to the exact rendered and structured version presented to the signer.

CLN-P3-SIG-002 Signature shall record signer, authenticated identity, capacity, timestamp, method, note version and outcome.

CLN-P3-SIG-003 A user shall not sign on behalf of another person unless a distinct legally approved recorder/attestation workflow represents both identities correctly.

CLN-P3-SIG-004 Signing shall lock the signed version against ordinary editing.

CLN-P3-SIG-005 A failed or interrupted signature attempt shall not mark the note signed.

CLN-P3-SIG-006 The system shall distinguish professional, legal and organizational attestation modes where required by policy.

CLN-P3-SIG-007 Credential or privilege expiry at signature time shall block signature where that authority is required.

9. Co-Signature and Supervision

CLN-P3-COS-001 Co-signature rules shall be configurable by note type, author title/category, training state, service, client space, encounter class and risk level.

CLN-P3-COS-002 The original author shall remain visible after co-signature.

- CLN-P3-COS-003** A co-signer shall review the content being attested and may accept, return for amendment or reject according to configured workflow.
- CLN-P3-COS-004** Returning a note shall not erase the submitted version or reviewer comments.
- CLN-P3-COS-005** Co-signature queues shall route by effective authority and relationship, not merely by a hardcoded title string.
- CLN-P3-COS-006** Overdue co-signatures shall escalate according to configured time and risk rules.
- CLN-P3-COS-007** A note awaiting mandatory co-signature shall display its provisional status and shall not falsely appear fully attested.

10. Late Entries, Addenda, Corrections and Entered-in-Error

Action	Use	Effect on Original
Late entry	Document a past event after the fact.	Original timeline retained; event and entry times both shown.
Addendum	Add supplementary information.	Original content unchanged.
Correction	Correct erroneous content.	Original readable; corrected version becomes current.
Entered in error	Declare record invalid for clinical use.	Original retained and prominently marked.
Administrative correction	Correct non-clinical metadata where permitted.	Full audit and no clinical meaning change.

- CLN-P3-AMD-001** Each post-finalization action shall require its own permission, reason and policy eligibility.
- CLN-P3-AMD-002** The system shall preserve the original and every successive amendment, correction or augmentation.
- CLN-P3-AMD-003** The current version shall clearly link to previous versions and vice versa.
- CLN-P3-AMD-004** A correction shall specify what was wrong, the corrected information, authorizing user and date/time.
- CLN-P3-AMD-005** Entered-in-error content shall be excluded from ordinary decision support and summaries where safe, while remaining accessible to authorized audit/review users.
- CLN-P3-AMD-006** If downstream modules consumed the original, Clinical shall publish a correction event and monitor reconciliation.
- CLN-P3-AMD-007** An addendum shall not be used to conceal a required correction.
- CLN-P3-AMD-008** A late entry shall not be backdated as though entered contemporaneously.

11. Copy-Forward, Copy-Paste and Reuse

- CLN-P3-CPY-001** Copy-forward shall be configurable by note type and section and may be disabled for high-risk content.
- CLN-P3-CPY-002** Copied content shall preserve source patient, encounter, note, author, source date/time and section provenance.
- CLN-P3-CPY-003** Cross-patient copy-forward shall be prohibited.
- CLN-P3-CPY-004** The interface shall make copied text identifiable during review and shall provide access to its source context.
- CLN-P3-CPY-005** The author shall affirm review and update of copied content before signature.
- CLN-P3-CPY-006** The system shall warn when copied content is stale, conflicts with current structured facts or includes unresolved placeholders.
- CLN-P3-CPY-007** Copying another author's assessment shall not change its provenance or falsely attribute original observation to the current author.
- CLN-P3-CPY-008** Copy-forward metrics shall support governance review without judging clinical appropriateness solely from volume.

12. Smart Text, Macros and Templates

- CLN-P3-MAC-001** Smart text and macros shall be versioned, attributable and governed.
- CLN-P3-MAC-002** Macros shall not insert affirmative normal findings unless the user actively confirms them according to policy.
- CLN-P3-MAC-003** Unresolved tokens and placeholders shall block signature.

CLN-P3-MAC-004 Auto-populated facts shall identify source and timestamp and shall not present stale values as current.

CLN-P3-MAC-005 Personal macros shall be distinguishable from institution-approved standard macros.

CLN-P3-MAC-006 Deactivating a macro shall not alter signed notes that used it.

13. Voice Dictation and Transcription

CLN-P3-VCE-001 Voice dictation shall use the approved Shared AI Gateway speech-to-text service or another approved shared platform service, not a private Clinical provider integration.

CLN-P3-VCE-002 The transcript shall remain draft until reviewed and accepted by an authorized user.

CLN-P3-VCE-003 The UI shall display recording/transcription state and shall prevent accidental capture into the wrong patient or encounter.

CLN-P3-VCE-004 The transcript shall preserve service transaction reference, capture time, author/requester, edits and acceptance status.

CLN-P3-VCE-005 Low-confidence segments shall be highlighted where the service supplies confidence data.

CLN-P3-VCE-006 Voice commands that suggest orders, diagnoses or structured observations shall create reviewable suggestions only.

CLN-P3-VCE-007 Audio retention shall follow configured privacy and retention policy; the SRD shall not assume audio is retained indefinitely.

CLN-P3-VCE-008 If transcription is unavailable, manual authoring shall remain available where safe.

14. AI-Assisted Documentation

CLN-P3-AI-001 AI may generate draft summaries, SOAP structures, discharge drafts, referral drafts, coding candidates and extracted CDE suggestions only through the Shared AI Gateway.

CLN-P3-AI-002 AI shall use only source information the requesting user is authorized to access for the stated purpose.

CLN-P3-AI-003 AI output shall visibly identify itself as generated or assisted until accepted into authored content.

CLN-P3-AI-004 The user shall be able to inspect source references where the service supports them and shall remain responsible for verifying accuracy and completeness.

CLN-P3-AI-005 AI shall not sign, co-sign, finalize, publish, place orders, assign diagnoses or activate protocols.

CLN-P3-AI-006 Clinical shall preserve AI request ID, service, prompt-template version where provided, output version, user edits, acceptance/rejection and final signer.

CLN-P3-AI-007 The system shall prevent unreviewed AI text from being represented as the user's final attested statement.

CLN-P3-AI-008 AI outage shall not block ordinary manual documentation.

15. Imported and Referenced Content

CLN-P3-IMP-001 Content imported from LIMS, Imaging, devices, Main, external documents or other services shall retain source module/system, source public ID, status, time and version.

CLN-P3-IMP-002 A note may reference an external result or document without copying its full content where access and retention requirements are satisfied.

CLN-P3-IMP-003 A referenced object's correction or withdrawal shall be visible and shall not leave a frozen misleading excerpt without status.

CLN-P3-IMP-004 Users shall distinguish authored interpretation from imported source content.

CLN-P3-IMP-005 External attachments shall be scanned, classified, access-controlled and linked to the correct patient and encounter before availability.

CLN-P3-IMP-006 Unverified patient-generated content shall be labelled by source and verification status.

16. Multidisciplinary and Collaborative Documentation

CLN-P3-COL-001 Different professionals may document concurrently in the same encounter without sharing one mutable author identity.

CLN-P3-COL-002 Each authored section or contribution shall preserve its author and attestation requirements.

CLN-P3-COL-003 A multidisciplinary document may assemble signed contributions while preserving section-level provenance and whole-document issuance rules.

CLN-P3-COL-004 One user shall not silently edit another user's signed contribution.

CLN-P3-COL-005 Handover and collaborative comments shall not automatically become permanent legal notes unless the configured workflow commits them as such.

CLN-P3-COL-006 Mentioning or assigning another user shall not grant them patient access beyond Phase 1 rules.

17. Confidentiality, Sensitive Notes and Disclosure

CLN-P3-SEC-001 Note types and individual notes may carry sensitivity labels and disclosure restrictions.

CLN-P3-SEC-002 Restricted note titles, counts, snippets and search facets shall not leak protected information.

CLN-P3-SEC-003 A user who can view a general chart shall not automatically view every sensitive note section.

CLN-P3-SEC-004 Print, download, export and external sharing shall require separate permissions and purpose controls.

CLN-P3-SEC-005 Break-glass access to sensitive documentation shall follow Phase 1 and shall be reviewable per note accessed.

CLN-P3-SEC-006 Redaction for disclosure shall not alter the source legal record and shall produce a separately governed derivative.

18. Documentation Validation and Safety Checks

- Patient and encounter still valid
- User still authorized
- Encounter state permits action
- Required sections complete
- No unresolved placeholders
- Required structured fields valid
- Dates and times plausible
- Required signer/co-signer resolved
- Imported/copy-forward content reviewed
- Sensitive labeling completed
- Referenced objects resolvable
- No stale version conflict

CLN-P3-VAL-001 Blocking rules and advisory warnings shall be distinguishable.

CLN-P3-VAL-002 Overrides shall require an eligible permission, configured reason and audit; mandatory legal or identity checks shall not be overridable unless policy explicitly permits.

CLN-P3-VAL-003 Validation shall run server-side at completion and signature even if the client previously passed validation.

19. Closed Encounters and Post-Closure Documentation

CLN-P3-CLS-001 Ordinary new notes shall not be added to a closed encounter.

CLN-P3-CLS-002 Permitted post-closure actions shall use late-entry, addendum, correction, result-follow-up or other expressly defined workflows.

CLN-P3-CLS-003 Post-closure documentation shall display the original encounter closure time and the new entry time.

CLN-P3-CLS-004 Encounter reopening shall not be required merely to receive a final result or create an authorized addendum unless configured policy requires it.

CLN-P3-CLS-005 A post-closure note shall not change the historical patient location or responsible team without a separate correction workflow.

20. Offline, Downtime and Recovery

CLN-P3-DWN-001 Offline documentation shall preserve patient, encounter, author, device, capture time, template version and local draft ID.

CLN-P3-DWN-002 Offline content shall remain unsigned unless an approved offline signature method provides authenticated attestation and later verification.

CLN-P3-DWN-003 Synchronization shall revalidate authorization, patient/encounter state, template validity and version conflicts.

CLN-P3-DWN-004 Conflicting offline content shall enter reconciliation and shall not silently overwrite online notes.

CLN-P3-DWN-005 Downtime paper or external documents shall be entered or scanned through an approved reconciliation workflow with source and time metadata.

21. Permissions Catalogue

Permission	Action
clinical.note.create	Create permitted note draft
clinical.note.view	View ordinary authorized notes
clinical.note.view_sensitive	View sensitive documentation
clinical.note.edit_draft	Edit own or permitted draft
clinical.note.complete	Submit note for attestation
clinical.note.sign	Sign permitted note
clinical.note.cosign	Co-sign or supervise
clinical.note.return	Return submitted note
clinical.note.addendum	Add supplementary content
clinical.note.late_entry	Create late entry
clinical.note.correct	Correct final content
clinical.note.mark_entered_in_error	Declare final record entered in error
clinical.note.copy_forward	Reuse permitted prior content
clinical.note.export	Export documentation
clinical.template.manage	Manage note types/templates
clinical.documentation.audit	View detailed provenance/audit

CLN-P3-PRM-001 Permissions shall be Main Module registered and bundle-configurable. Laravel policies shall evaluate atomic permission plus Phase 1 and Phase 2 context, not hardcoded titles.

22. Notifications, Queues and Escalation

- Notes awaiting mandatory signature
- Mandatory co-signature due or overdue
- Returned documentation requiring revision
- Stale drafts
- Failed correction propagation
- Entered-in-error review
- Unresolved transcription segment
- Template retired while draft remains open
- Closed-encounter documentation awaiting review
- Downstream document delivery failure

CLN-P3-NTF-001 Notifications shall reveal only minimum necessary patient information and shall reauthorize when opened.

CLN-P3-NTF-002 Queue routing shall use effective assignment, relationship and authority at delivery time.

CLN-P3-NTF-003 Escalation shall not automatically sign or approve documentation.

23. Audit and Event Catalogue

- Draft created, updated, recovered, abandoned or expired
- Template/content imported or copied
- Voice/AI request and response
- Completion requested
- Signature/co-signature attempted, completed or failed
- Note returned or rejected
- Addendum, correction, late entry or entered-in-error
- Sensitive note viewed, printed or exported

- Validation overridden
- Offline synchronization and conflict resolution
- clinical.note.created
- clinical.note.completed
- clinical.note.signed
- clinical.note.cosigned
- clinical.note.returned
- clinical.note.addendum.created
- clinical.note.corrected
- clinical.note.entered_in_error
- clinical.note.disclosure.created
- clinical.template.activated
- clinical.documentation.reconciliation.required

CLN-P3-EVT-001 Events shall include stable IDs, version, patient, encounter, tenant, actor, occurred-at, correlation ID and idempotency key without placing unrestricted note text in event headers.

24. Reports and Oversight

- Unsigned and incomplete notes by age and service
- Mandatory co-signatures due/overdue
- Late entries and addenda
- Corrections and entered-in-error trends
- Copy-forward use and source age
- AI-assisted documentation acceptance/rejection
- Voice transcription exception rates
- Sensitive note access and exports
- Stale drafts
- Template versions in use
- Post-closure documentation
- Failed downstream correction reconciliation

CLN-P3-RPT-001 Reports shall be scope-controlled, audited and shall not expose note content unless the report permission explicitly allows it.

25. Acceptance Scenarios

ID	Scenario	Expected Result
P3-AT-001	Nurse creates authorized ward note.	Draft bound to patient, encounter, location, author and template version.
P3-AT-002	User changes patient with unsaved note.	Warning shown; content cannot transfer to new patient.
P3-AT-003	Author signs complete progress note.	Exact version locked with attestation metadata.
P3-AT-004	Junior author requires co-signature.	Note remains provisional and routes to eligible supervisor.
P3-AT-005	Signed note contains error.	Correction creates linked current version; original remains readable.
P3-AT-006	Author adds information after closure.	Authorized late-entry/addendum workflow records both times.
P3-AT-007	User copies prior section.	Copied text is identifiable with source provenance and review confirmation.
P3-AT-008	Cross-patient copy is attempted.	Blocked.
P3-AT-009	Voice transcript has low-confidence phrase.	Phrase highlighted; transcript remains draft.
P3-AT-010	AI drafts discharge summary.	Output labelled as AI-assisted, reviewed and signed only by authorized user.
P3-AT-011	Template changes after note signature.	Historical note retains original template

		version and rendering.
P3-AT-012	Imported result is later corrected.	Note reference shows corrected source status; authored interpretation remains distinct.
P3-AT-013	Offline draft conflicts with online update.	Conflict routed to reconciliation without overwrite.
P3-AT-014	Sensitive note count requested by unauthorized user.	No count, title or snippet leaks.
P3-AT-015	Credential expires before signature.	Signature blocked where credential is required.

26. Mandatory Negative Tests

- Direct API signing without sign permission
- Signing another user's identity
- Editing signed text in place
- Removing prior corrected version
- Backdating late entry as contemporaneous
- Cross-patient draft recovery
- Cross-patient copy/paste through manipulated payload
- Unresolved macro token allowed at signature
- AI output auto-finalized
- Co-signer selected by title string without authority check
- Closed encounter ordinary note creation
- Sensitive note exposed in search facet
- Stale draft overwrites current version
- Template retirement changes signed note
- Correction event duplicated on retry
- Offline unsigned note represented as signed

27. Traceability

Area	Prefix	Evidence
Governance	CLN-P3-GOV	Record governance review
Types	CLN-P3-TYP	Registry tests
Templates	CLN-P3-TPL	Version tests
Authoring	CLN-P3-AUT	Workflow tests
Metadata	CLN-P3-MET	Schema/provenance tests
Drafts	CLN-P3-DRF	Recovery/concurrency tests
Signature	CLN-P3-SIG	Attestation tests
Co-sign	CLN-P3-COS	Routing tests
Amendment	CLN-P3-AMD	Version preservation tests
Copy	CLN-P3-CPY	Provenance tests
Voice	CLN-P3-VCE	Draft safety tests
AI	CLN-P3-AI	Human approval tests
Import	CLN-P3-IMP	Source correction tests
Collaboration	CLN-P3-COL	Authorship tests
Security	CLN-P3-SEC	Leakage tests
Downtime	CLN-P3-DWN	Reconciliation tests

28. Registered Phase 3 Gaps

Gap ID	Resolution Required	Blocking
CLN-P3-GAP-001	Approve canonical note-type registry and initial templates.	Yes
CLN-P3-GAP-002	Approve signature and reauthentication	Yes

	methods.	
CLN-P3-GAP-003	Approve co-signature rules by title, training grade and note type.	Yes
CLN-P3-GAP-004	Approve legal retention and disclosure policies by jurisdiction.	Yes
CLN-P3-GAP-005	Approve late-entry, correction, addendum and entered-in-error authorities.	Yes
CLN-P3-GAP-006	Approve copy-forward sections, provenance display and stale-content thresholds.	Yes
CLN-P3-GAP-007	Approve voice-audio retention policy.	Before voice release
CLN-P3-GAP-008	Approve AI-assisted documentation services and minimum source-citation behavior.	Before AI release
CLN-P3-GAP-009	Approve sensitive note categories and masking.	Yes
CLN-P3-GAP-010	Approve downtime signature and reconciliation process.	Yes
CLN-P3-GAP-011	Confirm FHIR/CDA document and note mappings.	Before interoperability release
CLN-P3-GAP-012	Define downstream correction propagation contracts.	Yes

29. Phase 3 Completion Gate

- Phases 1 and 2 incorporated without contradiction.
- Note types and template lifecycle approved.
- Draft, completion, signature and co-signature states approved.
- Authorship and attestation roles approved.
- Late entry, addendum, correction and entered-in-error rules approved.
- Copy-forward controls approved.
- Voice and AI draft-only rules approved.
- Sensitive documentation and disclosure controls approved.
- Concurrency, downtime and reconciliation rules approved.
- Permissions, events, reports, acceptance tests and gaps assigned.

Phase 4 shall reference these documentation controls for problem lists, diagnoses and care plans. It shall not create alternate signature, correction or copy-forward rules.

Appendix A. Documentation State Model

State	Editable	Visible as Final	Next Actions
DRAFT	Yes, authorized	No	Complete, abandon
COMPLETED	Restricted	No	Sign, return
AWAITING_COSIGN	No ordinary edit	Provisional	Co-sign, return
SIGNED	No	Yes	Addendum, correction, entered in error
CORRECTED	No	Yes, current version	Further governed correction
ENTERED_IN_ERROR	No	No ordinary clinical use	Replacement link/review
ABANDONED	No	No	Audit/retention only

Appendix B. Minimum Note Record

Group	Fields
Identity	Public ID, patient, encounter, note type
Context	Facility, client space, service, relationship

Content	Structured sections, narrative, references
Authorship	Author, recorder, verifier, signer, co-signer
Time	Event, authored, recorded, completed, signed
State	Status, version, correction chain
Template	Template ID/version
Provenance	Copy/import/AI/voice references
Security	Sensitivity and disclosure controls
Audit	Correlation, device/client, policy version

Appendix C. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phase 1 Draft.
- KashTre Clinical Module SRD v6.1 Phase 2 Draft.
- HL7 EHR System Functional Model amendment/correction guidance.
- HL7 FHIR Composition and attestation materials.
- HL7 Clinical Document Architecture overview.
- Joint Commission and ECRI copy-and-paste safety recommendations.

PHASE 4 DRAFT

PROBLEMS, DIAGNOSES, GOALS AND CARE PLANNING

Phase 4 defines how clinical concerns become governed problems and diagnoses, how goals and care plans are created and reviewed, and how terminology assistance remains subject to human clinical authority.

Document Control and Phase Authority

Field	Rule
Purpose	Define problem lists, encounter diagnoses, diagnostic certainty, clinical goals, care plans, interventions and review.
Dependencies	Phase 1 authorization, Phase 2 patient/encounter context and Phase 3 documentation/signature controls remain mandatory.
Boundaries	Orders and medication actions referenced by care plans are specified in later phases and are not executed merely by plan entry.
Terminology	ICD-11 and other approved terminologies assist standardized representation; coding does not replace clinical judgment.
Implementation	Functional requirements only. Detailed Laravel migrations, Eloquent models, policies, services, APIs and Vue interfaces belong in the EDD.
Release	Controlled draft, not approved for production.

Phase 4 outcomes

- Separate symptoms, concerns, problems, encounter diagnoses and longitudinal diagnoses.
- Preserve clinical and verification status independently.
- Support provisional, differential, confirmed, refuted and entered-in-error states.
- Create measurable goals and multidisciplinary care plans.
- Link care plans to conditions, goals, activities, responsible people and review schedules.
- Preserve authorship, correction and history.
- Define permissions, safety checks, audit, reports, tests and release gaps.

1. Governing Principles

CLN-P4-GOV-001 No coded diagnosis shall be created solely because text, AI or an external system suggested a code.

CLN-P4-GOV-002 Clinical meaning and classification shall be stored separately: the clinician records the clinical assertion, while approved terminology services provide standardized coding.

CLN-P4-GOV-003 Clinical status and verification status shall be separate attributes and shall not be inferred from each other.

CLN-P4-GOV-004 The longitudinal problem list shall not be an unqualified copy of all encounter diagnoses.

CLN-P4-GOV-005 Problems, diagnoses, goals and plans shall preserve patient, encounter where applicable, author, time, source, version and status history.

CLN-P4-GOV-006 Care plans express intended care. They shall not themselves prove that an activity occurred or automatically execute an order.

CLN-P4-GOV-007 A signed assessment or care-plan version shall follow Phase 3 correction and amendment controls.

Clinical authority rule: terminology, rules and AI may suggest. An appropriately authorized human clinician must assert, verify, refute or close the clinical record.

2. Conceptual Separation

Concept	Meaning	Not Equivalent To
Symptom/sign	Reported or observed manifestation.	Confirmed diagnosis
Clinical concern	Issue requiring attention before formal problem assertion.	Longitudinal problem
Problem	Health issue judged important enough for monitoring or management.	Every historical diagnosis
Encounter diagnosis	Diagnosis asserted in the context of a specific encounter.	Automatically active problem
Longitudinal diagnosis	Diagnosis maintained across encounters where clinically appropriate.	Billing-only code
Differential diagnosis	One of several candidates guiding	Confirmed diagnosis

	evaluation.	
Clinical status	Active, inactive, resolved, remission or other approved state.	Verification certainty
Verification status	Unconfirmed, provisional, differential, confirmed, refuted or entered in error.	Clinical activity
Goal	Desired measurable or observable outcome.	Intervention
Care plan	Coherent intention for managing one or more concerns over time.	Proof of completion
Activity/intervention	Planned action within a care plan.	Order unless separately committed
Outcome	Observed result used to assess progress.	Goal target

3. Source-of-Truth and Interoperability Boundaries

CLN-P4-OWN-001 Clinical shall own patient-specific problem, diagnosis, goal and care-plan assertions and their lifecycle.

CLN-P4-OWN-002 Approved terminology services shall own canonical classification releases, codes, descriptions, synonyms and mappings. Clinical shall retain stable references and terminology version.

CLN-P4-OWN-003 Finance or coding modules may consume diagnoses for coding and billing but shall not silently change the clinician's clinical assertion.

CLN-P4-OWN-004 LIMS and Imaging results may support diagnosis review but shall remain source records owned by their authoritative modules.

CLN-P4-OWN-005 Orders and tasks generated from a care plan shall be separately created, authorized and tracked by their owning workflow.

CLN-P4-OWN-006 FHIR Condition, CarePlan and Goal mappings shall be defined in the interoperability profile without forcing Clinical storage to duplicate external resource structures blindly.

4. Problem and Diagnosis Data Model

Field Group	Required Data
Identity	Public ID, tenant, patient, encounter association, category
Clinical assertion	Clinical term, narrative, body site where applicable, severity, stage/grade where applicable
Terminology	Coding system, canonical code/URI, display, release/version, postcoordination/extension details
Status	Clinical status, verification status, lifecycle status
Timing	Onset, recorded, asserted, resolved/abated and last-reviewed times
Authorship	Asserter, recorder, verifier, responsible clinician and authority context
Evidence	Linked symptoms, observations, results, procedures, notes and external references
Context	Facility, service, client space, source encounter and care relationship
Governance	Sensitivity, reason for change, version, correction chain and audit metadata

CLN-P4-DAT-001 Each record shall have a stable public ID and immutable patient ownership.

CLN-P4-DAT-002 Encounter association shall be mandatory for encounter diagnoses and optional or multiple-reference capable for longitudinal problems according to policy.

CLN-P4-DAT-003 Onset may be exact, approximate, ranged or unknown and shall not be fabricated from record-entry time.

CLN-P4-DAT-004 Absence of a coded term shall not prevent safe draft capture of clinical narrative, but finalization requirements shall be configurable by use case.

CLN-P4-DAT-005 Terminology version and selected expression shall be preserved so historical meaning remains reproducible.

5. Problem Categories

Category	Purpose
ACTIVE_PROBLEM	Current issue needing monitoring or management
HISTORICAL_PROBLEM	Relevant past issue no longer active
ENCOUNTER_DIAGNOSIS	Diagnosis for a specific encounter
ADMISSION_DIAGNOSIS	Assertion at admission or initial assessment
DISCHARGE_DIAGNOSIS	Final encounter/discharge assertion
DIFFERENTIAL	Candidate diagnosis set
COMPLICATION	Condition associated with care or disease progression
PREGNANCY_OR_HEALTH_STATE	Clinically relevant health state
PATIENT_REPORTED_CONCERN	Concern not yet clinician-verified
SCREENING_OR_RISK_CONDITION	Risk/status requiring monitoring under policy

CLN-P4-CAT-001 Categories shall be controlled and configurable, with approved interoperability mappings.

CLN-P4-CAT-002 The same clinical concept may have more than one contextual record only where purpose and provenance justify it; the UI shall prevent confusing duplicates.

CLN-P4-CAT-003 Patient-reported concerns shall remain visibly distinct from clinician-confirmed diagnoses.

6. Clinical and Verification Status

Dimension	Illustrative Values	Rule
Clinical status	ACTIVE, RECURRENCE, RELAPSE, INACTIVE, REMISSION, RESOLVED	Describes current clinical state.
Verification status	UNCONFIRMED, PROVISIONAL, DIFFERENTIAL, CONFIRMED, REFUTED, ENTERED_IN_ERROR	Describes certainty/validity.
Record lifecycle	DRAFT, FINAL, CORRECTED, RETIRED	Describes record governance.

CLN-P4-ST5-001 A condition with an abatement/resolution time shall use a compatible inactive, resolved or remission clinical status.

CLN-P4-ST5-002 Refuted shall not be used for data-entry mistakes; entered in error shall be used for invalid records.

CLN-P4-ST5-003 Resolution shall not erase prior active periods or linked care plans.

CLN-P4-ST5-004 Reactivation, recurrence or relapse shall create a historically traceable transition and may require a new episode link according to policy.

CLN-P4-ST5-005 A differential diagnosis shall not appear in displays as confirmed.

7. Problem List Workflow

Step	System Behavior
Identify concern	User selects existing coded concept or enters narrative for review.
Confirm context	Patient, encounter, location and authorizing relationship displayed.
Classify	Choose problem category, clinical status and verification status.
Add details	Onset, severity, body site, evidence and responsible owner.
Duplicate check	Show active, historical and similar existing concepts.
Validate	Apply permission, terminology, status and required-field rules.
Assert/sign	Authorized user attests according to Phase 3.
Monitor	Schedule review or link care plan where required.
Update	Record status change, review, correction or entered-in-error action.

CLN-P4-PRB-001 Adding a longitudinal problem shall require clinical.problem.create and eligible professional authority.

CLN-P4-PRB-002 The system shall search for existing active and historical matching or similar problems before creating another.

CLN-P4-PRB-003 A duplicate warning shall not merge clinically distinct episodes automatically.

CLN-P4-PRB-004 The user shall be able to link encounter diagnoses to a longitudinal problem without rewriting either record.

CLN-P4-PRB-005 Each active problem may have an owner, review date, priority and associated care plan according to policy.

CLN-P4-PRB-006 Problems overdue for review shall enter a work queue but shall not auto-resolve.

8. Encounter Diagnosis Workflow

CLN-P4-DIA-001 Encounter diagnosis entry shall require an active or otherwise eligible encounter and `clinical.diagnosis.create`.

CLN-P4-DIA-002 The diagnosis shall record role in the encounter, including principal/primary, secondary, provisional, differential, complication or other approved classification.

CLN-P4-DIA-003 The system shall not infer principal diagnosis merely from display order.

CLN-P4-DIA-004 Admission, working and discharge diagnoses shall remain distinct assertions with traceable relationships.

CLN-P4-DIA-005 A discharge diagnosis may require confirmation, terminology completion and signature before encounter closure according to configured policy.

CLN-P4-DIA-006 A diagnosis created from an AI or coding suggestion shall remain unasserted until the authorized clinician explicitly selects and confirms it.

CLN-P4-DIA-007 Coding staff may propose classification refinements without changing clinical meaning or verification status unless separately authorized clinically.

9. Terminology and ICD-11 Assistance

CLN-P4-TRM-001 Clinical shall support an approved terminology service with versioned coding systems and stable identifiers.

CLN-P4-TRM-002 ICD-11 search shall support official descriptions, synonyms and approved postcoordination where configured.

CLN-P4-TRM-003 The selected ICD-11 entity URI/code, release, language, display and extension/postcoordination expression shall be preserved.

CLN-P4-TRM-004 A local clinical term may map to one or more classifications, but the mapping shall identify author, status, version and confidence/review state.

CLN-P4-TRM-005 When the terminology service is unavailable, the system may preserve a narrative draft but shall not guess a code.

CLN-P4-TRM-006 Deprecated codes shall remain resolvable historically and replacement suggestions shall require review.

CLN-P4-TRM-007 Classification updates shall not rewrite previously signed diagnoses automatically.

9.1 Search presentation

- Preferred term and code
- Synonyms and inclusions where supplied
- Exclusions and coding notes where supplied
- Parent/child context
- Postcoordination options
- Terminology release/version
- Local mapping status
- Warning for deprecated or inactive concepts

10. Problem and Diagnosis Evidence

CLN-P4-EVD-001 A condition may link to evidence including observations, laboratory results, imaging results, procedures, notes and external documents.

CLN-P4-EVD-002 Evidence links shall preserve source object identity, status and version and shall not copy the evidence into an untraceable text field.

CLN-P4-EVD-003 A corrected or withdrawn result shall cause the diagnosis workspace to display the changed evidence status and may generate a review task.

CLN-P4-EVD-004 The system shall not automatically refute or confirm a diagnosis solely because an individual result changes.

CLN-P4-EVD-005 Clinical rationale may be documented narratively and shall follow Phase 3 authorship and correction rules.

11. Goal Registry and Patient-Specific Goals

Goal Element	Requirement
Description	Clear intended patient outcome
Category	Clinical, functional, behavioral, safety, education or other approved category
Priority	Configured priority
Status	PROPOSED, PLANNED, ACCEPTED, ACTIVE, ON_HOLD, ACHIEVED, SUSTAINING, CANCELLED, NOT_ACHIEVED, ENTERED_IN_ERROR
Target	Measure/observation, qualitative outcome or milestone
Due time	Target date, period or review cadence
Owner	Responsible practitioner/team and contributors
Addresses	Linked problem, diagnosis, risk or care plan
Participation	Patient/family preference and agreement where applicable
Outcome	Linked observations or assessments used for review

CLN-P4-GOL-001 Goals shall be patient-specific and shall not be confused with reusable template definitions.

CLN-P4-GOL-002 Quantitative targets shall use canonical units and Clinical unit policies through the Main Module Unit Engine.

CLN-P4-GOL-003 A goal status shall be changed only by an authorized user or governed workflow and shall record reason and evidence.

CLN-P4-GOL-004 Missed target dates shall not automatically mark a goal failed; they shall create a review exception.

CLN-P4-GOL-005 Patient or caregiver goals shall preserve source and shall not be misrepresented as clinician-authored.

12. Care Plan Types and Scope

- Encounter-specific plan
- Longitudinal condition plan
- Nursing care plan
- Multidisciplinary plan
- Discharge/transition plan
- Rehabilitation plan
- Nutrition plan
- Wound-care plan
- Maternal/newborn plan
- Pediatric plan
- Chronic-disease plan
- Palliative/supportive care plan
- Patient/caregiver self-management plan

CLN-P4-CP-001 Care-plan types shall be controlled and versioned.

CLN-P4-CP-002 A patient may have multiple plans when their scopes differ; the workspace shall show relationships and prevent contradictory hidden plans.

CLN-P4-CP-003 A master integrated plan may reference component plans without absorbing or rewriting their provenance.

CLN-P4-CP-004 A reusable protocol or plan definition shall remain separate from the instantiated patient care plan.

13. Care Plan Data and Lifecycle

State	Meaning	Typical Next Actions
DRAFT	Being authored	Submit, abandon
PROPOSED	Recommended but not accepted	Accept, reject, revise
ACTIVE	Current approved plan	Review, update, place on hold, complete
ON_HOLD	Temporarily paused	Resume, discontinue
COMPLETED	Intended plan period completed	Outcome review

REVOKED	Withdrawn before ordinary completion	Retain reason/history
ENTERED_IN_ERROR	Invalid record	Replacement/correction link
SUPERSEDED	Replaced by new governed version	Historical view

CLN-P4-CPL-001 Each care plan shall record patient, encounter/episode scope, period, category, status, intent, author, responsible owner, care team, addressed conditions, goals, activities, review schedule and provenance.

CLN-P4-CPL-002 Activation shall require validation of responsible ownership, goals, required activities, permissions and patient/encounter context.

CLN-P4-CPL-003 Plan versioning shall preserve prior signed/approved versions and identify the superseding version.

CLN-P4-CPL-004 Completing or revoking a plan shall not automatically mark all linked goals achieved or all linked problems resolved.

14. Care Plan Activities and Interventions

Activity Element	Requirement
Description/code	Planned action with approved terminology where available
Status	NOT_STARTED, SCHEDULED, IN_PROGRESS, ON_HOLD, COMPLETED, CANCELLED, NOT_DONE
Timing	Start, recurrence, due date or milestone
Performer	Role, team or named user resolved through assignments
Location	Client space or remote context where applicable
Prerequisites	Conditions, observations, consent or readiness
Linked workflow	Order, task, appointment, education or manual intervention reference
Outcome	Completion record, result or reason not done

CLN-P4-ACT-001 Adding an activity to a plan shall not place a medication, laboratory, imaging or procedure order unless the authorized user completes the separate order workflow.

CLN-P4-ACT-002 Activities assigned to roles or teams shall resolve to eligible users through Phase 1 assignment rules.

CLN-P4-ACT-003 A plan activity shall distinguish planned, scheduled and completed states.

CLN-P4-ACT-004 Not-done or cancelled states shall require configured reason codes where clinically significant.

CLN-P4-ACT-005 Completion evidence shall identify the performing user/system and actual time.

15. Multidisciplinary Planning and Participation

CLN-P4-MDT-001 Multidisciplinary plans shall identify author, responsible owner, contributors and section/activity-level responsibility.

CLN-P4-MDT-002 Each professional shall remain accountable only for contributions and attestations attributed to them.

CLN-P4-MDT-003 Care-team membership shall not by itself grant authority to add diagnoses or approve another profession's restricted intervention.

CLN-P4-MDT-004 Patient and caregiver participation, preferences, declined actions and education needs shall be recordable with source and consent context.

CLN-P4-MDT-005 A disagreement shall be documented without silently replacing another professional's signed assessment.

CLN-P4-MDT-006 Handover of plan ownership shall preserve the previous owner and require acceptance where configured.

16. Review, Outcomes and Closure

CLN-P4-REV-001 Active problems, goals and plans may carry review due dates and configurable recurrence.

CLN-P4-REV-002 A review shall record reviewer, date/time, evidence considered, current status, changes and next review.

CLN-P4-REV-003 Outcome assessment shall link to observations, results or authored assessments and shall not overwrite target definitions.

CLN-P4-REV-004 A care plan may close as completed, revoked or superseded with an explicit reason and outcome summary.

CLN-P4-REV-005 Encounter closure shall identify unresolved active problems, goals and plans, but shall not automatically close longitudinal items.

CLN-P4-REV-006 Transition/discharge workflows shall assign ongoing plan ownership or mark follow-up responsibility explicitly.

17. Corrections, Refutation and Entered-in-Error

CLN-P4-COR-001 Corrections shall follow Phase 3 and preserve the original assertion and every version.

CLN-P4-COR-002 Refuted means later clinical evidence ruled out a previously considered condition; entered in error means the record itself was invalid.

CLN-P4-COR-003 Changing a diagnosis from provisional to confirmed shall be a status history event, not a destructive replacement.

CLN-P4-COR-004 Correcting terminology without changing clinical meaning shall remain distinguishable from changing the clinical assertion.

CLN-P4-COR-005 A material correction shall notify affected plan owners and downstream consumers through idempotent events.

CLN-P4-COR-006 Decision support shall exclude entered-in-error records and apply refuted/historical statuses correctly.

18. Decision Support and AI Assistance

CLN-P4-AI-001 AI terminology and diagnosis assistance shall be accessed only through the Shared AI Gateway.

CLN-P4-AI-002 The service may return ranked candidate diagnoses, codes, missing-detail prompts or plan suggestions, but shall not assert them.

CLN-P4-AI-003 The user shall see that suggestions are advisory and must explicitly select, edit or reject them.

CLN-P4-AI-004 The accepting user shall independently possess the action permission and clinical privilege.

CLN-P4-AI-005 Clinical shall preserve the AI transaction reference, suggestion set, selected item, edits, rejection and final assertion metadata.

CLN-P4-AI-006 The system shall not train or modify clinical terminology from local acceptance behavior except through an approved governed service.

CLN-P4-AI-007 AI unavailability shall not block manual problem, diagnosis, goal or plan entry.

19. Permissions Catalogue

Permission	Purpose
<code>clinical.problem.view</code>	View ordinary authorized problem list
<code>clinical.problem.create</code>	Assert new problem
<code>clinical.problem.update</code>	Update status/details
<code>clinical.problem.resolve</code>	Resolve or inactivate
<code>clinical.problem.correct</code>	Correct final assertion
<code>clinical.problem.mark_entered_in_error</code>	Invalidate erroneous record
<code>clinical.diagnosis.create</code>	Create encounter diagnosis
<code>clinical.diagnosis.confirm</code>	Confirm provisional/differential diagnosis
<code>clinical.diagnosis.refute</code>	Record refutation
<code>clinical.diagnosis.correct</code>	Correct diagnosis
<code>clinical.goal.create</code>	Create patient goal
<code>clinical.goal.update</code>	Update goal and outcome
<code>clinical.care_plan.create</code>	Create care plan
<code>clinical.care_plan.activate</code>	Activate approved plan
<code>clinical.care_plan.update</code>	Revise active plan
<code>clinical.care_plan.review</code>	Perform formal review
<code>clinical.care_plan.close</code>	Complete/revoke/supersede plan
<code>clinical.terminology.mapping.review</code>	Review local/canonical mapping
<code>clinical.problem_config.manage</code>	Manage categories/status policy
<code>clinical.care_plan_config.manage</code>	Manage plan definitions/templates

CLN-P4-PRM-001 Permissions shall be registered in the Main Module and assigned through configurable title bundles and privileges. Laravel shall enforce policies contextually without direct hardcoded title comparisons.

20. User Interfaces and Work Queues

- Active problem list with filters and overdue-review indicators
- Encounter diagnosis panel with certainty and role
- Differential diagnosis workspace
- Terminology search and coding detail panel
- Goal dashboard
- Care-plan authoring workspace
- Multidisciplinary contribution view
- Activity and intervention board
- Review-due queue
- Unowned plan queue
- Conflicting/duplicate problem review
- Corrected evidence review
- Post-discharge ongoing-plan handover

CLN-P4-UI-001 Every workspace shall preserve the Phase 2 patient banner and active encounter context.

CLN-P4-UI-002 Clinical and verification statuses shall be displayed independently and not through colour alone.

CLN-P4-UI-003 Provisional and differential diagnoses shall be visibly distinguishable from confirmed diagnoses throughout lists, summaries and exports.

CLN-P4-UI-004 The interface shall show source, author, last review and terminology version without requiring audit-level permission for ordinary provenance.

21. Notifications, Audit and Events

21.1 Notifications and exceptions

- Problem review overdue
- Provisional diagnosis pending review
- Evidence corrected or withdrawn
- Goal review due or target date passed
- Care plan awaiting activation
- Plan activity overdue
- Care-plan owner unavailable
- Conflicting active plans
- Plan handover pending acceptance
- Downstream correction reconciliation failed

21.2 Audit actions

- Create, verify, confirm, refute, resolve, reactivate, correct or enter in error
- Terminology selection and mapping override
- Goal create/update/outcome review
- Plan create/activate/update/hold/complete/revoke/supersede
- Activity assignment, completion or not-done
- AI suggestion displayed, accepted, edited or rejected
- Sensitive problem/plan viewed or exported

21.3 Events

- clinical.problem.created
- clinical.problem.status_changed
- clinical.problem.corrected
- clinical.diagnosis.confirmed
- clinical.diagnosis.refuted
- clinical.goal.created
- clinical.goal.status_changed
- clinical.care_plan.activated

- clinical.care_plan.updated
- clinical.care_plan.closed
- clinical.care_plan.review_due
- clinical.condition.reconciliation_required

CLN-P4-EVT-001 Events shall carry stable IDs, version, patient, encounter where applicable, tenant, actor, occurred-at, correlation ID and idempotency key without unrestricted sensitive narrative.

22. Reports and Oversight

- Active problems by status, service and owner
- Provisional/differential diagnoses awaiting review
- Discharge diagnoses incomplete
- Duplicate and conflicting condition records
- Conditions using deprecated terminology
- Goals overdue for review
- Goals achieved/not achieved
- Active care plans without owner
- Plans and activities overdue
- Care-plan handovers pending
- AI suggestion acceptance/rejection
- Corrections, refutations and entered-in-error trends
- Terminology release and mapping usage

CLN-P4-RPT-001 Reports and exports shall remain permission-, scope- and sensitivity-controlled.

CLN-P4-RPT-002 Aggregates shall expose terminology version and inclusion criteria where needed for reproducibility.

23. Acceptance Scenarios

ID	Scenario	Expected Result
P4-AT-001	Clinician records provisional malaria diagnosis.	Verification status remains provisional and is not displayed as confirmed.
P4-AT-002	Clinician confirms after evidence review.	Status history preserved with verifier and evidence links.
P4-AT-003	Diagnosis later ruled out.	Marked refuted, not deleted or entered in error.
P4-AT-004	Wrong-patient diagnosis was recorded.	Entered-in-error workflow preserves original and triggers reconciliation.
P4-AT-005	Duplicate active problem is proposed.	Existing similar problems shown; no silent merge.
P4-AT-006	ICD service is unavailable.	Narrative draft may be saved; no code guessed.
P4-AT-007	Terminology release changes.	Signed historical diagnosis retains original code/version.
P4-AT-008	Goal target uses mg/dL while base policy uses mmol/L.	Shared Unit Engine verifies conversion and provenance.
P4-AT-009	Care plan includes lab monitoring.	Plan records intended activity; order requires separate authorization.
P4-AT-010	Plan activity target time passes.	Review exception created; activity not auto-failed.
P4-AT-011	Patient transfers wards.	Plan remains patient-linked; task routing recalculates by assignment.
P4-AT-012	AI suggests diagnosis to nurse without confirmation privilege.	Suggestion cannot be asserted/confirmed.
P4-AT-013	Evidence result is corrected.	Responsible clinician receives review task; diagnosis not auto-changed.
P4-AT-014	Encounter closes with active chronic problem.	Longitudinal problem remains active and ownership is explicit.

P4-AT-015

Two disciplines contribute to one plan.

Each contribution preserves professional authorship and authority.

24. Mandatory Negative Tests

- Hardcoded title grants diagnosis authority
- Clinical status inferred from verification status
- Differential displayed as confirmed
- Encounter diagnosis automatically copied to active problem list
- Principal diagnosis inferred from row order
- AI candidate automatically committed
- Deprecated code rewrites signed history
- Care-plan activity silently places an order
- Goal target calculated with unverified unit conversion
- Entered-in-error condition used by decision support
- User resolves problem outside patient/client-space scope
- Care-team member edits another discipline's signed contribution
- Closed plan overwritten rather than versioned
- Corrected result auto-refutes diagnosis
- Report reveals sensitive diagnosis without permission
- Repeated event creates duplicate plan activity

25. Traceability

Area	Prefix	Evidence
Governance	CLN-P4-GOV	Clinical governance review
Ownership	CLN-P4-OWN	Contract tests
Data	CLN-P4-DAT	Schema/provenance tests
Categories	CLN-P4-CAT	Registry tests
Statuses	CLN-P4-STs	Lifecycle tests
Problem list	CLN-P4-PRB	Workflow tests
Diagnosis	CLN-P4-DIA	Certainty tests
Terminology	CLN-P4-TRM	Version/API tests
Evidence	CLN-P4-EVD	Correction tests
Goals	CLN-P4-GOL	Target/unit tests
Care plans	CLN-P4-CPL	Version/state tests
Activities	CLN-P4-ACT	Order-boundary tests
Multidisciplinary	CLN-P4-MDT	Authorship tests
Review	CLN-P4-REV	Due/closure tests
Corrections	CLN-P4-COR	History tests
AI	CLN-P4-AI	Human-approval tests

26. Registered Phase 4 Gaps

Gap ID	Resolution Required	Blocking
CLN-P4-GAP-001	Approve problem and diagnosis categories and status value sets.	Yes
CLN-P4-GAP-002	Approve professional authority for asserting, confirming and refuting diagnoses.	Yes
CLN-P4-GAP-003	Approve ICD-11 release, API/service ownership and licensing approach.	Before coding release
CLN-P4-GAP-004	Approve local clinical terminology and mapping governance.	Yes
CLN-P4-GAP-005	Approve principal/admission/discharge diagnosis rules by encounter class.	Yes

CLN-P4-GAP-006	Approve problem duplicate-detection and merge-review behavior.	Yes
CLN-P4-GAP-007	Approve care-plan types, templates and lifecycle.	Yes
CLN-P4-GAP-008	Approve goal categories, measurable target rules and review cadence.	Yes
CLN-P4-GAP-009	Approve multidisciplinary ownership and handover rules.	Yes
CLN-P4-GAP-010	Define plan-to-order/task integration contracts.	Yes
CLN-P4-GAP-011	Approve AI diagnosis/care-plan assistance scope.	Before AI release
CLN-P4-GAP-012	Confirm FHIR Condition, CarePlan and Goal profiles.	Before interoperability release

27. Phase 4 Completion Gate

- Phases 1 to 3 incorporated without contradiction.
- Problem, diagnosis and certainty models approved.
- ICD-11 and terminology governance approved.
- Problem-list and encounter-diagnosis separation approved.
- Goal structure and unit handling approved.
- Care-plan types, lifecycle, ownership and review approved.
- Plan activities separated from executable orders.
- Correction, refutation and entered-in-error rules approved.
- Permissions, interfaces, events, reports and tests approved.
- Every blocking gap assigned an owner and acceptance test.

Phase 5 shall use these problems, diagnoses, goals and plan references when defining orders. It shall not allow care-plan activities or AI suggestions to bypass order authorization.

Appendix A. Condition State Examples

Example	Clinical Status	Verification Status
Working diagnosis under investigation	ACTIVE	PROVISIONAL
Candidate among alternatives	ACTIVE	DIFFERENTIAL
Established chronic disease	ACTIVE	CONFIRMED
Condition improved and no longer active	RESOLVED	CONFIRMED
Previously suspected but ruled out	INACTIVE	REFUTED
Record placed on wrong patient	Not clinically applicable	ENTERED_IN_ERROR

Appendix B. Minimum Care Plan Record

Group	Fields
Identity	Public ID, patient, encounter/episode scope, category
Intent/state	Intent, status, period, version
Addresses	Problems, diagnoses, risks
Goals	Goal references, targets, priority and outcomes
Activities	Interventions, timing, performers and linked workflows
Team	Author, owner, contributors, patient/caregiver participation
Review	Review schedule, reviewers and outcome summary
Provenance	Template/protocol, source, signatures and corrections
Security	Sensitivity and disclosure controls
Audit	Correlation, context and event history

Appendix C. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phases 1, 2 and 3.
- HL7 FHIR Condition, CarePlan and Goal materials.
- WHO ICD-11 browser, coding tools, API and implementation information.

PHASE 5

CLINICAL ORDERS AND CLOSED-LOOP REQUEST MANAGEMENT

Phase 5 defines safe creation, authorization, transmission, acknowledgement, fulfilment, modification, cancellation and closure of medication, laboratory, imaging, procedure, referral and clinical-task orders.

Document Control and Phase Authority

Field	Rule
Purpose	Define complete order-entry and closed-loop request-management requirements.
Dependencies	Phase 1 authorization, Phase 2 patient/encounter context, Phase 3 attestation and Phase 4 problem/care-plan rules remain mandatory.
Boundary	Detailed medication administration and eMAR are Phase 6; results acknowledgement is Phase 8.
Ownership	Clinical owns ordering intent and clinical follow-up. Fulfilling modules own their fulfilment records and authoritative output.
Implementation	Functional specification. Laravel engineering details belong in the companion EDD.
Release	Controlled draft, not approved for production.

Phase 5 outcomes

- One consistent request framework across order families.
- Explicit distinction between proposal, plan, order and fulfilment.
- No unsigned or unauthorised order transmitted as active.
- Closed-loop acknowledgement and status reconciliation.
- Safe order sets, standing, verbal and emergency orders.
- Deterministic safety checks with governed overrides.
- Complete permissions, audit, events, reports and tests.

1. Governing Principles

CLN-P5-GOV-001 An order shall be a patient-specific authorized request, distinct from a care-plan activity, suggestion, draft, result or fulfilment record.

CLN-P5-GOV-002 No AI output, protocol recommendation, copied text or unsigned draft shall become an active order without explicit human authorization.

CLN-P5-GOV-003 Every order shall be bound to the correct patient and encounter where required, with two-identifier confirmation for configured high-risk actions.

CLN-P5-GOV-004 Order authority shall require atomic permission, effective scope, client-space or service context, care relationship, credential and permitted encounter state.

CLN-P5-GOV-005 Clinical shall preserve placer intent while fulfilling systems preserve authoritative fulfilment and result state.

CLN-P5-GOV-006 Cancellation, discontinuation, replacement, correction and entered-in-error shall remain distinct actions.

CLN-P5-GOV-007 All transmissions and retries shall be idempotent and traceable.

Order safety rule: if required patient, requester, item, timing, destination, authorization or decision-support context cannot be verified, the order shall not be activated.

2. Order Concepts and Separation

Concept	Meaning	Not Equivalent To
Proposal	Suggested action requiring acceptance.	Active order
Plan	Intended future action.	Authorization to fulfil
Draft order	Editable uncommitted request.	Recipient work item
Active order	Authorized request available to fulfiller.	Proof of performance
Order set	Versioned group of candidate orders.	Automatic batch execution
Standing order	Pre-authorized rule usable only under defined criteria.	Blanket authority
Verbal/telephone order	Order communicated when requester cannot enter directly.	Anonymous order

Emergency order	Urgent order under configured emergency controls.	Audit bypass
Fulfilment	Work performed by receiving module/service.	Order intent
Result	Output of performed service.	Order completion by itself
Task	Operational assignment derived from a request.	Clinical order authority

3. Order Families

Family	Examples	Primary Fulfiller
MEDICATION	Inpatient, outpatient, infusion, PRN, taper	Pharmacy/eMAR workflows
LABORATORY	Haematology, chemistry, microbiology, pathology	LIMS
IMAGING	X-ray, ultrasound, CT, MRI	RIS/PACS
PROCEDURE	Bedside, endoscopy, surgery request	Clinical/Theatre/Surgery service
REFERRAL_CONSULT	Internal consult, external referral, second opinion	Receiving service/entity
THERAPY	Physiotherapy, occupational, respiratory, nutrition	Allied-health service
NURSING_SERVICE	Configured nursing intervention requests	Nursing workflow
DEVICE_SUPPLY	Patient-focused device or supply request	Inventory/device service
OTHER_CLINICAL_SERVICE	Approved configured service	Registered fulfiller

CLN-P5-FAM-001 Order families and service catalogues shall be controlled, versioned and mapped to authoritative fulfilling services.

CLN-P5-FAM-002 The UI shall present only order families permitted for the user, patient, encounter, facility and client space.

CLN-P5-FAM-003 A single request shall not conflate independent medications or services merely to reduce record count.

4. Common Order Record

Group	Required Fields
Identity	Public ID, tenant, patient, encounter, order family, placer identifier
Request	Requested code/item, display, intent, priority, status, reason and clinical question
Context	Facility, client space, service, care-plan/problem references
Requester	Author, recorder if different, signer, professional authority
Timing	Authored, requested occurrence, start, end, frequency or urgency
Destination	Fulfilling module/service, facility and recipient endpoint
Safety	Allergies, relevant observations, warnings, overrides and consent
Lifecycle	Status, versions, replaces/replaced-by, cancellation/discontinuation reason
Fulfilment	Filler identifier, acknowledgement and status references
Governance	Sensitivity, provenance, correlation ID and audit metadata

CLN-P5-DAT-001 Each order shall have stable placer and public identifiers; receiving filler identifiers shall be stored separately.

CLN-P5-DAT-002 Clinical reason and linked problem/diagnosis shall be captured where required and not inferred solely from order text.

CLN-P5-DAT-003 Order instructions shall be structured where safety depends on them, with narrative retained only as supplementary context.

CLN-P5-DAT-004 Terminology and catalogue version shall be preserved for historical interpretation.

5. Lifecycle and Status Model

Status	Meaning	Allowed Examples
DRAFT	Not authorized or transmitted	Submit, abandon
PROPOSED	Suggested for clinician decision	Accept, reject
PENDING_SIGNATURE	Complete but not authorized	Sign, return
ACTIVE	Authorized and available to fulfil	Acknowledge, start, cancel, discontinue
ON_HOLD	Temporarily paused	Resume, discontinue
IN_PROGRESS	Fulfiller reports work underway	Complete, fail, discontinue
COMPLETED	Requested service fulfilled	Result/follow-up may remain
CANCELLED	Withdrawn before fulfilment	Retain reason/history
DISCONTINUED	Stopped after activation/start	Retain effective stop
REVOKED	Authorization withdrawn under policy	Retain reason
FAILED	Could not be fulfilled	Review/reorder
ENTERED_IN_ERROR	Invalid request record	Correction/replacement link

CLN-P5-ST5-001 Status transitions shall be explicit, versioned, permission-controlled and audited.

CLN-P5-ST5-002 Clinical shall not mark an order completed merely because it was transmitted.

CLN-P5-ST5-003 A final result may arrive after order completion and shall be managed by Phase 8.

CLN-P5-ST5-004 Illegal or stale state transitions shall be rejected with a structured conflict response.

6. Order Entry Workflow

Step	Required Behavior
1. Start	Confirm patient, encounter and ordering context.
2. Select family	Show permitted catalogue and order sets.
3. Select item/service	Resolve canonical item/service and version.
4. Enter parameters	Capture indication, priority, timing, instructions and destination.
5. Safety checks	Run patient-, order- and context-specific deterministic checks.
6. Review warnings	Resolve blockers and document eligible overrides.
7. Preview	Show complete human-readable order and recipient.
8. Sign/authorize	Revalidate authority and attest exact version.
9. Transmit	Persist active order and durable outbox atomically.
10. Acknowledge	Receive recipient acceptance/rejection and filler ID.
11. Track	Display fulfilment, exceptions and follow-up.

CLN-P5-ENT-001 Draft saving may be permitted before all fields are complete, but activation shall require all mandatory data.

CLN-P5-ENT-002 The server shall rerun authorization and safety validation at signature, not rely on earlier client checks.

CLN-P5-ENT-003 The user shall see the fulfiller and destination before signing.

CLN-P5-ENT-004 Transmission failure shall not create a second order on retry.

7. Clinical Decision Support and Safety Checks

- Patient identity and encounter state
- Allergy and intolerance
- Duplicate and overlapping order
- Drug-drug, drug-condition and drug-laboratory checks
- Dose, route, frequency and duration completeness
- Age, weight, body-surface-area and organ-function context
- Pregnancy/lactation context where configured
- Contraindication and prerequisite checks
- Recent duplicate laboratory/imaging service
- Laterality, body site and specimen completeness
- Required consent or preparation
- Client-space/service capability

- Formulary, catalogue and availability status
- Linked order-set dependencies

CLN-P5-CDS-001 Safety checks shall distinguish hard stop, interruptive warning, advisory information and unavailable-check status.

CLN-P5-CDS-002 A hard stop shall not be bypassed unless policy explicitly defines an eligible override permission and reason.

CLN-P5-CDS-003 Every override shall preserve rule, severity, user, reason, evidence and order version.

CLN-P5-CDS-004 The UI shall avoid duplicate or irrelevant warnings while never silently suppressing critical rules.

CLN-P5-CDS-005 If a required safety dependency is unavailable, the system shall follow configured safe-failure rules and visibly identify the unavailable check.

8. Medication Order Foundation

CLN-P5-MED-001 Medication orders shall reference authoritative medication/item identifiers and preserve generic/brand, strength, form and route context.

CLN-P5-MED-002 Each medication order shall capture dose, dose unit, route, frequency/timing, start, duration/end, indication and PRN conditions where applicable.

CLN-P5-MED-003 Dose and rate units shall use Main Module canonical unit IDs and verified conversions.

CLN-P5-MED-004 Weight- or body-surface-area-based orders shall preserve the source measurement, time, formula, calculated value, rounding and author confirmation.

CLN-P5-MED-005 Taper, titration, infusion and conditional instructions shall use structured components and shall not rely solely on ambiguous free text.

CLN-P5-MED-006 High-alert and controlled medication orders shall require additional privilege and configured verification.

CLN-P5-MED-007 Phase 6 shall define dispensing, administration, omission, wastage and eMAR behavior without weakening these order controls.

9. Laboratory Orders

CLN-P5-LAB-001 Laboratory orders shall reference the LIMS service/test catalogue and required specimen definitions.

CLN-P5-LAB-002 The request shall capture test/panel, clinical question, priority, requested time, specimen type/site where required, collection responsibility and infection hazard/precautions where authorized.

CLN-P5-LAB-003 Panel expansion shall show constituent tests and preserve the panel version.

CLN-P5-LAB-004 Duplicate checking shall consider recent, pending and in-progress tests and shall not cancel a clinically justified repeat automatically.

CLN-P5-LAB-005 Order status and specimen status shall remain separate.

CLN-P5-LAB-006 Add-on testing shall reference an eligible existing specimen and use a separate authorized request.

10. Imaging Orders

CLN-P5-IMG-001 Imaging orders shall reference the RIS service catalogue and capture modality/procedure, body site, laterality, clinical question, priority and requested timing.

CLN-P5-IMG-002 Contrast, sedation, pregnancy, renal-function and implant/device checks shall be invoked where applicable.

CLN-P5-IMG-003 The system shall preserve relevant prior imaging references without substituting them for the clinical question.

CLN-P5-IMG-004 Changing modality or body site after authorization shall require governed modification or replacement and recipient reconciliation.

CLN-P5-IMG-005 Image acquisition, interpretation and report release remain owned by RIS/PACS workflows.

11. Procedure, Therapy and Nursing Service Orders

CLN-P5-SVC-001 Procedure requests shall capture procedure, indication, body site/laterality, priority, requested timing, performer/service, consent/preparation and prerequisites where applicable.

CLN-P5-SVC-002 A procedure order shall not itself document procedure completion.

CLN-P5-SVC-003 Therapy orders shall capture goals, precautions, frequency, duration and service responsibility.

CLN-P5-SVC-004 Nursing service orders shall not override nursing professional assessment or title-derived scope.

CLN-P5-SVC-005 Theatre and Surgery modules shall own scheduling and operative fulfilment where applicable, while Clinical preserves request and follow-up context.

12. Referrals and Consultations

CLN-P5-REF-001 A referral shall capture reason, clinical question, urgency, requested specialty/service, recipient, relevant records, consent/disclosure basis and expected response.

CLN-P5-REF-002 Internal consultation and external referral shall remain distinct pathways.

CLN-P5-REF-003 Referral acceptance shall create an authorized care relationship only according to Phase 1 and Phase 2 rules.

CLN-P5-REF-004 Sending a referral shall not automatically grant unrestricted chart access to every member of the receiving organization.

CLN-P5-REF-005 Decline, redirect and request-more-information states shall preserve reason and requester notification.

CLN-P5-REF-006 Referral closure shall identify response received, follow-up owner and unresolved actions.

13. Order Sets and Protocol-Based Ordering

CLN-P5-SET-001 Order sets shall be versioned definitions containing candidate orders, grouping, defaults, dependencies and explanatory guidance.

CLN-P5-SET-002 Instantiating an order set shall create reviewable drafts, not automatically active orders.

CLN-P5-SET-003 The user shall select or deselect optional components and review every included order before signature.

CLN-P5-SET-004 Defaults shall never silently insert patient-specific dose, laterality or timing without validation.

CLN-P5-SET-005 Set updates shall not alter previously signed orders.

CLN-P5-SET-006 Each resulting order shall retain order-set ID/version and independent lifecycle.

14. Standing and Conditional Orders

CLN-P5-STD-001 A standing order shall define authorizing policy, eligible patient population, setting, trigger criteria, permitted users, action, limits, effective period and review date.

CLN-P5-STD-002 Use of a standing order shall create a patient-specific order record attributed to the invoking user and standing-order authority.

CLN-P5-STD-003 The system shall verify all criteria at invocation and shall not allow use after expiry or outside scope.

CLN-P5-STD-004 Conditional orders shall define machine-readable conditions and required confirmation at execution points.

CLN-P5-STD-005 Standing orders shall be versioned, approved and periodically reviewed.

15. Verbal, Telephone and Emergency Orders

CLN-P5-VER-001 Verbal or telephone orders shall be limited to configured circumstances and permissions.

CLN-P5-VER-002 The recorder shall capture stated requester, communication method, read-back confirmation, time, reason and exact order content.

CLN-P5-VER-003 The responsible requester shall authenticate the order within the configured timeframe; overdue authentication shall escalate.

CLN-P5-VER-004 The record shall always distinguish requester from recorder.

CLN-P5-VER-005 Emergency processing shall not remove identity, safety, read-back or audit requirements unless a formally approved downtime policy specifies an alternative.

CLN-P5-VER-006 An unauthenticated verbal order shall remain visibly provisional even if urgent fulfilment is allowed by policy.

16. Modification, Replacement, Cancellation and Discontinuation

CLN-P5-CHG-001 Material changes to an active order shall create a new version or replacement order according to family policy.

CLN-P5-CHG-002 Replacement shall link prior and new orders and shall prevent both remaining active unintentionally.

CLN-P5-CHG-003 Cancellation applies before ordinary fulfilment; discontinuation stops an active or started order prospectively.

CLN-P5-CHG-004 The system shall show whether the fulfiller accepted the cancellation/discontinuation and whether work had already started.

CLN-P5-CHG-005 Entered in error shall not be used to conceal a legitimate cancelled or discontinued order.

CLN-P5-CHG-006 Every change shall require reason, authority, effective time and downstream reconciliation.

17. Transmission, Acknowledgement and Closed Loop

Stage	Required Evidence
Committed	Order persisted with authorization and outbox record
Transmitted	Message delivered or queued to registered endpoint
Received	Recipient technical acknowledgement
Accepted	Fulfiller accepts clinical work
Rejected	Fulfiller rejects with actionable reason
In progress	Fulfiller reports performance underway
Completed	Service reports fulfilment complete
Resulted/responded	Expected output available where applicable
Reviewed	Responsible Clinical user reviewed output in Phase 8
Closed	No unresolved workflow obligation remains

CLN-P5-LOOP-001 Technical receipt shall not be treated as clinical acceptance.

CLN-P5-LOOP-002 The order workspace shall display placer status and authoritative fulfiller status separately when they differ.

CLN-P5-LOOP-003 Unacknowledged, rejected, failed and overdue orders shall enter exception queues with escalation.

CLN-P5-LOOP-004 Closure shall not occur while a required result, consultation response or follow-up remains unresolved.

18. AI-Assisted Ordering

CLN-P5-AI-001 AI may extract or suggest candidate orders only through the Shared AI Gateway.

CLN-P5-AI-002 Suggestions shall be presented as drafts with source context and shall not be transmitted automatically.

CLN-P5-AI-003 Each selected suggestion shall pass the same catalogue, authorization, safety and signing checks as manual entry.

CLN-P5-AI-004 The system shall preserve AI request ID, suggestion, user edits, acceptance/rejection and final signer.

CLN-P5-AI-005 AI outage shall not block manual ordering.

19. Permissions Catalogue

Permission	Purpose
clinical.order.create	Create draft order
clinical.order.sign	Authorize permitted order
clinical.order.modify	Modify or replace active order
clinical.order.cancel	Cancel before fulfilment
clinical.order.discontinue	Stop active order
clinical.order.override_warning	Apply eligible warning override
clinical.order.verbal_enter	Record verbal/telephone order
clinical.order.verbal_authenticate	Authenticate own communicated order
clinical.order.standing.invoke	Invoke eligible standing order
clinical.order.standing.manage	Manage standing-order definitions
clinical.order_set.manage	Manage order sets
clinical.referral.create	Create referral/consultation
clinical.referral.accept	Accept clinical referral
clinical.order.exception.manage	Resolve transmission/status exceptions
clinical.order.audit	View detailed order provenance

CLN-P5-PRM-001 Permissions shall be registered in the Main Module and assigned through configurable title bundles and privileges. Laravel policies shall not rely on hardcoded title names.

20. Operational Interfaces and Queues

- Order composer
- Order review and signature
- Current orders by family and status
- Medication order view
- Laboratory order and specimen status
- Imaging order status
- Procedure and therapy requests
- Referral inbox and response workspace
- Orders awaiting signature/co-signature
- Verbal orders awaiting authentication
- Rejected/unacknowledged transmission queue
- Overdue fulfilment queue
- Cancellation/discontinuation reconciliation
- Order-set and standing-order administration

CLN-P5-UI-001 Every interface shall retain Phase 2 patient banner and encounter context.

CLN-P5-UI-002 Draft, proposal, active, cancelled, discontinued and entered-in-error orders shall be visually distinct without relying only on colour.

CLN-P5-UI-003 The UI shall show requester, signer, fulfiller, indication, priority and current state without requiring audit-level access.

21. Audit, Events and Reports

- Draft create/update/abandon
- Safety check and override
- Signature and authorization failure
- Transmission, retry and acknowledgement
- Acceptance/rejection
- Status update and fulfilment
- Modification/replacement
- Cancellation/discontinuation
- Verbal read-back/authentication
- Standing-order invocation
- AI suggestion acceptance/rejection
- Sensitive order view/export
- clinical.order.created
- clinical.order.activated
- clinical.order.transmitted
- clinical.order.acknowledged
- clinical.order.accepted
- clinical.order.rejected
- clinical.order.status_changed
- clinical.order.replaced
- clinical.order.cancelled
- clinical.order.discontinued
- clinical.order.exception_raised
- clinical.referral.accepted
- Orders by family/status/service
- Unsigned orders
- Verbal orders awaiting authentication
- Unacknowledged or rejected orders
- Overdue fulfilment

- Duplicate-order overrides
- Safety-alert overrides
- Cancelled after work started
- Standing-order usage
- Order-set usage and deselection
- AI suggestion acceptance
- Transmission and reconciliation failures

CLN-P5-EVT-001 Events shall use stable IDs, tenant, patient, encounter, version, occurred-at, correlation ID and idempotency key and shall avoid unrestricted sensitive narrative.

CLN-P5-RPT-001 Reports and exports shall be permission-, scope- and sensitivity-controlled.

22. Acceptance Scenarios

ID	Scenario	Expected Result
P5-AT-001	Clinician creates routine laboratory order.	Patient/context validated, order signed, transmitted once and filler acknowledgement tracked.
P5-AT-002	Care plan proposes imaging.	Draft suggestion created; no active order until authorized.
P5-AT-003	Duplicate test exists.	Warning shows prior/pending test; justified repeat requires reason.
P5-AT-004	Medication dose uses mg/kg.	Current weight, unit conversion, calculation and rounding preserved.
P5-AT-005	Imaging laterality missing.	Activation blocked where laterality is required.
P5-AT-006	Order set selected.	Candidate orders shown individually for review; optional items not silently included.
P5-AT-007	Standing order invoked outside criteria.	Blocked with rule explanation.
P5-AT-008	Nurse records urgent verbal order.	Requester/recorder/read-back preserved and authentication queued.
P5-AT-009	Cancellation sent after fulfilment started.	Recipient status shown; reconciliation required.
P5-AT-010	Transmission times out and retries.	Same order/idempotency key used; no duplicate recipient order.
P5-AT-011	AI suggests CBC.	Suggestion remains draft and passes normal checks before signing.
P5-AT-012	Referral accepted.	Scoped care relationship created according to Phase 1/2 rules.
P5-AT-013	Result expected after order completion.	Order closes only when configured follow-up obligation is resolved.
P5-AT-014	User lacks controlled-drug privilege.	Restricted medication order cannot be signed.
P5-AT-015	Unit Engine unavailable for required dose conversion.	Activation blocked; no guessed conversion.

23. Mandatory Negative Tests

- Hardcoded title bypasses permission
- Draft transmitted as active
- AI suggestion auto-orders
- Care-plan activity treated as executable order
- Wrong-patient payload substitution
- Encounter closed but ordinary order signed
- Duplicate retry creates second filler order
- Technical receipt treated as clinical acceptance

- Medication order with free-text-only dose
- Molar/mass conversion without context
- Imaging order without required site/laterality
- Standing order used after expiry
- Verbal order loses requester identity
- Cancellation deletes original order
- Entered-in-error used instead of discontinuation
- Recipient rejection hidden from requester
- Report exposes sensitive order outside scope

24. Traceability

Area	Prefix	Evidence
Governance	CLN-P5-GOV	Architecture review
Families	CLN-P5-FAM	Catalogue tests
Data	CLN-P5-DAT	Schema/provenance tests
Status	CLN-P5-STS	Lifecycle tests
Entry	CLN-P5-ENT	Workflow tests
Safety	CLN-P5-CDS	Rule/override tests
Medication	CLN-P5-MED	Dose/unit tests
Laboratory	CLN-P5-LAB	LIMS contract tests
Imaging	CLN-P5-IMG	RIS contract tests
Services	CLN-P5-SVC	Procedure boundary tests
Referral	CLN-P5-REF	Access/acceptance tests
Order sets	CLN-P5-SET	Instantiation tests
Standing	CLN-P5-STD	Criteria tests
Verbal	CLN-P5-VER	Read-back tests
Changes	CLN-P5-CHG	Reconciliation tests
Closed loop	CLN-P5-LOOP	Acknowledgement tests
AI	CLN-P5-AI	Human approval tests

25. Registered Phase 5 Gaps

Gap ID	Resolution Required	Blocking
CLN-P5-GAP-001	Approve common order status and intent value sets.	Yes
CLN-P5-GAP-002	Approve authoritative service catalogues and owning modules.	Yes
CLN-P5-GAP-003	Approve professional ordering privileges by family.	Yes
CLN-P5-GAP-004	Approve hard stops, warnings and override authorities.	Yes
CLN-P5-GAP-005	Define LIMS, RIS/PACS, Inventory, Theatre and Surgery contracts.	Yes
CLN-P5-GAP-006	Approve medication dose/rate unit policies and calculation rules.	Yes
CLN-P5-GAP-007	Approve order-set governance and initial seed sets.	Yes
CLN-P5-GAP-008	Approve standing-order criteria and review model.	Yes
CLN-P5-GAP-009	Approve verbal/telephone authentication timelines and escalation.	Yes
CLN-P5-GAP-010	Approve referral disclosure and care-relationship rules.	Yes
CLN-P5-GAP-011	Define acknowledgement, timeout, retry and reconciliation SLAs.	Yes

26. Phase 5 Completion Gate

- Phases 1 to 4 incorporated without contradiction.
- Order families, intents and statuses approved.
- Common order record and authorization workflow approved.
- Safety checks and override governance approved.
- Medication, laboratory, imaging, procedure and referral requirements approved.
- Order sets, standing and verbal orders approved.
- Modification, cancellation and discontinuation semantics approved.
- Closed-loop acknowledgement and exception queues approved.
- Permissions, audit, events, reports and tests approved.
- Every blocking gap assigned an owner and acceptance test.

Phase 6 shall use these medication-order records as authoritative administration instructions. It shall not permit eMAR to invent, broaden or silently reinterpret the signed order.

Appendix A. Minimum Order Record

Group	Fields
Identity	Public ID, placer ID, patient, encounter, family
Intent	Draft/proposal/plan/order and priority
Request	Canonical item/service, parameters, reason and instructions
Requester	Author, recorder, signer and authority context
Timing	Authored, occurrence/start/end and recurrence
Destination	Fulfiller, endpoint and facility
Safety	Warnings, checks, overrides and evidence
Lifecycle	Status history, replacement/cancellation links
Fulfilment	Filler ID, acknowledgement and authoritative status
Audit	Version, correlation, idempotency and provenance

Appendix B. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phases 1 to 4.
- HL7 FHIR ServiceRequest and MedicationRequest materials.
- ONC SAFER Computerized Provider Order Entry with Decision Support guidance.
- AHRQ computerized provider order entry materials.

PHASE 6 DRAFT

MEDICATION MANAGEMENT, DISPENSING INTEGRATION AND eMAR

Phase 6 defines medication reconciliation, pharmacy verification integration, electronic medication administration, infusions, PRN review, omissions, refusals, wastage, controlled medicines and adverse medication-event workflows.

Document Control and Phase Authority

Field	Rule
Purpose	Define safe medication use from verified order through supply, administration, monitoring and reconciliation.
Dependencies	Phases 1 to 5 remain mandatory, especially patient identity, authority, documentation, unit conversion and medication-order controls.
Boundary	Pharmacy and Inventory remain authoritative for dispensing and stock; Clinical owns bedside administration records and clinical follow-up.
Vaccines	Immunization may use a dedicated profile/workflow but shall follow equivalent patient, product, administration and safety controls.
Implementation	Functional requirements only. Laravel implementation belongs in the companion EDD.
Release	Controlled draft, not approved for production.

Phase 6 outcomes

- One accurate active medication profile.
- Closed-loop reconciliation at transitions.
- Order-faithful eMAR generation.
- Positive patient and product identification.
- Structured administration, infusion and PRN documentation.
- Controlled omission, refusal, waste and correction workflows.
- High-alert and controlled-medicine safeguards.
- Complete audit, events, reports, tests and release gaps.

1. Governing Principles

CLN-P6-GOV-001 MedicationAdministration shall record an event that occurred or was not done; it shall not replace the authorizing MedicationRequest.

CLN-P6-GOV-002 The eMAR shall be a projection of active medication orders, dispensing/availability state and administration events, not an independently editable order source.

CLN-P6-GOV-003 Administration shall require the correct patient, medication/product, dose, unit, route, time, indication where applicable and authorized administrator.

CLN-P6-GOV-004 A nurse or other administrator shall not broaden, reinterpret or silently repair an ambiguous medication order.

CLN-P6-GOV-005 Medication safety rules shall apply at ordering, verification, supply, preparation, administration and monitoring points according to configuration.

CLN-P6-GOV-006 Medication events, corrections and waste records shall preserve immutable provenance and link to the order and dispense/supply where available.

CLN-P6-GOV-007 High-alert and controlled medicines shall use additional safeguards defined by approved institutional policy.

Bedside rule: when patient identity, order validity, product identity, dose, route, timing, privilege or a required safety check cannot be verified, administration shall be blocked or placed into an explicitly governed emergency pathway.

2. Medication Concepts and Boundaries

Concept	Meaning	Owner/Rule
Medication order	Authorization and instructions for medication use.	Clinical ordering, Phase 5
Medication dispense	Provision of medication supply.	Pharmacy/Inventory
Medication administration	Actual or not-done administration event.	Clinical eMAR
Medication statement	Reported medication use from patient or	Reconciliation source, not automatic

	another source.	order
Medication reconciliation	Comparison and resolution of medication lists at transitions.	Clinical workflow
Dose due	Calculated administration opportunity from an active order.	eMAR projection
Preparation	Compounding/drawing-up action before administration.	Recorded when policy requires
Waste	Medication prepared/dispensed but not administered and disposed.	Separate witnessed event where required
Omission	Scheduled dose not administered.	Not-done event with reason
Refusal	Patient/caregiver declines offered medicine.	Not-done event plus follow-up
Adverse reaction	Suspected harmful response.	Clinical event linked to allergy/safety workflow

CLN-P6-BND-001 Clinical shall consume authoritative medication/item and package identifiers from Main/Inventory and authoritative dispense status from Pharmacy.

CLN-P6-BND-002 Stock quantity shall not be inferred from eMAR events alone.

CLN-P6-BND-003 A patient-reported medication shall not appear as an active administered inpatient order until reconciled and authorized.

3. Medication Profile and Reconciliation

3.1 Medication sources

- Current active orders
- Prior encounter orders
- Patient or caregiver report
- External prescription/dispense records
- Pharmacy history
- Referral or transfer documents
- Medication containers or verified list
- Device-administered medication history

CLN-P6-REC-001 Reconciliation shall record source, reliability/verification state, medication, strength, form, route, dose, frequency, last dose, indication and adherence where available.

CLN-P6-REC-002 The system shall compare sources and identify duplicates, omissions, dose differences, route differences, interactions and unresolved uncertainty.

CLN-P6-REC-003 Each item shall receive an explicit decision such as CONTINUE, MODIFY, HOLD, STOP, SUBSTITUTE, DEFER_REVIEW or NOT_CURRENT.

CLN-P6-REC-004 A reconciliation decision that creates or changes therapy shall invoke the Phase 5 order workflow and authority checks.

CLN-P6-REC-005 Admission, internal transfer and discharge reconciliation shall remain distinct completed activities with responsible clinician and timestamp.

CLN-P6-REC-006 Unresolved discrepancies shall remain visible and route to an accountable owner; they shall not disappear when the encounter closes.

4. Pharmacy Verification and Dispensing Integration

State	Meaning
NOT_REQUIRED	Policy does not require pharmacy verification
PENDING	Awaiting pharmacist review
VERIFIED	Pharmacist accepted for dispensing/administration
CLARIFICATION_REQUIRED	Order requires prescriber response
REJECTED	Cannot be verified under policy
DISPENSED	Supply issued

PARTIALLY_DISPENSED	Only part supplied
NOT_AVAILABLE	Supply unavailable
RETURNED	Unused supply returned
RECALLED	Product/lot recall applies

CLN-P6-PHX-001 The eMAR shall show pharmacy verification and availability independently from order status.

CLN-P6-PHX-002 Where verification is mandatory, administration shall be blocked until VERIFIED unless an approved emergency rule applies.

CLN-P6-PHX-003 Clarification messages shall preserve question, sender, recipient, response, timestamps and resulting order version.

CLN-P6-PHX-004 Dispense events shall retain product, lot/batch, expiry, quantity, unit, dispenser and source identifiers where supplied.

CLN-P6-PHX-005 A substitution shall require an approved equivalence and shall not change the ordered clinical ingredient, dose or route silently.

CLN-P6-PHX-006 Recall notifications shall identify affected patients, products/lots and administrations without rewriting completed history.

5. eMAR Generation and Dose Scheduling

CLN-P6-MAR-001 The eMAR shall derive dose opportunities from the current signed medication order version and configured scheduling rules.

CLN-P6-MAR-002 Each due event shall preserve order ID/version, scheduled time/window, dose, route, PRN/conditional status and patient location.

CLN-P6-MAR-003 Order modification, hold, discontinuation or transfer shall recalculate future opportunities prospectively while preserving prior events.

CLN-P6-MAR-004 The system shall prevent duplicate due events during retries, daylight/time-zone changes or order replacement.

CLN-P6-MAR-005 Overdue status shall not itself record an omission; an authorized user or governed rule must resolve the event.

CLN-P6-MAR-006 The eMAR shall distinguish scheduled, due, overdue, administered, partial, refused, omitted, held, unavailable, cancelled and not-applicable states.

5.1 Administration windows

CLN-P6-MAR-010 Early and late windows shall be configurable by medication, frequency, care setting and risk class.

CLN-P6-MAR-011 Administration outside the ordinary window shall require a reason and may require additional authorization.

CLN-P6-MAR-012 The interface shall show actual time and scheduled time separately.

6. Positive Patient and Product Identification

CLN-P6-ID-001 The administrator shall confirm at least two approved patient identifiers before administration.

CLN-P6-ID-002 Barcode or equivalent scanning shall verify patient wristband and medication product where configured.

CLN-P6-ID-003 A mismatch between scanned patient, active chart, medication, strength, form, route, lot or order shall block administration.

CLN-P6-ID-004 Room, bed or ward number shall not serve as a patient identifier.

CLN-P6-ID-005 Manual barcode override shall require an eligible permission, reason and audit and shall never override a true patient/product mismatch.

CLN-P6-ID-006 Bedside verification shall re-check the current order version and not rely on a stale client display.

7. Administration Workflow

Step	Required Behavior
1. Select due dose	Open from authorized patient eMAR/worklist.
2. Confirm patient	Verify two identifiers and current encounter.
3. Confirm order	Display medication, dose, route, time, indication and

	warnings.
4. Confirm product	Scan/select exact product, strength, form, lot and expiry where available.
5. Prepare	Record calculations, dilution, preparation and witness if required.
6. Safety check	Revalidate allergies, parameters, hold criteria and privileges.
7. Administer	Record actual dose, unit, route, site, method and time.
8. Complete	Authenticate event and update eMAR projection.
9. Monitor	Create required reassessment/follow-up tasks.
10. Reconcile supply	Send consumption/waste/return events to authoritative services.

CLN-P6-ADM-001 Administration shall require clinical.medication.administer and any additional medication/route privilege.

CLN-P6-ADM-002 The administrator shall attest to the exact event record; pre-charting as administered before the event shall be prohibited.

CLN-P6-ADM-003 Actual dose may differ from ordered dose only where the order explicitly permits a range/titration or an authorized exception workflow applies.

CLN-P6-ADM-004 The event shall record performer, witness where required, patient, encounter, order, dispense/product, occurrence, recorded time, dose, route, site and notes/reason.

CLN-P6-ADM-005 A failed save shall not display the dose as administered; retry shall remain idempotent.

7.1 Point-of-Care Consumption Confirmation

The Confirm Consumption step forms part of the medication-administration workflow. It confirms the physical products used without turning the nurse into an Inventory or billing decision-maker.

CLN-P6-CNS-001 Upon successful commitment of a Medication Administration Event, the Clinical Module shall automatically create a draft Consumption Session and display a persistent Confirm Consumption interface.

CLN-P6-CNS-002 The session shall inherit tenant, facility, patient, encounter, client space, Medication Administration Event, administering user, occurrence time and clinical activity without duplicate entry.

CLN-P6-CNS-003 The interface shall pre-populate every medication product recorded as physically used during administration.

CLN-P6-CNS-004 Each medication line shall show product, strength, formulation, physical stock quantity, inventory unit, source store, batch or lot and expiry where available.

CLN-P6-CNS-005 Clinical dose and physical stock quantity shall be displayed distinctly where they differ, for example ceftriaxone 2 g administered from two 1 g vials.

CLN-P6-CNS-006 The user shall be able to scan, search for or select additional items used during administration, including gloves, syringes, needles, swabs, diluents and other non-prescribed supplies.

CLN-P6-CNS-007 Repeated scanning shall increment quantity unless batch, lot, serial or traceability rules require separate item lines.

CLN-P6-CNS-008 The user shall either confirm that all additional items are listed or explicitly select No additional items were used.

CLN-P6-CNS-009 The Clinical interface shall not ask the user to determine Approved Pool eligibility, prior payment, coverage, pricing, financial responsibility or chargeability.

CLN-P6-CNS-010 Confirmation shall submit one grouped Consumption Session to Inventory using stable session, source-event, correlation and idempotency identifiers.

CLN-P6-CNS-011 Inventory shall create authoritative item-level Consumption Events and apply Approved Pool, physical-stock and reconciliation rules.

CLN-P6-CNS-012 The Main Module shall determine the downstream financial outcome for quantities not satisfied by the Approved Pool or another recognized entitlement.

CLN-P6-CNS-013 Successful medication administration shall not wait for downstream financial evaluation.

CLN-P6-CNS-014 Failure of Inventory or Main processing shall not delete, reverse or invalidate a valid Medication Administration Event.

CLN-P6-CNS-015 An unresolved Consumption Session shall retain its contents and enter an accountable reconciliation queue.

CLN-P6-CNS-016 The interface shall offer Save for Reconciliation rather than silently discarding an incomplete session.

CLN-P6-CNS-020 Administered, wasted, returned and remaining quantities shall be separate dispositions within the session and shall not cause duplicate consumption.

CLN-P6-CNS-021 Breakage, contamination or damage shall be represented as a separate Inventory disposition and not as medication administered.

CLN-P6-CNS-022 Shelf expiry not attributable to a patient shall use the Inventory stock-disposition workflow and shall not require a false patient association.

8. Partial Doses, Split Products and Multi-Component Administration

CLN-P6-PAR-001 Partial administration shall record ordered dose, administered dose, remaining disposition and reason.

CLN-P6-PAR-002 The system shall support one administration composed from multiple products or strengths while preserving each product and lot.

CLN-P6-PAR-003 The system shall support one dispensed product contributing to an administered dose and a waste quantity without duplicating consumption.

CLN-P6-PAR-004 Unit conversions shall use the Main Module Unit Engine and store original, normalized and rule provenance.

CLN-P6-PAR-005 Dose rounding shall follow medication-specific policy and show the calculated and administered values.

9. PRN and Conditional Medication

CLN-P6-PRN-001 A PRN order shall specify indication, minimum interval, maximum dose/quantity and reassessment requirements where applicable.

CLN-P6-PRN-002 Before PRN administration, the user shall document or confirm the indication and relevant assessment.

CLN-P6-PRN-003 The system shall check last administration, cumulative dose and overlapping products before allowing the event.

CLN-P6-PRN-004 Required effectiveness reassessment shall generate a task with due time and accountable role/user.

CLN-P6-PRN-005 Failure to complete reassessment shall enter an exception queue and shall not alter the administration record.

10. Infusions, Titration and Pumps

CLN-P6-INF-001 Infusion administration shall capture solution/product components, concentration, route/access, rate, rate unit, volume, start and expected end.

CLN-P6-INF-002 Titration orders shall define permitted range, increment, interval, target parameter and maximum/minimum limits.

CLN-P6-INF-003 Every rate change shall record old rate, new rate, time, reason/target observation, performer and order version.

CLN-P6-INF-004 Pause, resume, bag change, line change, completion and early termination shall be distinct events.

CLN-P6-INF-005 Pump/device integration shall stage device data for patient/order validation; device data shall not auto-finalize an administration event without approved validation.

CLN-P6-INF-006 Cumulative infused volume and dose shall be reproducible from events and shall identify gaps or conflicting device/manual records.

11. High-Alert Medicines and Independent Checks

CLN-P6-HAM-001 High-alert status and required safeguards shall come from an approved, versioned institutional policy linked to authoritative medication records.

CLN-P6-HAM-002 Safeguards may include restricted access, standardized concentrations, mandatory pharmacy verification, dose limits, independent checks, smart-pump library use and enhanced monitoring.

CLN-P6-HAM-003 An independent double check shall require a second eligible user to verify configured elements independently and record their own authenticated decision.

CLN-P6-HAM-004 The first user shall not sign for the second user, and simultaneous shared credentials shall be prohibited.

CLN-P6-HAM-005 The system shall define whether an emergency override is permitted for each high-alert class and what retrospective review is required.

CLN-P6-HAM-006 Changing or retiring the high-alert list shall not rewrite historical administrations.

12. Controlled Medicines and Wastage

CLN-P6-CTL-001 Controlled-medicine processes shall be configurable to applicable law and institutional policy and shall not assume one jurisdiction.

CLN-P6-CTL-002 Receipt, issue, administration, return, waste and discrepancy events shall preserve quantity, unit, product/lot, custody actors, location and timestamps.

CLN-P6-CTL-003 Waste shall record quantity wasted, reason, method/location and witness where required.

CLN-P6-CTL-004 A waste event shall not be represented as a patient-administered dose.

CLN-P6-CTL-005 Balance discrepancies shall enter an exception workflow and shall not be silently adjusted.

CLN-P6-CTL-006 Witness eligibility shall be checked at witness time, including client-space assignment and required privilege.

13. Omission, Refusal, Hold and Unavailability

Outcome	Required Meaning
REFUSED	Medication offered but declined by patient/authorized decision-maker
OMITTED_CLINICAL	Not given for a clinical reason
OMITTED_OPERATIONAL	Not given for an operational reason
HELD	Temporarily withheld under valid order/instruction
NOT_AVAILABLE	Required product unavailable
PATIENT_UNAVAILABLE	Patient not available at due time
NIL_BY_MOUTH_OR_CONTRAINDICATED	Configured clinical condition prevents administration
NOT_DONE_OTHER	Approved reason with required narrative

CLN-P6-NDG-001 Not-done events shall require a controlled reason code and free text where configured.

CLN-P6-NDG-002 The system shall distinguish patient refusal from medication unavailability and clinical withholding.

CLN-P6-NDG-003 A refusal or significant omission shall trigger configured notification, prescriber/pharmacy review or reassessment.

CLN-P6-NDG-004 Repeated omissions shall be detectable and reportable.

CLN-P6-NDG-005 A held order and a held dose shall remain distinct.

14. Self-Administration, Patient-Owned and Device-Administered Medicines

CLN-P6-SELF-001 Self-administration requires an active order/policy, patient assessment, product verification, storage plan and documentation responsibility.

CLN-P6-SELF-002 Patient-owned medicines shall be identified and verified before use and shall remain distinct from pharmacy-dispensed stock.

CLN-P6-SELF-003 The system shall record whether the event was administered by staff, patient, caregiver or device.

CLN-P6-SELF-004 Device-generated administrations shall remain linked to device identity and validation status.

CLN-P6-SELF-005 Inability to verify a patient-owned product shall block its ordinary inpatient administration.

15. Adverse Reactions, Errors and Escalation

CLN-P6-ADE-001 A suspected adverse reaction may be recorded from the administration event and linked to allergy/intolerance and safety-reporting workflows.

CLN-P6-ADE-002 The record shall distinguish adverse drug reaction, side effect, medication error, near miss and therapeutic failure where policy supports these concepts.

CLN-P6-ADE-003 Clinical response, observations, treatment, notifications and outcome shall be documented without altering the original administration event.

CLN-P6-ADE-004 A medication error report shall not be hidden by correcting the eMAR entry.

CLN-P6-ADE-005 Serious events shall trigger configured urgent escalation and governance review.

16. Corrections and Entered-in-Error

CLN-P6-COR-001 A completed administration event shall not be edited destructively.

CLN-P6-COR-002 Correction shall preserve original and corrected values, reason, author, time and clinical impact assessment.

CLN-P6-COR-003 Entered-in-error shall be used only when the event record itself is invalid, not when a legitimate dose was omitted or discontinued.

CLN-P6-COR-004 Corrections shall reconcile eMAR, pharmacy, inventory, controlled-drug balance and downstream reports where affected.

CLN-P6-COR-005 A corrected event shall remain linked to the original order and dispense records.

17. Transfers, Leave and Discharge

CLN-P6-TRN-001 Patient movement shall recalculate future eMAR worklists and responsible administrators without changing prior administration context.

CLN-P6-TRN-002 Transfer handover shall identify doses due during transit, infusions in progress, patient-owned medicines, controlled medicines and unresolved discrepancies.

CLN-P6-TRN-003 Leave medication shall use an approved supply and documentation workflow with clear responsibility for self-administration.

CLN-P6-TRN-004 Discharge reconciliation shall produce the intended discharge medication list through authorized orders and distinguish continue, change, stop and new medicines.

CLN-P6-TRN-005 Encounter closure shall not erase outstanding medication reconciliation, adverse-event follow-up or controlled-medicine discrepancies.

18. Downtime, Offline and Recovery

CLN-P6-DWN-001 Downtime eMAR shall use an approved current medication snapshot with issuance time and expiry.

CLN-P6-DWN-002 Offline administration shall record local event ID, patient, order, product, dose, route, occurrence, performer and device.

CLN-P6-DWN-003 Synchronization shall revalidate order version, duplicate events, patient location, authority and product status.

CLN-P6-DWN-004 A possible duplicate administration shall enter urgent reconciliation and shall never be silently posted twice.

CLN-P6-DWN-005 Downtime controlled-medicine records shall be reconciled against electronic balances after restoration.

18.1 Standalone Patient-Linked Consumption

CLN-POC-CNS-001 Clinical shall provide a patient-linked Record Consumption action for supplies used during care without a Medication Administration Event.

CLN-POC-CNS-002 The action shall be available from authorized patient, procedure, treatment, observation and nursing interfaces.

CLN-POC-CNS-003 Clinical shall pre-populate patient, encounter, client space, user and clinical activity when launched from an active workflow.

CLN-POC-CNS-004 The user shall scan or select items and confirm physical quantities without making coverage or billing decisions.

19. Permissions Catalogue

Permission	Purpose
clinical.medication_profile.view	View authorized medication profile
clinical.medication_reconciliation.perform	Perform reconciliation
clinical.medication_reconciliation.verify	Verify reconciliation
clinical.medication.administer	Record permitted administration
clinical.medication.administer_high_alert	Administer configured high-alert medication
clinical.medication.infusion.manage	Start/change/pause/complete infusion
clinical.medication.prn_administer	Administer PRN medicine

clinical.medication.omission_record	Record not-done outcome
clinical.medication.waste_record	Record medication waste
clinical.medication.waste_witness	Witness controlled/high-risk waste
clinical.medication.double_check	Perform independent check
clinical.medication.correct	Correct final event
clinical.medication.mark_entered_in_error	Invalidate erroneous administration record
clinical.medication.exception.manage	Resolve eMAR/integration exceptions
clinical.medication_config.manage	Manage eMAR and safeguard policy
clinical.medication.audit	View detailed medication provenance

CLN-P6-PRM-001 Permissions shall be registered centrally and assigned through configurable title bundles and privileges. Laravel policies shall evaluate context and never rely solely on title names.

Additional consumption permissions

- clinical.consumption.confirm
- clinical.consumption.record_patient
- clinical.consumption.defer
- clinical.consumption.correct
- clinical.consumption.view
- clinical.consumption.reconcile
- clinical.consumption.scan_override

20. Operational Interfaces and Queues

- Patient medication profile
- Admission/transfer/discharge reconciliation
- Ward eMAR and due-medication worklist
- Medication administration wizard
- Infusion dashboard
- PRN reassessment queue
- Pharmacy clarification queue
- Not-available medication queue
- Omission/refusal review
- High-alert double-check queue
- Controlled-medicine waste and discrepancy queue
- Recall impact list
- Adverse medication-event follow-up
- Administration correction/reconciliation

CLN-P6-UI-001 Every patient-specific page shall retain the Phase 2 patient banner and encounter context.

CLN-P6-UI-002 Order, pharmacy, supply, due-dose and administration states shall be displayed separately.

CLN-P6-UI-003 The interface shall show scheduled time, actual time, administrator, product and dose without requiring audit-level permission.

CLN-P6-UI-004 Critical warnings shall not rely on colour alone.

20.1 Confirm Consumption Interface

CLN-P6-UI-010 The Confirm Consumption interface shall open automatically after a Medication Administration Event is saved successfully.

CLN-P6-UI-011 Desktop shall use a persistent side drawer and smaller devices a full-screen sheet with equivalent functionality.

CLN-P6-UI-012 The patient, encounter, client space and source administration context shall remain visible.

CLN-P6-UI-013 The interface shall provide barcode scanning, item search, quantity controls and approved frequently used item selectors.

CLN-P6-UI-014 The primary action shall be Confirm Consumption.

CLN-P6-UI-015 Closing an incomplete interface shall preserve the session and require a controlled reconciliation deferral reason.

CLN-P6-UI-016 Prices, charging responsibility and payment decisions shall not interrupt the bedside confirmation workflow.

21. Audit, Events and Reports

- Reconciliation source/decision
- Pharmacy verification/clarification
- eMAR due-event generation
- Patient/product scan and override
- Preparation/double-check
- Administration/partial administration
- Infusion rate change
- PRN assessment/reassessment
- Omission/refusal/hold
- Waste/witness
- Adverse reaction/error
- Correction/reconciliation
- Downtime synchronization
- clinical.medication.reconciled
- clinical.medication.due
- clinical.medication.administered
- clinical.medication.not_done
- clinical.medication.infusion_changed
- clinical.medication.waste_recorded
- clinical.medication.discrepancy_raised
- clinical.medication.adverse_event_recorded
- clinical.medication.corrected
- clinical.medication.reassessment_due
- Due/overdue administrations
- Omissions and refusals
- Medication unavailability
- Late/early administrations
- High-alert checks and overrides
- PRN reassessments overdue
- Infusions and rate changes
- Controlled-medicine waste/discrepancies
- Reconciliation completion/discrepancies
- Adverse events and near misses
- Barcode overrides/mismatches
- Corrections and downtime duplicates

CLN-P6-EVT-001 Events shall carry stable IDs, tenant, patient, encounter, order/event version, occurred-at, correlation ID and idempotency key without unnecessary sensitive narrative.

CLN-P6-RPT-001 Reports shall remain permission-, scope- and sensitivity-controlled.

21.1 Consumption Acceptance Scenarios

- P6-CNS-AT-001: Two 1 g ceftriaxone vials are recorded for a 2 g administration; the drawer opens with two vials pre-populated.
- P6-CNS-AT-002: The nurse adds gloves, syringe and swabs; one grouped session reaches Inventory.
- P6-CNS-AT-003: No incidental supplies were used; explicit No additional items were used completes the session.
- P6-CNS-AT-004: An Approved Pool quantity covers the medication; Inventory reduces the pool without duplicate physical-stock deduction or new billing.
- P6-CNS-AT-005: A floor-stock supply is outside the Approved Pool; Inventory reduces the originating Satellite Store and sends the usage fact for Main evaluation.
- P6-CNS-AT-006: Inventory is unavailable after administration; the administration remains valid and the session enters reconciliation.
- P6-CNS-AT-007: The same confirmation is replayed; idempotency prevents duplicate consumption.
- P6-CNS-AT-008: Part of a product is administered and part wasted; dispositions remain separate.
- P6-CNS-AT-009: A vial breaks before administration; breakage is recorded without falsely charting administration.
- P6-CNS-AT-010: Wound care records gloves and dressings through standalone Record Consumption without a medication order.

22. Acceptance Scenarios

ID	Scenario	Expected Result
P6-AT-001	Nurse scans correct patient and medicine for due dose.	Current order verified and administration recorded once.
P6-AT-002	Scanned wristband belongs to another patient.	Administration blocked and mismatch audited.
P6-AT-003	Pharmacy verification is required but pending.	Administration blocked unless approved emergency pathway applies.
P6-AT-004	Order changed after eMAR screen loaded.	Commit revalidates and rejects stale instructions.
P6-AT-005	PRN opioid requested too soon.	Interval/cumulative-dose rule blocks or warns according to approved policy.
P6-AT-006	Partial tablet administered.	Administered and waste/remaining quantities preserved without duplicate consumption.
P6-AT-007	Infusion rate titrated.	Old/new rate, target evidence, performer and time recorded.
P6-AT-008	High-alert drug requires independent check.	Second eligible user independently verifies configured elements.
P6-AT-009	Patient refuses medicine.	Refusal recorded with reason and configured follow-up.
P6-AT-010	Scheduled product unavailable.	Not-available event and pharmacy notification created; not charted as administered.
P6-AT-011	Controlled medicine waste witnessed.	Waste quantity and both authenticated actors recorded.
P6-AT-012	Administration entered on wrong patient.	Entered-in-error/correction and cross-system reconciliation initiated.
P6-AT-013	Patient transfers with infusion running.	Handover preserves infusion state and receiving responsibility.
P6-AT-014	Offline administration synchronizes twice.	Idempotency prevents duplicate administration.
P6-AT-015	Discharge reconciliation has unresolved discrepancy.	Closure blocked or exception assigned according to policy.

23. Mandatory Negative Tests

- Hardcoded nurse title bypasses permission/client-space check
- eMAR invents dose absent from signed order
- Pre-charted dose marked administered
- Wrong patient/product scan overridden
- Stale order administered
- Pharmacy rejection ignored
- High-alert second check completed by same credential
- Waste recorded as administered dose
- Refusal coded as unavailable
- Held dose treated as held order
- Unverified patient-owned medicine administered
- Device event auto-finalized to wrong patient
- Unit conversion guessed during outage
- Correction deletes original administration
- Offline retry duplicates dose
- Transfer rewrites historical administration location
- Report exposes controlled-medicine details outside scope

24. Traceability

Area	Prefix	Evidence
Governance	CLN-P6-GOV	Medication governance review
Boundaries	CLN-P6-BND	Module contract tests
Reconciliation	CLN-P6-REC	Transition tests
Pharmacy	CLN-P6-PHX	Verification/dispense tests
eMAR	CLN-P6-MAR	Schedule projection tests
Identity	CLN-P6-ID	Barcode/mismatch tests
Administration	CLN-P6-ADM	Bedside workflow tests
Partial doses	CLN-P6-PAR	Quantity/unit tests
PRN	CLN-P6-PRN	Interval/reassessment tests
Infusions	CLN-P6-INF	Rate/cumulative tests
High alert	CLN-P6-HAM	Independent-check tests
Controlled	CLN-P6-CTL	Custody/balance tests
Not done	CLN-P6-NDG	Reason/escalation tests
Self administration	CLN-P6-SELF	Product verification tests
Adverse events	CLN-P6-ADE	Escalation tests
Corrections	CLN-P6-COR	History/reconciliation tests
Transitions	CLN-P6-TRN	Handover tests
Downtime	CLN-P6-DWN	Duplicate/recovery tests

25. Registered Phase 6 Gaps

Gap ID	Resolution Required	Blocking
CLN-P6-GAP-001	Approve medication reconciliation sources, decisions and transition requirements.	Yes
CLN-P6-GAP-002	Define Pharmacy/Inventory verification, dispense, return and recall contracts.	Yes
CLN-P6-GAP-003	Approve eMAR scheduling windows and overdue rules.	Yes
CLN-P6-GAP-004	Approve barcode identifiers and manual-override policy.	Yes
CLN-P6-GAP-005	Approve high-alert list ownership and safeguards.	Yes
CLN-P6-GAP-006	Approve independent double-check elements and eligible titles/privileges.	Yes
CLN-P6-GAP-007	Approve controlled-medicine custody, witness and reconciliation rules for applicable jurisdictions.	Yes
CLN-P6-GAP-008	Approve infusion/titration models and pump integration boundary.	Yes
CLN-P6-GAP-009	Approve PRN indications, cumulative limits and reassessment rules.	Yes
CLN-P6-GAP-010	Approve omission/refusal escalation and reason codes.	Yes
CLN-P6-GAP-011	Approve adverse-reaction and medication-error integration.	Yes
CLN-P6-GAP-012	Confirm FHIR MedicationAdministration, MedicationDispense and MedicationStatement profiles.	Before interoperability release

26. Phase 6 Completion Gate

- Phases 1 to 5 incorporated without contradiction.

- Medication profile and reconciliation model approved.
- Pharmacy/Inventory boundaries approved.
- eMAR projection, scheduling and patient/product identification approved.
- Administration, partial-dose, PRN and infusion workflows approved.
- High-alert and controlled-medicine safeguards approved.
- Omission, refusal, waste, adverse-event and correction rules approved.
- Transfer, discharge and downtime reconciliation approved.
- Permissions, interfaces, events, reports and tests approved.
- Every blocking gap assigned an owner and acceptance test.

Phase 7 shall consume medication administrations, infusion state and relevant monitoring requirements when defining observations and scoring. It shall not reinterpret medication events without preserving their source status and provenance.

Appendix A. Minimum Medication Administration Record

Group	Fields
Identity	Public ID, patient, encounter, administration category
Authorization	Medication order ID/version and standing/emergency authority if used
Product	Canonical medication/product, strength, form, lot, expiry and dispense reference
Occurrence	Scheduled time/window, actual start/end and recorded time
Dosage	Dose, unit, route, site, method, rate and volume
Actors	Administrator, recorder, witness/checker and device
Outcome	Completed, partial, not done, reason and follow-up
Safety	Patient/product verification, warnings, overrides and assessments
Provenance	Correction chain, source, correlation and idempotency
Context	Facility, client space, patient location and care relationship

Appendix B. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phases 1 to 5.
- HL7 FHIR MedicationAdministration materials.
- WHO Medication Without Harm initiative.
- ISMP high-alert medication guidance for acute care settings.

- KashTre Inventory Endstore Systems Requirements Document v6.0.

PHASE 7

OBSERVATIONS, CDEs, MONITORING, SCORING AND UNITS

Phase 7 defines atomic Clinical Data Elements, observation capture, device staging, unit safety, reference ranges, trends, scores, monitoring schedules and deterioration escalation.

Document Control and Phase Authority

Field	Rule
Purpose	Define safe capture, normalization, interpretation, display and monitoring of clinical observations.
Dependencies	Phases 1 to 6 remain mandatory, including patient context, documentation, diagnosis, orders, medication events and Shared Unit Engine rules.
Ownership	Clinical owns CDE policy, observations, clinical validation, scores and monitoring. Main Module owns canonical units and conversions.
External results	LIMS and Imaging remain authoritative for their released results; Clinical may display or reference them without relabelling ownership.
Implementation	Functional specification only. Laravel schema and services belong in the EDD.
Release	Controlled draft, not approved for production.

Phase 7 outcomes

- One governed atomic CDE registry.
- Safe manual, calculated and device-originated observations.
- No competing Clinical unit master.
- Reproducible normalization and scoring.
- Clear reference, critical and physiological limits.
- Due, overdue, missed and escalated monitoring.
- Trend visualization with provenance.
- Complete permissions, events, reports, tests and gaps.

1. Governing Principles

CLN-P7-GOV-001 An observation shall be an event recording a measurement or clinical assertion about a patient or other authorized subject.

CLN-P7-GOV-002 Every observation shall identify what was observed, subject, encounter/context where required, value, unit where applicable, status, effective time, performer/source and provenance.

CLN-P7-GOV-003 Clinical shall maintain CDE definitions and unit-use policies but shall not maintain a competing writable canonical unit or conversion master.

CLN-P7-GOV-004 Original value and unit shall be preserved whenever normalization or display conversion occurs.

CLN-P7-GOV-005 Calculated scores and derived observations shall preserve inputs, algorithm version, calculation time and rounding.

CLN-P7-GOV-006 A warning or score shall support clinical judgment and shall not automatically diagnose, order treatment or execute a protocol.

CLN-P7-GOV-007 Device data shall not become trusted chart data until patient, encounter, CDE, unit, time and device provenance are validated under policy.

Observation safety rule: the system shall never silently change a value's meaning through guessed units, incompatible conversion, stale ranges, wrong-patient association or hidden algorithm changes.

2. Observation and CDE Concepts

Concept	Meaning	Rule
CDE	Governed definition of one clinical data element.	Stable code and versioned policy
Observation	Patient-specific measurement or assertion.	Event with status and provenance

Observation group	Logical set captured together.	Components remain individually traceable
Component	Part of a multi-component observation.	Example systolic/diastolic BP
Panel/organizer	Groups related observations.	Does not duplicate values
Derived observation	Calculated from one or more inputs.	Preserves dependency versions
Score	Rule-based aggregate or assessment result.	Inputs and matrix version retained
Reference range	Expected interval for a defined population/context.	Not a physiological hard limit
Critical limit	Threshold requiring configured urgent response.	Triggers escalation
Physiological limit	Plausibility boundary for capture validation.	Outside value requires correction/override
Monitoring plan	Schedule and rules for repeated observations.	Creates due work, not fabricated values

3. CDE Registry

Field Group	Required Data
Identity	Public ID, stable code, canonical name, aliases, category
Value	Data type, cardinality, components, coded value set or text constraints
Units	Base unit public ID, permitted input/display units, precision and rounding
Clinical interpretation	Reference ranges, critical limits, physiological limits and age/sex/context qualifiers
Capture	Allowed manual, device, imported, calculated and voice methods
Display	Label, chartability, graph type, decimals and grouping
Governance	Version, owner, status, effective dates, approval and audit
Interoperability	Canonical terminology code/system and approved profile/mapping

CLN-P7-CDE-001 CDE codes and public IDs shall remain stable when labels change.

CLN-P7-CDE-002 A CDE definition shall support NUMERIC, BOOLEAN, TEXT, CODED, DATE_TIME, QUANTITY and MULTI_COMPONENT patterns as approved.

CLN-P7-CDE-003 CDE activation shall require valid value rules, unit policy where applicable, interpretation policy, capture methods and provenance requirements.

CLN-P7-CDE-004 Retired CDE versions shall remain resolvable for historical observations and calculations.

CLN-P7-CDE-005 Changing a CDE shall not reinterpret previously final observations automatically.

3.1 Observation-First Work and System-Rendered Forms

Active Form Definitions, client-space Clinical Profiles, orders, protocols, care plans, medication-monitoring rules and authorized patient-specific instructions define what observations are needed and when. Clinical consolidates those requirements into a patient-specific observation worklist. Users record the observations once. Clinical then populates and renders all applicable forms and other views automatically.

CLN-P7-GOV-008 Forms shall define observation requirements and presentation rules, while atomic Observations shall remain the authoritative patient-specific clinical facts.

CLN-P7-GOV-009 Clinical shall translate applicable observation requirements into due work rather than require users to identify required form fields manually.

CLN-P7-GOV-010 The ordinary bedside interface shall present the observations due for the patient, their permitted capture windows and why they are required.

CLN-P7-GOV-011 Saving an Observation shall update every applicable form, flowsheet, chart, score and report without additional entry by the user.

CLN-P7-GOV-012 Failure to render or refresh a form shall not invalidate, roll back, delete or conceal a successfully committed Observation.

3.2 Clinical Form Definition Registry

A Clinical Form Definition is a governed, versioned specification describing the observations and approved non-observation data required by a form and the rules by which authoritative Clinical data are rendered. Supported examples may include a Labour Care Guide or partograph, vital-signs chart, fluid-balance chart, neurological chart, postoperative chart, anaesthetic record, intraoperative monitoring chart, newborn chart, diabetic monitoring chart and wound-assessment chart.

CLN-P7-FRM-001 Each Form Definition shall have a stable public ID, code, name, purpose, clinical domain, version, owner, status, effective period and approval history.

CLN-P7-FRM-002 Each Form Definition shall identify the CDEs, Observation Groups, calculated indicators and other controlled data elements that it consumes.

CLN-P7-FRM-003 Each form element shall be classified as required, optional, conditional, calculated, display-only or non-observation clinical content.

CLN-P7-FRM-004 Each scheduled observation requirement shall identify frequency, start and end conditions, permitted window, responsible role and missed-observation policy.

CLN-P7-FRM-005 A Form Definition shall specify rendering rules including sections, labels, sequence, rows, columns, time axes, plotted values, graphs, calculated fields and printable presentation where applicable.

CLN-P7-FRM-006 A form may include controlled non-observation content such as instructions, assessment, plan, declarations, attestations and signatures.

CLN-P7-FRM-007 Semantic requirements shall be separated from visual layout so that a rendering change does not silently change clinical meaning.

CLN-P7-FRM-008 Addition, removal or reinterpretation of required CDEs, schedules, calculations or alert rules shall create a new approved Form Definition version.

CLN-P7-FRM-009 Historical forms shall remain reproducible using the applicable Form Definition version and source Observations.

CLN-P7-FRM-010 A Form Definition shall not redefine a governed CDE unit, value set, reference range, provenance or status independently of the applicable CDE policy.

3.3 Clinical Form and Chart Designer

CLN-P7-FRM-020 Clinical shall provide an authorized configuration interface for creating, reviewing, approving, activating, deprecating and previewing Form Definitions.

CLN-P7-FRM-021 Form configuration shall support selection of existing CDEs and Observation Groups, requirement classification, schedules, calculations, chart rules, ordered sections, conditional visibility, completion rules and interactive and printable previews.

CLN-P7-FRM-022 If a required observation does not exist in the CDE Registry, the workflow shall require creation and approval of the CDE before form activation.

CLN-P7-FRM-023 A source PDF, image or Word document may be uploaded as a controlled reference, statutory source or rendering guide.

CLN-P7-FRM-024 Uploading a source document shall not by itself create an executable form or unreviewed CDE definitions.

CLN-P7-FRM-025 Authorized reviewers shall be able to compare the digital rendering with the approved source or statutory reference where one exists.

3.4 Client-Space Observation and Form Defaults

CLN-P7-CSP-001 Each client space may have a versioned Clinical Profile containing default Observation Groups, monitoring schedules, forms, flowsheets and escalation policies.

CLN-P7-CSP-002 Client-space Clinical Profiles shall be configured through an authorized clinical-governance workflow and shall not be inferred from the client-space display name.

CLN-P7-CSP-003 A client-space form assignment shall specify an activation mode of automatically active, available for selection, required to commence, conditional, order-activated, protocol-activated or care-pathway activated.

CLN-P7-CSP-004 A Labour Ward profile may activate maternal and fetal observations and an approved Labour Care Guide; a Postoperative or Postnatal Ward profile shall activate its own prospective defaults without continuing inapplicable labour monitoring.

CLN-P7-CSP-005 Client-space defaults shall apply only to patients satisfying configured patient, encounter, status and clinical-context criteria.

CLN-P7-CSP-006 Changes to a client-space Clinical Profile shall apply prospectively and shall not restructure historical observations, due work or forms.

CLN-P7-CSP-007 An overflow bed or temporary treatment position shall inherit the effective Clinical Profile of its governing client space unless an authorized exception is recorded.

3.5 Effective Patient Observation Plan

The Effective Patient Observation Plan is the system-derived union of universal baseline requirements, client-space defaults, active forms, orders, protocols, care pathways, care plans, medication-monitoring requirements and authorized patient-specific additions, less approved holds, exemptions and resolved duplications.

CLN-P7-PLAN-001 The Effective Observation Plan shall identify each required CDE or Observation Group, frequency, next due time, permitted window, responsible role and contributing sources.

CLN-P7-PLAN-002 Applicability shall be evaluated using patient, encounter, client space, clinical status, active orders, active forms, protocols, care plans and medication context.

CLN-P7-PLAN-003 Equivalent requirements for the same CDE and clinically equivalent time shall be consolidated into one due observation opportunity.

CLN-P7-PLAN-004 The system shall not create duplicate blood-pressure, temperature or other observation tasks merely because the same CDE is required by multiple active sources.

CLN-P7-PLAN-005 Where active sources specify different compatible frequencies, the clinically stricter frequency shall apply unless an approved precedence or exception rule states otherwise.

CLN-P7-PLAN-006 Consolidation shall retain all contributing sources so the system can explain why an Observation is due and which obligations it satisfies.

CLN-P7-PLAN-007 One eligible Observation shall satisfy every applicable requirement for which its CDE, patient, encounter, occurrence time, status, context and quality are valid.

CLN-P7-PLAN-008 Completion or removal of one source shall not end monitoring still required by another active source.

CLN-P7-PLAN-009 Recalculation shall be idempotent and shall not create duplicate due events.

3.6 Observation-to-Form Population and Rendering

CLN-P7-REN-001 Clinical shall render a form by retrieving qualifying authoritative Observations and other approved Clinical records.

CLN-P7-REN-002 Observation matching shall consider patient, encounter, CDE, occurrence time or period, status, quality, clinical context, capture method and Form Definition rules.

CLN-P7-REN-003 A saved Observation shall automatically update every applicable active form and flowsheet.

CLN-P7-REN-004 Observations recorded through a worklist, quick capture, device validation or a permitted empty form location shall use the same authoritative Observation model.

CLN-P7-REN-005 Clinical shall not copy an Observation into each consuming form as an independently editable value.

CLN-P7-REN-006 A rendered form shall retain a traceable reference to every Observation used to populate it.

CLN-P7-REN-007 An authorized user shall be able to navigate from a rendered value to its source Observation and provenance.

CLN-P7-REN-008 A form position without a qualifying Observation shall display pending, due, overdue, not obtained, not applicable or another approved state rather than fabricate a value.

CLN-P7-REN-009 Where multiple Observations could populate one form location, deterministic selection and conflict-display rules shall apply.

CLN-P7-REN-010 Preliminary, corrected, entered-in-error, device-unvalidated, late and estimated Observations shall retain their status when rendered.

CLN-P7-REN-011 A rendering failure shall enter technical reconciliation without rolling back valid Observations.

3.7 Patient-Specific Add Form

CLN-P7-ADD-001 The patient workspace shall provide Add Form to authorized users.

CLN-P7-ADD-002 The catalogue shall show eligible approved Form Definitions filtered by tenant, facility, client space, encounter, specialty, clinical context and user authority.

CLN-P7-ADD-003 Adding a form shall create a patient- and encounter-linked activation record referring to the exact Form Definition version.

CLN-P7-ADD-004 Adding a form shall evaluate its observation requirements against the Effective Observation Plan and add only genuinely new or stricter work.

CLN-P7-ADD-005 The added form shall immediately populate from existing qualifying Observations.

CLN-P7-ADD-006 Adding a form shall not modify the client-space Clinical Profile or defaults for other patients.

CLN-P7-ADD-007 The system shall warn when an equivalent active form already exists and may require a configured indication or reason.

CLN-P7-ADD-008 The activation record shall govern applicability, rendering, lifecycle and provenance but shall not own duplicate copies of Observations.

CLN-P7-ADD-009 Supported activation states shall include AVAILABLE, ACTIVE, COMPLETED, DISCONTINUED, SUPERSEDED and ENTERED_IN_ERROR.

CLN-P7-ADD-010 The form shall remain historically visible after its active lifecycle ends.

4. Observation Groups and Flowsheets

- Vital signs
- Neurological observations
- Fluid balance inputs and outputs
- Respiratory observations
- Pain assessment
- Falls and pressure-injury assessments
- Maternal and fetal monitoring
- Newborn observations
- Pediatric observations
- ICU/HDU flowsheets
- Wound observations
- Nutrition and anthropometry
- Device and ventilator parameters
- Medication-related monitoring

CLN-P7-GRP-001 Observation-group definitions shall specify constituent CDEs, required/optional status, order, conditional fields, default capture method and completion rules.

CLN-P7-GRP-002 Capturing a group shall create traceable member observations with a shared group identifier and capture session.

CLN-P7-GRP-003 A group may be incomplete when policy permits, but missing required elements shall be visible and shall affect score calculation appropriately.

CLN-P7-GRP-004 Flowsheets shall present time-series data without losing original values, units, status or source.

CLN-P7-GRP-005 Template changes shall not restructure historical groups silently.

5. Observation Status Lifecycle

Status	Meaning
REGISTERED	Known but no usable result yet
PRELIMINARY	Available but not final/validated
FINAL	Completed and clinically available
AMENDED	Final record supplemented or changed
CORRECTED	Erroneous final value replaced with traceable correction
CANCELLED	Legitimate observation process cancelled

ENTERED_IN_ERROR	Invalid record excluded from ordinary clinical use
UNKNOWN	Status cannot be established and requires review

CLN-P7-ST5-001 Status transitions shall be explicit, permission-controlled and audited.

CLN-P7-ST5-002 Preliminary and unvalidated observations shall be visually distinct from final observations.

CLN-P7-ST5-003 Entered-in-error observations shall not drive scores, trends, alerts or decision support.

CLN-P7-ST5-004 Corrected observations shall preserve original value, source and correction reason.

6. Manual Observation Capture

Step	Required Behavior
1. Open capture	Confirm patient, encounter, location and authorized workflow.
2. Select group/CDE	Use active definition/version.
3. Enter value	Show unit prominently and enforce data type.
4. Validate	Check plausibility, permitted unit, time and contextual rules.
5. Interpret	Display applicable range/limit and warnings.
6. Review	Show full group, missing fields and any calculated score.
7. Commit	Revalidate authority and persist observations atomically as configured.
8. Escalate	Create alerts/tasks for critical or overdue findings.

CLN-P7-MAN-001 Manual capture shall require clinical.observation.create and applicable client-space/care relationship.

CLN-P7-MAN-002 The patient banner and encounter shall remain visible throughout capture.

CLN-P7-MAN-003 Observation effective time shall default transparently and be editable only within policy; recorded time shall remain system-generated.

CLN-P7-MAN-004 Pre-charting a measurement before it occurs shall be prohibited.

CLN-P7-MAN-005 A failed group save shall not present partial values as a completed set unless partial commit is expressly designed and labelled.

7. Unit Policy and Shared Unit Engine

CLN-P7-UOM-001 Each quantitative CDE shall reference a canonical base unit in the Main Module Unit Engine.

CLN-P7-UOM-002 Clinical unit policy shall define permitted input units, permitted display units, preferred unit, precision and context-specific conversion requirements.

CLN-P7-UOM-003 The system shall store original value/unit, normalized value/base unit, conversion rule ID/version and rounding mode.

CLN-P7-UOM-004 Dimensional incompatibility shall block conversion and explain the mismatch.

CLN-P7-UOM-005 Contextual conversion, including molar-to-mass or product-specific relationships, shall require the appropriate analyte/product context.

CLN-P7-UOM-006 When conversion is unavailable or invalid, manual commit shall be blocked or device/import data staged as an exception. The numeric value shall not be relabelled.

CLN-P7-UOM-007 Unit switching in the UI shall be explicit and shall immediately update labels, displayed ranges and precision using verified conversions.

CLN-P7-UOM-008 Cumulative calculations shall use normalized precision appropriate to the CDE and shall avoid binary floating-point assumptions in the EDD.

8. Reference Ranges and Limits

Rule Type	Purpose	Response
Reference range	Expected range for population/method/context	Interpretation indicator
Critical limit	Value requiring urgent clinical response	Alert and escalation
Physiological limit	Plausibility boundary	Block or exceptional confirmation
Device technical limit	Device/reportable constraint	Technical exception

Target range	Patient-specific intended range	Care-plan/monitoring comparison
CLN-P7-RNG-001 Ranges shall be effective-dated and may depend on age, sex, pregnancy, method, specimen, posture, oxygen scale, care setting or other approved context.		
CLN-P7-RNG-002 The exact range/limit version applied to an observation shall be reproducible.		
CLN-P7-RNG-003 A value outside a reference range shall not automatically be labelled critical.		
CLN-P7-RNG-004 Physiological-limit overrides shall require permission, reason and repeat/verification action where configured.		
CLN-P7-RNG-005 Missing applicable range shall be shown as unavailable, not normal.		

9. Device and Telemetry Ingestion

Stage	Behavior
RECEIVED	Raw device payload durably stored
MAPPED	Device metric mapped to CDE/unit
PATIENT_MATCHED	Patient and encounter association verified
VALIDATED	Technical and clinical validation completed
COMMITTED	Observation created
REJECTED	Payload/value rejected with reason
QUARANTINED	Requires manual resolution

- CLN-P7-DEV-001** Device records shall preserve device public ID, model, metric/channel, gateway, calibration/status metadata where supplied, raw payload hash and timestamps.
- CLN-P7-DEV-002** Device-to-patient association shall be explicit, effective-dated and independently auditable.
- CLN-P7-DEV-003** A bed association alone shall not establish patient identity.
- CLN-P7-DEV-004** Unknown CDE, unrecognized unit, implausible value, stale timestamp, duplicate payload or patient mismatch shall prevent automatic commit.
- CLN-P7-DEV-005** Automatic validation eligibility shall be configurable by device, metric, care setting and risk level.
- CLN-P7-DEV-006** Sampled high-frequency data and charted summary observations shall remain distinct, with derivation rules preserved.
- CLN-P7-DEV-007** Device disconnection, calibration failure or technical alarm shall not be represented as a patient observation value.

10. Voice and AI-Assisted Observation Capture

- CLN-P7-AI-001** Voice and text extraction shall use the Shared AI Gateway and shall return proposed observations only.
- CLN-P7-AI-002** Each proposed observation shall show extracted CDE, value, unit, time, source phrase and confidence where available.
- CLN-P7-AI-003** An authorized user shall confirm or edit every proposed observation before commit.
- CLN-P7-AI-004** AI shall not infer a missing unit, patient, time or body site silently.
- CLN-P7-AI-005** Accepted observations shall pass the same unit, range, identity and authorization checks as manual entry.
- CLN-P7-AI-006** AI outage shall not block manual capture.

11. Calculated and Derived Observations

- CLN-P7-CAL-001** A calculated observation shall record algorithm code/version, input observation IDs/versions, coefficients, unit rules, precision, rounding and result.
- CLN-P7-CAL-002** Inputs shall be final or otherwise explicitly eligible under the algorithm policy.
- CLN-P7-CAL-003** Correction of an input shall mark dependent calculations for recalculation/review without overwriting prior calculated versions.
- CLN-P7-CAL-004** The interface shall distinguish measured, asserted, device-originated and calculated observations.
- CLN-P7-CAL-005** A calculation failure or missing input shall never yield a plausible default score/value.

12. Scoring and Assessment Engines

CLN-P7-SCR-001 Scoring definitions shall be versioned dictionaries containing required inputs, ranges, weights, modifiers, missing-input behavior, thresholds and response guidance.

CLN-P7-SCR-002 Supported examples may include NEWS2, APGAR, GCS and locally approved tools, each subject to licensing and governance.

CLN-P7-SCR-003 The score record shall preserve component observations, component scores, aggregate score, matrix version, calculation time and any override.

CLN-P7-SCR-004 Incomplete scores shall be labelled incomplete and shall follow tool-specific guidance; the system shall not substitute zero for missing values unless the definition requires it.

CLN-P7-SCR-005 Score thresholds may trigger alerts and response workflows but shall not independently diagnose or place orders.

CLN-P7-SCR-006 A patient-specific threshold adaptation shall require an authorized, documented and effective-dated rule.

12.1 NEWS2 boundary

CLN-P7-SCR-010 If NEWS2 is enabled, its official parameter, oxygen-scale, supplemental-oxygen, threshold and response policies shall be implemented from the approved governed version rather than re-created from memory.

CLN-P7-SCR-011 New confusion shall be captured explicitly where NEWS2 policy requires it.

13. Monitoring Schedules and Work Generation

Concept	Rule
Schedule	Defines CDE/group, frequency, start/end and responsible role
Due event	Expected observation opportunity
Window	Permitted early/late period
Overdue	Due event remains unresolved after window
Missed	Authorized resolution that observation was not obtained
Deferred	Rescheduled according to policy
Escalated	Failure or value routed for response

CLN-P7-MON-001 Monitoring may be created by an order, care plan, protocol, medication requirement or authorized manual action.

CLN-P7-MON-002 A monitoring schedule shall not fabricate observations; it creates due work only.

CLN-P7-MON-003 Changes to frequency shall apply prospectively and preserve previous due events.

CLN-P7-MON-004 Due events shall route by current patient location, client-space assignment and eligible role.

CLN-P7-MON-005 Missed or deferred monitoring shall require controlled reasons and may trigger escalation.

CLN-P7-MON-006 Patient transfer shall hand over unresolved due and overdue monitoring without duplication.

13.1 Form-Derived Work and Consolidation

CLN-P7-MON-007 Monitoring schedules may additionally be contributed by client-space Clinical Profiles and active Form Definitions.

CLN-P7-MON-008 Due work shall name the observation to be performed rather than require the user to navigate separately to every consuming form.

CLN-P7-MON-009 A due event shall show patient, observation or group, due time, permitted window, priority and responsible context.

CLN-P7-MON-010 Due-event detail shall identify contributing sources such as client-space default, form, protocol, care plan, medication rule or clinician order.

CLN-P7-MON-011 One completed eligible Observation may resolve multiple contributing requirements.

CLN-P7-MON-012 A form-rendering requirement shall not create an additional task when an applicable due event already exists.

14. Critical Results and Deterioration Escalation

- CLN-P7-ESC-001** Critical threshold crossing shall create a persistent alert and accountable response task according to configured policy.
- CLN-P7-ESC-002** The alert shall identify observation, value/unit, threshold/version, patient, encounter, time, source and required response without obscuring provenance.
- CLN-P7-ESC-003** Acknowledgement, clinical assessment, action and resolution shall be distinct states.
- CLN-P7-ESC-004** Failure to acknowledge within the configured interval shall escalate to the next eligible person/team.
- CLN-P7-ESC-005** A corrected observation shall update the alert context and preserve the original alert history.
- CLN-P7-ESC-006** Alert suppression shall be patient-specific, time-limited, authorized and auditable and shall not delete triggering observations.

15. Trend Charts and Clinical Display

- CLN-P7-TRD-001** Trend charts shall display observation time, value, unit, interpretation, source and status.
- CLN-P7-TRD-002** Values displayed in a common unit shall be converted through the Shared Unit Engine while preserving original unit access.
- CLN-P7-TRD-003** A chart shall not connect points across clinically invalid gaps or incompatible methods without a visible distinction.
- CLN-P7-TRD-004** Corrected, preliminary, device-unvalidated and entered-in-error values shall use distinct visual and textual states.
- CLN-P7-TRD-005** Reference ranges and patient-specific targets shall be version-aware and displayed only when applicable.
- CLN-P7-TRD-006** Colour shall not be the sole indication of abnormality or criticality.

16. Fluid Balance and Cumulative Measures

- CLN-P7-FLD-001** Intake and output observations shall record source/type, quantity, canonical unit, occurrence period and performer.
- CLN-P7-FLD-002** Medication infusion volumes may be included through verified administration references and shall not be double-counted by manual entry.
- CLN-P7-FLD-003** Cumulative balance shall identify included events, exclusions, period, late entries and recalculation version.
- CLN-P7-FLD-004** Corrections to source events shall trigger recalculation while preserving prior totals.
- CLN-P7-FLD-005** Estimated values shall be visibly marked and shall not be indistinguishable from measured quantities.

17. Corrections and Entered-in-Error

- CLN-P7-COR-001** Final observations shall not be destructively edited.
- CLN-P7-COR-002** Correction shall retain original value/unit/status, corrected value, reason, actor, time and impact on scores/alerts.
- CLN-P7-COR-003** Entered-in-error shall exclude the observation from ordinary trends, scores and decision support while preserving audit access.
- CLN-P7-COR-004** Dependent calculations, alerts, care-plan outcomes and medication titration references shall be placed into review after material correction.
- CLN-P7-COR-005** Downstream consumers shall receive idempotent correction events.

18. Downtime and Offline Capture

- CLN-P7-DWN-001** Offline capture shall preserve local event ID, patient, encounter, CDE/version, value/unit, effective time, performer and device.
- CLN-P7-DWN-002** Synchronization shall validate duplicate observations, current patient association, unit policy, CDE status and authorization.

CLN-P7-DWN-003 Late synchronized data shall retain capture and commit times and shall not impersonate contemporaneous online entry.

CLN-P7-DWN-004 Conflicts and possible duplicates shall enter reconciliation rather than being silently merged.

CLN-P7-DWN-005 Critical offline values shall follow an approved local escalation process and shall not wait for connectivity.

19. Permissions Catalogue

Permission	Purpose
clinical.observation.view	View authorized observations
clinical.observation.create	Record manual observation
clinical.observation.validate	Validate staged/device observation
clinical.observation.override_limit	Override eligible plausibility rule
clinical.observation.correct	Correct final observation
clinical.observation.mark_entered_in_error	Invalidate erroneous record
clinical.observation.device_reconcile	Resolve device mapping/patient exceptions
clinical.monitoring.create	Create monitoring schedule
clinical.monitoring.modify	Change schedule
clinical.monitoring.resolve_missed	Resolve missed/deferred event
clinical.score.calculate	Complete authorized assessment/score
clinical.score.override	Apply eligible score override
clinical.alert.acknowledge	Acknowledge deterioration alert
clinical.alert.resolve	Record clinical resolution
clinical.cde_config.manage	Manage CDE definitions/policies
clinical.score_config.manage	Manage score matrices
clinical.observation.audit	View detailed provenance

CLN-P7-PRM-001 Permissions shall be centrally registered and contextually enforced through Laravel policies, never by hardcoded title comparisons.

19.1 Form and Profile Permissions

- **clinical.form_definition.create** — Create a draft Form Definition
- **clinical.form_definition.update** — Modify an eligible draft Form Definition
- **clinical.form_definition.review** — Review a Form Definition
- **clinical.form_definition.approve** — Approve a Form Definition version
- **clinical.form_definition.activate** — Activate an approved Form Definition
- **clinical.form_definition.deprecate** — Deprecate a Form Definition prospectively
- **clinical.form_assignment.manage** — Manage client-space form assignments and Clinical Profiles
- **clinical.patient_form.add** — Add an approved form for a patient or encounter
- **clinical.patient_form.discontinue** — Discontinue an active patient form
- **clinical.patient_form.mark_entered_in_error** — Invalidate an erroneous patient form activation
- **clinical.form_render.view** — View authorized rendered forms
- **clinical.form_render.print** — Print or export an authorized rendered form

CLN-P7-PRM-002 Form-governance permissions shall be assigned through approved bundles and privileges and shall not be hardcoded to one job-title name.

20. Operational Interfaces and Queues

- Observation quick capture
- Flowsheets
- Vital-signs workspace
- Device validation queue
- Unit/conversion exception queue
- Due and overdue monitoring
- Scoring workspace
- Critical deterioration queue
- Trend viewer

- Fluid-balance chart
- CDE registry administration
- Range/limit administration
- Score matrix administration
- Correction and dependent-calculation reconciliation

CLN-P7-UI-001 Every patient-specific interface shall retain the Phase 2 patient banner and encounter context.

CLN-P7-UI-002 Input unit shall be prominent at the field, not hidden in help text.

CLN-P7-UI-003 Measured, calculated, imported and device-originated data shall be distinguishable.

CLN-P7-UI-004 Preliminary, corrected, critical and entered-in-error states shall not rely on colour alone.

20.1 Additional Form Interfaces and Queues

- Effective patient observation worklist
- Client-space Clinical Profile configuration
- Form Definition Registry
- Clinical Form and Chart Designer
- Form review and approval queue
- Patient Forms and Clinical Charts
- Add Form catalogue
- Form-rendering exception queue
- Duplicate observation-requirement review
- Form-version impact review

21. Audit, Events and Reports

- CDE/range/score policy activation
- Manual capture and limit override
- Device receive/map/validate/reject
- Unit conversion and failure
- Score calculation/override
- Monitoring schedule and missed event
- Critical alert and acknowledgement
- Trend/export access
- Correction and dependent reconciliation
- Offline synchronization
- clinical.observation.created
- clinical.observation.finalized
- clinical.observation.corrected
- clinical.observation.entered_in_error
- clinical.observation.critical
- clinical.device_observation.quarantined
- clinical.score.calculated
- clinical.monitoring.due
- clinical.monitoring.overdue
- clinical.deterioration.acknowledged
- clinical.calculation.reconciliation_required
- Observations due/overdue/missed
- Critical values and response times
- Physiological-limit overrides
- Device rejection/quarantine
- Unit conversion failures
- Incomplete scores
- Score threshold distribution
- CDE/range versions in use
- Corrections and dependent recalculations
- Offline duplicates

- Fluid-balance completeness

CLN-P7-EVT-001 Events shall include stable IDs, tenant, patient, encounter, version, occurred-at, correlation ID and idempotency key and shall minimize sensitive payload.

CLN-P7-RPT-001 Reports and exports shall remain permission-, scope- and sensitivity-controlled.

22. Acceptance Scenarios

ID	Scenario	Expected Result
P7-AT-001	Nurse records temperature in permitted input unit.	Original and normalized values stored with conversion provenance.
P7-AT-002	User attempts mass-to-volume conversion without context.	Conversion blocked with dimensional/context explanation.
P7-AT-003	BP entered as systolic/diastolic components.	Components and organizer/group remain traceable.
P7-AT-004	Value exceeds physiological limit.	Commit blocked or exceptional confirmation required by policy.
P7-AT-005	Critical oxygen saturation recorded.	Persistent alert and accountable escalation task created.
P7-AT-006	Device payload maps to wrong patient by stale bed link.	Automatic commit blocked and payload quarantined.
P7-AT-007	Duplicate device message retried.	Idempotency prevents duplicate observation.
P7-AT-008	Score lacks one required input.	Score marked incomplete; no zero substituted.
P7-AT-009	Input to final score is corrected.	Dependent score and alerts enter review; history preserved.
P7-AT-010	Monitoring frequency changes.	Future due events recalculate without rewriting past ones.
P7-AT-011	Patient transfers with overdue observations.	Outstanding work handed to eligible receiving scope once.
P7-AT-012	Voice extracts value but no unit.	Proposed observation requires human unit confirmation.
P7-AT-013	Unit Engine unavailable.	Observation requiring conversion blocked or staged, never relabelled.
P7-AT-014	Infusion volume already linked from eMAR.	Fluid chart prevents manual duplicate counting.
P7-AT-015	Offline critical value captured.	Local escalation occurs and later sync preserves both timestamps.

22.1 Observation-First Form Acceptance Scenarios

P7-FRM-AT-001 A patient enters Labour Ward. Expected result: The applicable profile activates maternal and fetal observation work and the approved Labour Care Guide without requiring entry into a blank form.

P7-FRM-AT-002 Blood pressure is required by routine vital signs, an active form and a protocol at the same time. Expected result: One due blood-pressure task is created; one valid Observation satisfies and populates all applicable outputs.

P7-FRM-AT-003 A client-space form requires four-hourly blood pressure while a patient-specific order requires hourly blood pressure. Expected result: Hourly work is generated and both contributing requirement sources remain visible.

P7-FRM-AT-004 The nurse records temperature, pulse, blood pressure, fetal heart rate and contractions from the observation worklist. Expected result: The system stores atomic Observations and automatically updates the Labour Care Guide and all applicable flowsheets and scores.

P7-FRM-AT-005 A clinician adds a form after relevant Observations already exist. Expected result: The form prepopulates immediately and only genuinely new or stricter work is added.

P7-FRM-AT-006 A patient transfers from Labour Ward to Postnatal Ward. Expected result: Destination defaults activate prospectively, inapplicable labour work ends under policy and the Labour Care Guide remains historically available.

P7-FRM-AT-007 Two active forms require temperature in the same clinically equivalent window. Expected result: Only one observation task is produced and one eligible result populates both forms.

P7-FRM-AT-008 The rendering service is temporarily unavailable. Expected result: Observation capture succeeds; rendering enters reconciliation and regenerates without data loss.

P7-FRM-AT-009 A source blood-pressure Observation is corrected. Expected result: Every consuming form and score reflects the correction with provenance while the original remains preserved.

P7-FRM-AT-010 A user selects an empty location in a rendered form and records a value. Expected result: The same authoritative Observation workflow is invoked and every applicable consuming view updates.

23. Mandatory Negative Tests

- Hardcoded title bypasses observation permission
- Wrong-patient observation via modified payload
- Bed assignment used as sole device-patient identity
- Free-text unit accepted as canonical
- Incompatible conversion returns unchanged number as success
- Entered-in-error value included in score
- Missing score input treated as zero
- Score version change rewrites history
- Preliminary value displayed as final
- Critical alert acknowledged and treated as resolved
- Device technical alarm charted as patient value
- Monitoring schedule fabricates observation
- Transfer duplicates overdue task
- Fluid volume counted from eMAR and manual entry
- Correction deletes original value
- Offline retry duplicates observation
- Sensitive observation leaks through trend/report

24. Traceability

Area	Prefix	Evidence
Governance	CLN-P7-GOV	Clinical governance review
CDE	CLN-P7-CDE	Registry/version tests
Groups	CLN-P7-GRP	Flowsheet tests
Status	CLN-P7-STS	Lifecycle tests
Manual	CLN-P7-MAN	Capture tests
Units	CLN-P7-UOM	Conversion tests
Ranges	CLN-P7-RNG	Context/range tests
Devices	CLN-P7-DEV	Mapping/provenance tests
AI	CLN-P7-AI	Human confirmation tests
Calculations	CLN-P7-CAL	Dependency tests
Scores	CLN-P7-SCR	Gold-case tests
Monitoring	CLN-P7-MON	Scheduling tests
Escalation	CLN-P7-ESC	Response tests
Trends	CLN-P7-TRD	Display tests
Fluid	CLN-P7-FLD	Duplicate/recalc tests
Correction	CLN-P7-COR	History tests
Downtime	CLN-P7-DWN	Sync tests

25. Registered Phase 7 Gaps

Gap ID	Resolution Required	Blocking
CLN-P7-GAP-001	Approve canonical CDE registry and initial observation groups.	Yes

CLN-P7-GAP-002	Approve base/permitted units and precision for every quantitative CDE.	Yes
CLN-P7-GAP-003	Approve reference, critical and physiological limit governance.	Yes
CLN-P7-GAP-004	Approve device registry, association and auto-validation policy.	Yes
CLN-P7-GAP-005	Approve monitoring schedules, due windows and escalation routing.	Yes
CLN-P7-GAP-006	Approve score catalogue, licensed matrices and gold test cases.	Yes
CLN-P7-GAP-007	Approve NEWS2 deployment scope and official implementation version.	Before NEWS2 release
CLN-P7-GAP-008	Approve fluid-balance source and duplicate-prevention rules.	Yes
CLN-P7-GAP-009	Approve voice/AI observation extraction scope.	Before AI release
CLN-P7-GAP-010	Define LIMS/RIS observation-reference boundaries.	Yes
CLN-P7-GAP-011	Approve offline critical-value escalation.	Yes
CLN-P7-GAP-012	Confirm FHIR Observation, vital-sign and device profiles.	Before interoperability release

26. Phase 7 Completion Gate

- Phases 1 to 6 incorporated without contradiction.
- CDE registry, observation groups and statuses approved.
- Shared Unit Engine policy and conversion provenance approved.
- Ranges, critical limits and physiological limits approved.
- Manual, device, voice and calculated capture rules approved.
- Scoring definitions and calculation evidence approved.
- Monitoring schedules and escalation workflows approved.
- Trend, fluid-balance, correction and downtime rules approved.
- Permissions, interfaces, events, reports and tests approved.
- Every blocking gap assigned an owner and acceptance test.

Phase 8 shall consume released observations and external results using the statuses, correction history, unit provenance and critical-escalation rules defined here.

Appendix A. Minimum Observation Record

Group	Fields
Identity	Public ID, patient, encounter, CDE code/version
Value	Typed value/components, original unit and normalized value/base unit
Context	Facility, client space, body site/method and source group
Time	Effective time/period, issued/final time and recorded time
Status	Preliminary/final/corrected/entered-in-error and reason
Actors	Performer, recorder, validator and device
Interpretation	Range/limit version, flags and target comparison
Derivation	Input IDs, algorithm/score version and rounding
Provenance	Conversion rule, raw source, correction chain and audit
Workflow	Monitoring due event, alert and follow-up references

Appendix B. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phases 1 to 6.

- HL7 FHIR Observation and vital-sign profile materials.
- HL7 Point-of-Care Device Implementation Guide materials.
- Royal College of Physicians NEWS2 materials.

PHASE 8

RESULTS, DIAGNOSTIC REPORTS AND CLOSED-LOOP FOLLOW-UP

Phase 8 defines receipt, validation, presentation, notification, acknowledgement, correction, clinical review and follow-up of laboratory, imaging, pathology and other diagnostic results.

Document Control and Phase Authority

Field	Rule
Purpose	Define safe diagnostic-result ingestion and closed-loop clinical follow-up.
Dependencies	Phases 1 to 7 remain mandatory, especially ordering, observation status, unit provenance, patient identity and documentation controls.
Ownership	LIMS, RIS/PACS, Pathology and other diagnostic systems own authoritative fulfilment and report release. Clinical owns receipt, chart presentation, review and follow-up.
Boundary	Clinical shall not edit an external authoritative report as though it created it.
Implementation	Functional specification only. Laravel engineering details belong in the EDD.
Release	Controlled draft, not approved for production.

Phase 8 outcomes

- One result inbox and patient result workspace.
- Reliable order-result matching and exception handling.
- Clear preliminary, final, amended and corrected states.
- Critical-result escalation with accountable response.
- Separate delivery, acknowledgement, review, action and closure states.
- Safe patient communication and disclosure.
- Complete permissions, events, reports, tests and gaps.

1. Governing Principles

CLN-P8-GOV-001 A diagnostic result shall preserve its authoritative source, patient, encounter, order/request, report, observations, status, effective time, issued time and provenance.

CLN-P8-GOV-002 Technical receipt shall not equal clinical review, and acknowledgement shall not equal action or closure.

CLN-P8-GOV-003 Clinical shall display the source report faithfully and shall not silently alter the source conclusion, interpretation, units or status.

CLN-P8-GOV-004 Preliminary, final, amended, corrected, cancelled and entered-in-error results shall be distinguishable throughout the chart and inbox.

CLN-P8-GOV-005 Every result requiring follow-up shall have an accountable owner or governed exception route.

CLN-P8-GOV-006 Critical results shall use configured urgent communication, acknowledgement and escalation workflows.

CLN-P8-GOV-007 Result corrections shall preserve prior versions and trigger review of dependent decisions without destructively rewriting history.

Closed-loop rule: a result is not complete merely because it arrived. The required recipient must review it, document appropriate action or no-action rationale, and resolve follow-up obligations.

2. Diagnostic Result Concepts

Concept	Meaning	Rule
Atomic result	Single measured or asserted value.	Represented with source status and unit
Diagnostic report	Coherent report containing observations, narrative, codes, images or attachments.	Source-authoritative package
Preliminary result	Available before final authorization.	Clearly provisional
Final result	Authorized source version.	Eligible for ordinary clinical review
Amended report	Final report supplemented.	Prior version retained
Corrected report	Erroneous content replaced.	Correction reason/version retained

Critical result	Result meeting source or Clinical urgent threshold.	Escalation required
Abnormal result	Outside applicable range or interpreted abnormal.	Not automatically critical
Result delivery	Technical availability to Clinical.	Not clinical acknowledgement
Acknowledgement	Named recipient accepts awareness.	Not proof of action
Clinical review	Authorized interpretation in patient context.	Records reviewer and decision
Follow-up	Required action, communication or monitoring.	Tracked to closure

3. Source and Ownership Boundaries

CLN-P8-OWN-001 The fulfilling diagnostic module shall remain authoritative for report identifiers, performer, specimen/study, method, source status, issued time, conclusion and correction history.

CLN-P8-OWN-002 Clinical shall store source identifiers and a faithful versioned representation or reference according to the interface contract.

CLN-P8-OWN-003 Clinical may add its own review, interpretation, action and communication records without modifying the source report.

CLN-P8-OWN-004 Images remain authoritative in PACS or the designated imaging repository; Clinical shall store governed references and viewer links.

CLN-P8-OWN-005 The Phase 5 order remains the authoritative placer request while the fulfilling system owns filler and performance status.

CLN-P8-OWN-006 Unsolicited or external results shall retain source and reconciliation state and shall not be falsely represented as locally ordered.

4. Result Ingestion Pipeline

Stage	Required Behavior
RECEIVED	Durably record message/file and correlation metadata
TECHNICALLY_VALIDATED	Validate schema, syntax, signature and required fields
PATIENT_MATCHED	Resolve patient with assigning authority
ORDER_MATCHED	Resolve placer/filler request where applicable
CLINICALLY_MAPPED	Map report/result codes, units and status
COMMITTED	Persist report, observations and source version
NOTIFIED	Route according to urgency and ownership
ACKNOWLEDGED	Receive application-level technical acknowledgement
QUARANTINED	Hold unresolved or unsafe content
REJECTED	Reject with actionable reason

CLN-P8-ING-001 The system shall preserve raw message/file hash, source endpoint, message ID, received time, correlation ID and interface version.

CLN-P8-ING-002 Patient mismatch, ambiguous order match, unknown code, incompatible unit, missing source status or duplicate conflict shall prevent ordinary automatic posting.

CLN-P8-ING-003 Retries shall use idempotency and shall not create duplicate reports or notifications.

CLN-P8-ING-004 Technical acknowledgement shall identify accepted, error or rejected processing outcome according to the interface contract.

CLN-P8-ING-005 Quarantined results shall enter an accountable exception queue with severity and time target.

5. Order and Result Matching

CLN-P8-MAT-001 Matching shall use stable patient identifiers with assigning authority plus placer/filler order identifiers where supplied.

CLN-P8-MAT-002 Demographic similarity alone shall not auto-post a result to a patient when a stable match is absent or conflicting.

CLN-P8-MAT-003 One order may produce multiple reports/results and one report may group multiple observations according to the source model.

CLN-P8-MAT-004 Cancelled or discontinued order status shall not cause a legitimately performed result to be discarded; it shall be posted or quarantined according to policy with discrepancy visibility.

CLN-P8-MAT-005 Unmatched results shall not disappear from operational monitoring and shall require documented resolution.

CLN-P8-MAT-006 Manual matching shall require a dedicated permission, two-identifier confirmation, reason and audit.

6. Diagnostic Report Record

Group	Required Data
Identity	Clinical public ID, source report ID/version, tenant, patient and encounter
Request	Placer/filler order references and requested service
Performance	Diagnostic service, performer/interpreter, specimen/study and effective period
Content	Atomic results, conclusion, conclusion codes, recommendations, media and formatted report
Status	Source status, status reason, Clinical ingestion state and correction chain
Time	Performed/effective, issued, received, posted and notified times
Interpretation	Reference ranges, flags, criticality and source narrative
Governance	Sensitivity, provenance, digital signature where supplied, raw source hash and audit

CLN-P8-DAT-001 Atomic results shall reference Phase 7 CDE/Observation mappings where approved while preserving source code and display.

CLN-P8-DAT-002 The formatted source report and structured results shall remain linked, and discrepancies shall be visible.

CLN-P8-DAT-003 A report conclusion shall not be generated from atomic results by Clinical unless an authorized derived workflow explicitly creates a separate interpretation.

CLN-P8-DAT-004 Historical report versions shall remain accessible to authorized users.

7. Status and Version Lifecycle

Source Status	Clinical Handling
REGISTERED/PENDING	No usable result yet; track order only
PARTIAL	Some results available; show incompleteness
PRELIMINARY	Display prominently as provisional
FINAL	Eligible for standard review workflow
AMENDED	New information added to prior final
CORRECTED	Erroneous content replaced; prior version retained
APPENDED	Additional source content attached
CANCELLED	Diagnostic report cancelled by source
ENTERED_IN_ERROR	Invalid source report excluded from ordinary clinical use
UNKNOWN	Quarantine or display with explicit uncertainty under policy

CLN-P8-STs-001 A later version shall not silently replace a version already clinically reviewed.

CLN-P8-STs-002 Amended and corrected reports shall reopen review/follow-up when material under configured policy.

CLN-P8-STs-003 Entered-in-error results shall be excluded from ordinary trends, scores and decision support while preserving audit access.

CLN-P8-STs-004 Result version comparison shall identify changed values, narrative, interpretation, conclusion and recommendations where technically possible.

8. Result Presentation and Patient Chart

CLN-P8-UI-001 The patient banner and applicable encounter context shall remain visible on result pages.

CLN-P8-UI-002 The result display shall show source, service, status, issued time, performer/interpreter, order, specimen/study, value/unit, range and interpretation as applicable.

CLN-P8-UI-003 Preliminary, corrected, critical, externally sourced and entered-in-error results shall not rely on colour alone.

CLN-P8-UI-004 The UI shall distinguish source interpretation from the reviewing clinician's note or action.

CLN-P8-UI-005 Cumulative result charts shall use Phase 7 unit conversion and provenance rules and shall not overwrite original units.

CLN-P8-UI-006 Imaging viewer links shall reauthorize access and shall not expose unrestricted tokens or identifiers.

CLN-P8-UI-007 Sensitive report titles, counts, snippets and notifications shall not leak protected content.

9. Results Inbox and Assignment

Queue	Inclusion
My New Results	Unreviewed results assigned to the user
Team Results	Results assigned to an eligible care team
Critical Results	Critical results awaiting acknowledgement/action
Corrected Results	Material amendments/corrections needing re-review
Unmatched Results	Patient/order reconciliation exceptions
No Responsible Owner	Results lacking an eligible recipient
Post-Discharge Results	Results issued after encounter closure
Follow-Up Due	Reviewed results with outstanding actions
Communication Due	Patient/recipient communication required

CLN-P8-INB-001 Inbox routing shall use current responsible clinician/team, order requester, care relationship, client-space assignment and configured fallback hierarchy.

CLN-P8-INB-002 A result appearing in a queue shall not itself grant chart access.

CLN-P8-INB-003 Reassignment shall preserve previous owner, reason, time and acceptance where required.

CLN-P8-INB-004 Absence, transfer or employment-status change shall reroute unresolved results through approved cover rules.

CLN-P8-INB-005 Bulk acknowledgement shall be prohibited for critical results and restricted for other results by explicit policy.

10. Critical Results Workflow

State	Meaning
TRIGGERED	Critical rule/source flag identified
ROUTED	Sent to eligible primary recipient
DELIVERED	Notification reached configured channel
ACKNOWLEDGED	Named recipient accepts awareness
ASSESSED	Clinical assessment recorded
ACTION_TAKEN	Action or no-action rationale recorded
ESCALATED	Next recipient/level invoked
RESOLVED	Required response and follow-up complete

CLN-P8-CRT-001 Criticality may originate from source flags or Clinical rules; the origin and rule version shall be preserved.

CLN-P8-CRT-002 Critical notification shall identify patient safely, result, value/unit or finding, source, time and required response subject to confidentiality policy.

CLN-P8-CRT-003 Acknowledgement shall require an authenticated eligible user and shall record channel, time and result version.

CLN-P8-CRT-004 Failure to acknowledge within the configured interval shall escalate through an approved hierarchy.

CLN-P8-CRT-005 Acknowledgement shall not resolve the alert without clinical assessment and action/no-action rationale as configured.

CLN-P8-CRT-006 Corrected critical results shall preserve the original alert and create or update follow-up according to materiality rules.

11. Clinical Review and Action

CLN-P8-REV-001 Review shall require clinical.result.review and appropriate patient scope/relationship.

CLN-P8-REV-002 The reviewer shall attest to the exact report version reviewed.

CLN-P8-REV-003 Review outcomes shall include ACKNOWLEDGED_NO_ACTION, ACTION_REQUIRED, ACTION_COMPLETED, REFERRED, REPEAT_ORDERED, PATIENT_CONTACT_REQUIRED or other approved states.

CLN-P8-REV-004 Creating an order, diagnosis, care-plan update or note from a result shall invoke the owning workflow and permission checks.

CLN-P8-REV-005 A changed or corrected source report shall not inherit the previous review automatically unless policy determines the change immaterial and records that determination.

CLN-P8-REV-006 Review evidence shall include reviewer, result version, time, decision and linked action records.

12. Follow-Up Tasks and Closure

CLN-P8-FUP-001 A result may generate follow-up tasks for repeat testing, treatment, referral, monitoring, documentation or communication.

CLN-P8-FUP-002 Each task shall identify accountable owner, due time, priority, patient, result version and completion evidence.

CLN-P8-FUP-003 A result shall not be marked closed while mandatory tasks remain unresolved.

CLN-P8-FUP-004 Overdue follow-up shall escalate according to risk, time and organizational policy.

CLN-P8-FUP-005 Task reassignment shall require acceptance where configured and shall not remove prior accountability history.

CLN-P8-FUP-006 Encounter closure shall not close pending results or follow-up obligations automatically.

13. Post-Discharge and Delayed Results

CLN-P8-PDR-001 Results issued after discharge shall remain linked to the original order and encounter.

CLN-P8-PDR-002 The system shall route post-discharge results to the configured responsible clinician/team and fallback service.

CLN-P8-PDR-003 Patient movement, discharge or user assignment expiry shall not make an unresolved result ownerless.

CLN-P8-PDR-004 Delayed preliminary-to-final conversion shall trigger review according to the final report version and materiality policy.

CLN-P8-PDR-005 External follow-up responsibility shall be documented with recipient, transfer method, acceptance and remaining local obligation.

14. Patient and Caregiver Communication

CLN-P8-COM-001 Communication policy shall define which results may be released automatically, which require clinician review and which require direct discussion.

CLN-P8-COM-002 Communication records shall identify recipient, relationship/authority, channel, date/time, result version, content summary, interpreter/support and outcome.

CLN-P8-COM-003 Portal publication or message delivery shall not be treated as proof the patient understood the result.

CLN-P8-COM-004 Sensitive, unexpected, critical or complex results shall follow approved communication pathways.

CLN-P8-COM-005 Incorrect recipient or failed delivery shall enter a security-safe exception workflow.

CLN-P8-COM-006 Patient questions or refusal of follow-up shall be documented and routed appropriately.

15. Corrections, Amendments and Retractions

CLN-P8-COR-001 Clinical shall preserve every received source version and the source correction/amendment reason where supplied.

CLN-P8-COR-002 A material correction shall reopen clinical review, critical escalation, patient communication and follow-up as configured.

CLN-P8-COR-003 The system shall identify downstream diagnoses, scores, notes, orders and care plans that referenced the superseded result and create review tasks.

CLN-P8-COR-004 Clinical users shall not edit the authoritative result to resolve an interface or source error.

CLN-P8-COR-005 Local annotation shall remain separate from the source result and follow Phase 3 documentation rules.

CLN-P8-COR-006 Entered-in-error or retracted results shall not disappear from authorized audit and reconciliation views.

16. AI-Assisted Result Review

CLN-P8-AI-001 AI summaries, comparisons, abnormality highlights and suggested follow-up shall be provided only through the Shared AI Gateway.

CLN-P8-AI-002 AI shall use only report versions and chart context the requesting user is authorized to access.

CLN-P8-AI-003 AI output shall remain advisory and shall not acknowledge, close, diagnose, order, communicate or resolve a result automatically.

CLN-P8-AI-004 The user shall be able to inspect source references where the service supports them.

CLN-P8-AI-005 Clinical shall preserve AI transaction, input result versions, suggestion, edits, acceptance/rejection and final human action.

CLN-P8-AI-006 AI outage shall not block ordinary manual review and follow-up.

17. Downtime, Offline and Recovery

CLN-P8-DWN-001 Critical results received during Clinical downtime shall follow an approved source-to-clinician fallback communication process.

CLN-P8-DWN-002 Offline acknowledgement/action shall preserve patient, result/source version, recipient, time, channel and action for later reconciliation.

CLN-P8-DWN-003 Recovery shall reconcile source reports, message acknowledgements, critical communications and Clinical follow-up without duplicating alerts or tasks.

CLN-P8-DWN-004 Late-arriving results shall retain source effective and issued times plus received/posted times.

CLN-P8-DWN-005 A possible duplicate or conflicting version shall be quarantined rather than silently merged.

18. Permissions Catalogue

Permission	Purpose
<code>clinical.result.view</code>	View ordinary authorized result
<code>clinical.result.view_sensitive</code>	View sensitive result
<code>clinical.result.review</code>	Record clinical review
<code>clinical.result.critical_acknowledge</code>	Acknowledge critical result
<code>clinical.result.critical_resolve</code>	Resolve critical-result response
<code>clinical.result.reassign</code>	Reassign accountable owner
<code>clinical.result.follow_up_create</code>	Create follow-up task
<code>clinical.result.follow_up_complete</code>	Complete follow-up
<code>clinical.result.patient_communicate</code>	Record patient communication
<code>clinical.result.manual_match</code>	Resolve patient/order matching
<code>clinical.result.annotation_create</code>	Add separate Clinical annotation
<code>clinical.result.exception_manage</code>	Resolve ingestion/interface exception
<code>clinical.result.export</code>	Export result/report
<code>clinical.result_config.manage</code>	Manage routing, thresholds and materiality
<code>clinical.result.audit</code>	View detailed provenance

CLN-P8-PRM-001 Permissions shall be centrally registered and contextually enforced through Laravel policies, not hardcoded title checks.

19. Operational Interfaces and Queues

- Patient result summary
- Result detail and source report viewer
- Cumulative results/trends

- My New Results
- Critical Results
- Corrected/Amended Results
- Post-Discharge Results
- Follow-Up Due/Overdue
- Patient Communication Due
- Unmatched/Quarantined Results
- No Responsible Owner
- Interface acknowledgement/retry monitoring
- Routing and threshold administration

CLN-P8-OPS-001 Queues shall show age, urgency, owner, status and next required action without leaking restricted content.

CLN-P8-OPS-002 Opening any queue item shall reauthorize patient and result access.

CLN-P8-OPS-003 Technical, source, review and follow-up states shall be displayed separately.

20. Audit, Events and Reports

- Message receipt/validation/acknowledgement
- Patient/order match and manual override
- Report commit/version update
- Notification delivery/failure
- Critical acknowledgement/escalation/resolution
- Clinical review/action
- Follow-up task and reassignment
- Patient communication
- Sensitive view/export
- Correction and dependent review
- Downtime reconciliation
- clinical.result.received
- clinical.result.posted
- clinical.result.critical
- clinical.result.acknowledged
- clinical.result.reviewed
- clinical.result.corrected
- clinical.result.follow_up_required
- clinical.result.follow_up_overdue
- clinical.result.patient_communicated
- clinical.result.quarantined
- clinical.result.reconciliation_required
- Unreviewed results by age/risk
- Critical acknowledgement/response time
- Corrected results awaiting re-review
- Post-discharge unresolved results
- Follow-up overdue
- Communication due/failed
- Results without owner
- Unmatched/quarantined results
- Interface errors and duplicates
- Manual match overrides
- Sensitive access/exports

CLN-P8-EVT-001 Events shall include stable IDs, tenant, patient, encounter, report version, occurred-at, correlation ID and idempotency key while minimizing sensitive content.

CLN-P8-RPT-001 Reports and exports shall remain permission-, scope- and sensitivity-controlled.

21. Acceptance Scenarios

ID	Scenario	Expected Result
P8-AT-001	Final lab report matches order and patient.	Source report posted once, routed to responsible reviewer and linked to order.
P8-AT-002	Result has conflicting patient identifiers.	Quarantined; no ordinary chart posting.
P8-AT-003	Technical message accepted.	Application acknowledgement recorded but result not marked clinically reviewed.
P8-AT-004	Critical potassium result arrives.	Urgent alert routes, requires named acknowledgement and action.
P8-AT-005	Primary clinician unavailable.	Escalation routes to configured cover without losing accountability.
P8-AT-006	Preliminary report becomes final.	Version/status update triggers final review policy.
P8-AT-007	Final report is materially corrected.	Previous review not silently inherited; dependent actions enter review.
P8-AT-008	Imaging report has PACS link.	Viewer reauthorizes access; source report remains authoritative.
P8-AT-009	Result arrives after discharge.	Original encounter retained and accountable post-discharge queue created.
P8-AT-010	Reviewer orders repeat test.	Phase 5 workflow creates separately authorized order.
P8-AT-011	Patient portal publication succeeds.	Communication state records delivery, not understanding or clinical closure.
P8-AT-012	Unsolicited external result arrives.	Source-labelled reconciliation workflow used; no false local-order link.
P8-AT-013	AI summarizes report.	Advisory output cannot review or close result.
P8-AT-014	Same result message retries.	Idempotency prevents duplicate report, alert and task.
P8-AT-015	Downtime critical result communicated by phone.	Fallback action recorded and reconciled after restoration.

22. Mandatory Negative Tests

- Hardcoded title grants result review
- Technical ACK treated as clinical acknowledgement
- Acknowledgement treated as follow-up closure
- Wrong-patient result posted from demographic similarity
- Duplicate retry creates second critical alert
- Preliminary result displayed as final
- Correction overwrites prior report
- Entered-in-error value remains in trends/decision support
- Clinical user edits source conclusion
- Queue item grants chart access
- Bulk critical acknowledgement
- Discharged patient loses result owner
- Portal delivery treated as patient understanding
- AI auto-closes result
- Sensitive report title leaks in notification
- PACS link bypasses authorization
- No-owner result disappears from monitoring

23. Traceability

Area	Prefix	Evidence
Governance	CLN-P8-GOV	Clinical governance review
Ownership	CLN-P8-OWN	Module contract tests
Ingestion	CLN-P8-ING	Interface/idempotency tests
Matching	CLN-P8-MAT	Patient/order tests
Data	CLN-P8-DAT	Schema/version tests
Status	CLN-P8-STS	Lifecycle tests
UI	CLN-P8-UI	Display/access tests
Inbox	CLN-P8-INB	Routing tests
Critical	CLN-P8-CRT	Escalation tests
Review	CLN-P8-REV	Review/action tests
Follow-up	CLN-P8-FUP	Closure tests
Post-discharge	CLN-P8-PDR	Ownership tests
Communication	CLN-P8-COM	Recipient tests
Correction	CLN-P8-COR	Dependent-review tests
AI	CLN-P8-AI	Human-control tests
Downtime	CLN-P8-DWN	Reconciliation tests

24. Registered Phase 8 Gaps

Gap ID	Resolution Required	Blocking
CLN-P8-GAP-001	Approve diagnostic-source systems and authoritative report contracts.	Yes
CLN-P8-GAP-002	Approve patient/order matching rules and manual-match authority.	Yes
CLN-P8-GAP-003	Approve result status mappings by LIMS, RIS/PACS and Pathology.	Yes
CLN-P8-GAP-004	Approve routing hierarchy, cover and no-owner workflow.	Yes
CLN-P8-GAP-005	Approve critical-result thresholds, channels and acknowledgement times.	Yes
CLN-P8-GAP-006	Approve clinical review outcomes and closure evidence.	Yes
CLN-P8-GAP-007	Approve correction materiality and dependent-review rules.	Yes
CLN-P8-GAP-008	Approve post-discharge responsibility and escalation.	Yes
CLN-P8-GAP-009	Approve patient result-release and communication policy.	Yes
CLN-P8-GAP-010	Approve downtime critical-result fallback.	Yes
CLN-P8-GAP-011	Approve AI result-summary scope.	Before AI release
CLN-P8-GAP-012	Confirm FHIR DiagnosticReport, Observation, ImagingStudy and DocumentReference profiles.	Before interoperability release

25. Phase 8 Completion Gate

- Phases 1 to 7 incorporated without contradiction.
- Diagnostic-source ownership and ingestion contracts approved.
- Patient/order matching and quarantine rules approved.
- Result/report data and status lifecycle approved.
- Inbox ownership, critical escalation and review states approved.
- Follow-up and post-discharge closure rules approved.

- Patient communication and sensitive-result controls approved.
- Correction, dependent review and downtime reconciliation approved.
- Permissions, events, reports, acceptance and negative tests approved.
- Every blocking gap assigned an owner and acceptance test.

Phase 9 shall use the result-review and follow-up evidence defined here when governing broader transitions, discharge and continuity of care. It shall not close encounters while result obligations remain unresolved.

Appendix A. Minimum Diagnostic Report Record

Group	Fields
Identity	Public ID, source report ID/version, patient, encounter
Request	Placer/filler order references
Performance	Service, performer/interpreter, specimen/study, effective time
Content	Atomic results, conclusion, codes, media and formatted report
Status	Source status/reason, ingestion state and correction chain
Interpretation	Ranges, abnormal/critical flags and source recommendations
Time	Issued, received, posted, notified and reviewed
Workflow	Owner, acknowledgement, review, action, follow-up and communication
Security	Sensitivity and disclosure controls
Provenance	Source endpoint, raw hash, signature, correlation and audit

Appendix B. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phases 1 to 7.
- HL7 FHIR DiagnosticReport and Diagnostics Module materials.
- ONC SAFER Test Results Reporting and Follow-Up guidance.
- HL7 acknowledgement guidance for reliable interface processing.

PHASE 9

HANDOVER, TRANSITIONS, DISCHARGE AND CONTINUITY OF CARE

Phase 9 defines shift and service handover, internal transfer, discharge readiness, discharge documentation, external transfer, continuity-of-care communication and post-transition accountability.

Document Control and Phase Authority

Field	Rule
Purpose	Define safe transfer of information, responsibility and unresolved work across care transitions.
Dependencies	Phases 1 to 8 remain mandatory, including patient context, documentation, medication reconciliation, observations, orders and result follow-up.
Boundary	Administrative bed allocation and transport may be owned outside Clinical; Clinical owns clinical readiness, handover content and continuity obligations.
Discharge record	Discharge summary and instructions shall follow Phase 3 attestation and correction rules.
Implementation	Functional specification only. Laravel implementation belongs in the companion EDD.
Release	Controlled draft, not approved for production.

Phase 9 outcomes

- Structured, attributable handover.
- Explicit transfer of responsibility with acceptance.
- Discharge-readiness checks across all prior phases.
- Versioned discharge summary and patient instructions.
- Closed-loop communication to receiving providers.
- Post-discharge ownership for pending results and follow-up.
- Complete permissions, audit, reports, tests and release gaps.

1. Governing Principles

CLN-P9-GOV-001 A transition shall transfer clinical information and accountability without erasing the responsibilities or history of the sending team.

CLN-P9-GOV-002 Handover completion shall require an identified sender, intended receiver, patient, encounter, effective time, content version and acceptance state where configured.

CLN-P9-GOV-003 A patient movement event shall not by itself prove that clinical handover or responsibility acceptance occurred.

CLN-P9-GOV-004 Generated summaries shall preserve source status and shall never represent preliminary, unreviewed or entered-in-error information as final clinical fact.

CLN-P9-GOV-005 Encounter closure shall be blocked or exception-controlled when mandatory transition obligations remain unresolved.

CLN-P9-GOV-006 Patient and caregiver communication shall be understandable, accessible and recorded separately from professional handover.

CLN-P9-GOV-007 AI may draft transition content but shall not accept responsibility, sign, discharge or close an encounter.

Transition safety rule: information delivery is not responsibility transfer. The receiving person or service must accept the handover when the configured workflow requires acceptance.

2. Transition Concepts

Concept	Meaning	Key Distinction
Shift handover	Responsibility changes between staff shifts.	Patient location may not change
Service handover	Responsibility changes between specialties/teams.	Encounter may remain active
Internal transfer	Patient moves between client spaces/units.	Movement and clinical acceptance are separate
External transfer	Care transfers to another organization.	Requires disclosure and receiving

		acceptance
Discharge	Current encounter ends with onward-care arrangements.	Not merely physical departure
Leave	Temporary absence during active encounter.	Not discharge
Discharge readiness	Clinical and operational prerequisites assessed.	Not authorization by itself
Discharge summary	Attested professional continuity document.	Distinct from patient instructions
Patient instructions	Patient/caregiver-facing plan and warnings.	Not the entire legal record
Continuity task	Unresolved action with an owner and due time.	Survives encounter closure where required

3. Transition Types and Triggers

- Change of shift
- Temporary cover or on-call handover
- Transfer between ward, ICU, theatre, recovery or clinic
- Change of responsible specialty/service
- Inter-facility transfer
- Discharge home
- Discharge to rehabilitation, community or residential care
- Self-discharge or leaving before completion
- Temporary leave
- Virtual-to-physical or physical-to-virtual transition
- Death and mortuary transition under separate policy

CLN-P9-TYP-001 Transition types shall be controlled, versioned and mapped to applicable encounter and movement events.

CLN-P9-TYP-002 Each transition type shall define required content, sender/receiver roles, acceptance rule, readiness checks, documents, communication and escalation.

CLN-P9-TYP-003 A transition workflow may be initiated by movement, encounter status, scheduled discharge, clinician decision or authorized external request.

4. Handover Record

Group	Required Data
Identity	Public ID, patient, encounter, transition type and version
Context	Sending and receiving facility, client space, service and team
Actors	Sender, receiver, responsible clinicians and recorder
Time	Prepared, sent, received, accepted and effective times
Clinical state	Current condition, acuity, trajectory and recent significant changes
Safety	Allergies, alerts, infection/isolation, risks and escalation status
Treatment	Active medication, infusions, oxygen, devices and procedures
Work	Pending orders, results, tasks, monitoring and consultations
Plan	Goals, immediate priorities, contingency plans and review times
Governance	Source references, sensitivity, attestation and audit

CLN-P9-HND-001 Handover content shall be assembled from authorized current source data and identify stale, provisional, corrected or missing information.

CLN-P9-HND-002 The author shall be able to add an attested narrative assessment without overwriting source facts.

CLN-P9-HND-003 The handover shall display the data-as-of time and content version.

CLN-P9-HND-004 A later update shall create a new version and shall not silently alter an already accepted handover.

CLN-P9-HND-005 Restricted content shall follow minimum-necessary and receiving-authority rules.

5. Structured Communication Model

5.1 Minimum communication domains

- Situation and reason for handover
- Background and relevant history
- Current assessment and trajectory
- Recent vital signs, scores and significant results
- Active medication, infusions and recent administrations
- Pending investigations and expected results
- Required actions with owner and timing
- Risks, escalation status and contingency plan
- Patient/caregiver concerns and communication needs
- Receiver confirmation and questions

CLN-P9-COM-001 The system may support SBAR or another approved structured model through versioned templates.

CLN-P9-COM-002 The template shall not force users to duplicate source data as untraceable free text when a reference can be preserved.

CLN-P9-COM-003 The receiver shall be able to request clarification and the response shall remain linked to the handover.

CLN-P9-COM-004 Read-back or repeat-back confirmation shall be supported for configured high-risk transitions.

6. Shift and Temporary-Cover Handover

CLN-P9-SHF-001 Shift handover lists shall be generated from current authorized patient assignment, acuity, alerts, tasks and due work.

CLN-P9-SHF-002 The system shall identify patients with critical alerts, active infusions, high-alert medications, pending critical results, overdue monitoring or unresolved deterioration responses.

CLN-P9-SHF-003 Temporary cover shall have an effective period and shall not grant permanent care relationships.

CLN-P9-SHF-004 Acceptance shall transfer current operational ownership without deleting the prior team's authored records or unresolved accountability history.

CLN-P9-SHF-005 Patients not successfully handed over shall remain visible in an exception queue.

7. Internal Transfer

Stage	Required Evidence
REQUESTED	Destination and reason identified
CLINICALLY_READY	Sending team completes readiness checks
DESTINATION_ACCEPTED	Receiving service/space accepts
IN_TRANSIT	Movement underway where used
ARRIVED	Patient reaches destination
HANDOVER_ACCEPTED	Clinical responsibility accepted
COMPLETED	Movement, responsibility and work routing reconciled
CANCELLED/FAILED	Reason and current responsibility preserved

CLN-P9-INT-001 Transfer shall coordinate Phase 2 movement status with a distinct clinical handover state.

CLN-P9-INT-002 Receiving acceptance shall identify the accepting user/team and effective time.

CLN-P9-INT-003 Medication due events, infusions, monitoring schedules, pending results, orders and tasks shall transfer without duplication or loss.

CLN-P9-INT-004 The sending team shall retain responsibility until the configured acceptance point.

CLN-P9-INT-005 A rejected or delayed transfer shall preserve current patient location and accountable team and trigger contingency action where required.

8. Discharge Readiness

Domain	Readiness Check
Clinical stability	Discharge decision and condition appropriate to destination
Documentation	Required notes and summaries complete or exception-controlled
Diagnoses	Admission/discharge diagnoses and problem status reviewed
Medication	Discharge reconciliation and prescriptions completed
Orders/results	Pending, final, corrected and post-discharge results assigned
Monitoring	Outstanding observation requirements resolved or transferred
Care plan	Goals, ongoing plans and ownership updated
Referrals	Appointments, consultations and external referrals arranged
Patient needs	Education, equipment, transport, social and accessibility needs addressed
Communication	Patient/caregiver and receiving provider information prepared
Administration	Disposition and destination confirmed through owning workflow

CLN-P9-RDY-001 Readiness checks shall be versioned and configurable by encounter class, service and destination.

CLN-P9-RDY-002 Blocking failures and overrideable exceptions shall be distinguishable.

CLN-P9-RDY-003 An override shall require eligible permission, reason, accountable owner and due follow-up where applicable.

CLN-P9-RDY-004 The readiness checklist shall not auto-complete from missing or stale data.

9. Discharge Workflow

Step	Required Behavior
1. Initiate	Record expected discharge destination/date and responsible clinician
2. Assess readiness	Evaluate mandatory domains and unresolved exceptions
3. Reconcile medication	Complete Phase 6 discharge reconciliation
4. Resolve continuity	Assign pending results, orders, tasks and plans
5. Prepare documents	Create discharge summary and patient instructions
6. Review/attest	Apply Phase 3 signature, co-signature and correction rules
7. Communicate	Send to patient/caregiver and receiving providers as approved
8. Complete discharge	Record disposition, departure and encounter status
9. Confirm delivery	Track receipt/processing where contract supports it
10. Monitor follow-up	Escalate unresolved post-discharge obligations

CLN-P9-DSC-001 Discharge completion shall require clinical.encounter.discharge and the applicable Phase 2 closure authority.

CLN-P9-DSC-002 Actual departure time, discharge decision time, document issue time and encounter closure time shall remain distinct.

CLN-P9-DSC-003 Discharge shall not cancel pending legitimate diagnostic work or follow-up obligations.

CLN-P9-DSC-004 A patient leaving before completion shall use a distinct workflow and preserve attempted assessment, advice, risks and follow-up.

CLN-P9-DSC-005 Death shall use a separate governed workflow and shall not be represented as ordinary discharge home.

10. Discharge Summary

Section	Minimum Content
Encounter	Reason for admission/contact, dates, responsible service and disposition
Clinical course	Significant history, findings, progress and complications
Diagnosis	Admission, working and discharge diagnoses with verification state
Procedures	Important procedures and interventions

Medication	Admission changes, discharge medication and stopped medicines
Results	Significant results and clearly assigned pending results
Allergies/safety	Allergies, alerts and relevant precautions
Plan	Follow-up, monitoring, referrals, care plans and escalation
Patient communication	Information provided and outstanding education needs
Contacts	Receiving provider/service and support contacts where approved

CLN-P9-SUM-001 The discharge summary shall be a versioned, attested document governed by Phase 3.

CLN-P9-SUM-002 Auto-populated content shall preserve source links and status and shall not include entered-in-error data.

CLN-P9-SUM-003 Pending results shall include accountable reviewer, communication plan and follow-up expectation.

CLN-P9-SUM-004 A corrected source item after summary issue shall generate a review task and may require an amended summary.

CLN-P9-SUM-005 The summary shall identify its information cut-off and issue time.

11. Patient and Caregiver Instructions

CLN-P9-PAT-001 Patient instructions shall use approved plain-language templates and the patient's preferred language and accessibility needs where available.

CLN-P9-PAT-002 Instructions shall include medication directions, warning signs, activity/diet/wound guidance, follow-up, appointments and contact/escalation routes as applicable.

CLN-P9-PAT-003 The system shall record recipient, interpreter/support person, method, date/time and teach-back or understanding assessment where configured.

CLN-P9-PAT-004 Professional-only confidential content shall not be copied into patient instructions without disclosure review.

CLN-P9-PAT-005 Printing or portal delivery shall not prove understanding.

12. External Transfer and Receiving Provider Communication

CLN-P9-EXT-001 External transfer shall identify receiving organization/service, destination, accepting clinician where applicable and transport requirements.

CLN-P9-EXT-002 Disclosure shall be limited to authorized continuity information and shall preserve consent/legal basis where required.

CLN-P9-EXT-003 The transfer package shall include a controlled summary plus approved supporting documents/references.

CLN-P9-EXT-004 Transmission state shall distinguish sent, delivered, processed, rejected and accepted.

CLN-P9-EXT-005 A successful network transmission shall not be treated as acceptance of clinical responsibility.

CLN-P9-EXT-006 Failed or rejected delivery shall enter an urgent exception workflow when the patient transition depends on it.

13. Continuity Tasks and Follow-Up

CLN-P9-FUP-001 Every unresolved action at transition shall have an owner, due time, priority, source, patient and completion evidence.

CLN-P9-FUP-002 Continuity tasks may include pending-result review, repeat testing, medication monitoring, referral, appointment, document completion, patient contact or equipment/service confirmation.

CLN-P9-FUP-003 Encounter closure shall not delete or conceal continuity tasks.

CLN-P9-FUP-004 Ownership transfer shall preserve previous owner, acceptance and fallback escalation.

CLN-P9-FUP-005 Task completion shall reference the action performed and shall not be satisfied by acknowledgement alone.

CLN-P9-FUP-006 Overdue post-transition tasks shall remain reportable and escalate according to risk.

14. Medication Continuity

CLN-P9-MED-001 Discharge medication shall derive from completed reconciliation and authorized Phase 5 orders, not from a copied inpatient eMAR list.

CLN-P9-MED-002 The transition record shall distinguish new, continued, changed, held and stopped medicines with reasons where applicable.

CLN-P9-MED-003 The last administered dose and next due dose shall be available for high-risk transfers where clinically relevant.

CLN-P9-MED-004 Patient-owned, leave and controlled medicines shall have clear custody and supply disposition.

CLN-P9-MED-005 Medication changes made after summary issue shall trigger updated communication and reconciliation where material.

15. Pending Results and Diagnostic Continuity

CLN-P9-RES-001 The transition record shall identify pending, preliminary, expected and recently corrected results relevant to continuity.

CLN-P9-RES-002 Each pending result shall retain the Phase 8 accountable reviewer and escalation route after discharge or transfer.

CLN-P9-RES-003 Receiving-provider notification shall not automatically remove the sender's obligation unless acceptance of responsibility is recorded.

CLN-P9-RES-004 Material corrected results after transition shall reopen review and communication under Phase 8.

CLN-P9-RES-005 No-owner pending results shall block closure or enter an approved high-priority exception.

16. AI-Assisted Transition Documentation

CLN-P9-AI-001 AI may draft handovers, discharge summaries and patient instructions only through the Shared AI Gateway.

CLN-P9-AI-002 AI shall use only authorized source data and shall preserve source-version references where supported.

CLN-P9-AI-003 AI output shall remain visibly draft until reviewed and attested by an authorized human.

CLN-P9-AI-004 AI shall not invent negative findings, resolve pending work, assign acceptance or infer that follow-up occurred.

CLN-P9-AI-005 Clinical shall preserve AI transaction, input cut-off/version, output, edits, rejection/acceptance and final signer.

CLN-P9-AI-006 AI outage shall not block manual transition workflows.

17. Corrections, Cancellation and Reopening

CLN-P9-COR-001 Signed handovers and discharge summaries shall not be destructively edited.

CLN-P9-COR-002 A materially incorrect transition document shall use Phase 3 correction/amendment and notify affected recipients.

CLN-P9-COR-003 Cancelled transfers shall preserve prepared handover content and reason without changing patient location or responsibility incorrectly.

CLN-P9-COR-004 Encounter reopening shall require dedicated authority and shall not erase the original discharge/closure event.

CLN-P9-COR-005 A readmission shall ordinarily create a new encounter rather than reopening the prior encounter.

18. Downtime and Degraded Operations

CLN-P9-DWN-001 Downtime handover shall use an approved printable/offline dataset with issue time, page identifiers and patient identifiers.

CLN-P9-DWN-002 Verbal and paper handovers shall record sender, receiver, time, content version and acceptance for later reconciliation.

CLN-P9-DWN-003 Discharge during downtime shall use approved minimum documentation and create a mandatory electronic reconciliation task.

CLN-P9-DWN-004 Recovery shall reconcile movements, encounter status, medication, pending results, tasks and transmitted documents without duplication.

CLN-P9-DWN-005 Critical continuity information shall not wait for system restoration when an approved alternate communication channel exists.

19. Permissions Catalogue

Permission	Purpose
<code>clinical.handover.create</code>	Prepare handover
<code>clinical.handover.send</code>	Send handover
<code>clinical.handover.accept</code>	Accept responsibility
<code>clinical.handover.return</code>	Request clarification/return
<code>clinical.transfer.readiness_record</code>	Record clinical transfer readiness
<code>clinical.transfer.accept</code>	Accept internal/external clinical transfer
<code>clinical.discharge.initiate</code>	Start discharge workflow
<code>clinical.discharge.readiness_override</code>	Override eligible readiness item
<code>clinical.discharge.summary_create</code>	Prepare discharge summary
<code>clinical.discharge.summary_sign</code>	Attest discharge summary
<code>clinical.discharge.complete</code>	Complete clinical discharge
<code>clinical.continuity_task.manage</code>	Create/reassign/complete continuity task
<code>clinical.transition.patient_communicate</code>	Record patient instructions/communication
<code>clinical.transition.external_send</code>	Transmit continuity package
<code>clinical.transition.exception_manage</code>	Resolve transition exceptions
<code>clinical.transition.audit</code>	View detailed transition provenance

CLN-P9-PRM-001 Permissions shall be centrally registered and contextually enforced through Laravel policies, never hardcoded title comparisons.

20. Operational Interfaces and Queues

- Shift handover list
- Patient handover editor
- Internal transfer tracker
- Discharge readiness board
- Discharge summary workspace
- Patient instruction generator
- External transfer package
- Pending-result continuity queue
- Post-discharge follow-up queue
- No-owner continuity tasks
- Rejected/failed transmission queue
- Unsigned or incomplete summaries
- Transfer/discharge reconciliation exceptions

CLN-P9-UI-001 Every patient interface shall retain the Phase 2 patient banner and encounter context.

CLN-P9-UI-002 Preparation, transmission, delivery, acceptance and completion states shall be displayed separately.

CLN-P9-UI-003 The interface shall show source data cut-off, stale information and unresolved obligations prominently.

CLN-P9-UI-004 Critical transition information shall not rely on colour alone.

21. Audit, Events and Reports

- Handover create/update/send/accept/return
- Transfer readiness/acceptance
- Discharge initiation/readiness/override/completion
- Summary generation/signature/amendment
- Patient instructions and teach-back
- External package delivery/processing/acceptance
- Continuity task transfer/completion
- Pending-result ownership
- Downtime reconciliation
- AI draft acceptance/rejection
- `clinical.handover.created`

- clinical.handover.sent
- clinical.handover.accepted
- clinical.transfer.accepted
- clinical.discharge.initiated
- clinical.discharge.ready
- clinical.discharge.completed
- clinical.discharge.summary_issued
- clinical.continuity_task.created
- clinical.continuity_task.overdue
- clinical.transition.delivery_failed
- clinical.transition.reconciliation_required
- Handovers awaiting acceptance
- Transfer delays/rejections
- Discharge readiness exceptions
- Unsigned/incomplete summaries
- Patients discharged with pending results
- Post-discharge tasks overdue
- Failed continuity transmissions
- Readiness overrides
- Self-discharge/left-before-completion
- Medication reconciliation incomplete
- No-owner obligations
- Downtime transitions awaiting reconciliation

CLN-P9-EVT-001 Events shall include stable IDs, tenant, patient, encounter, document/task version, occurred-at, correlation ID and idempotency key while minimizing sensitive content.

CLN-P9-RPT-001 Reports and exports shall remain permission-, scope- and sensitivity-controlled.

22. Acceptance Scenarios

ID	Scenario	Expected Result
P9-AT-001	Night nurse prepares shift handover.	Current patients, risks, due work and source cut-off displayed.
P9-AT-002	Receiving nurse has not accepted transfer.	Sending team remains accountable under configured rule.
P9-AT-003	Patient moves before handover accepted.	Movement and clinical-responsibility states remain visibly discrepant and escalated.
P9-AT-004	Active infusion crosses unit transfer.	Infusion, rate, product, monitoring and responsibility hand over once.
P9-AT-005	Discharge has unassigned pending pathology.	Closure blocked or high-priority exception created.
P9-AT-006	Discharge medication list is generated.	Uses completed reconciliation and authorized discharge orders, not copied eMAR.
P9-AT-007	Summary includes preliminary result.	Status shown explicitly with assigned final-result reviewer.
P9-AT-008	External transfer package delivered.	Delivery recorded separately from receiving clinical acceptance.
P9-AT-009	Patient receives portal instructions.	Delivery recorded; understanding remains separate.
P9-AT-010	Material result correction arrives after discharge.	Review, communication and summary-amendment workflow reopens as configured.
P9-AT-011	Patient leaves before completion.	Distinct workflow records advice, risks and follow-up.
P9-AT-012	AI drafts summary.	Draft requires human review/signature

		and cannot close encounter.
P9-AT-013	Transfer is cancelled.	Current location/responsibility preserved and prepared handover retained historically.
P9-AT-014	Discharge occurs during downtime.	Minimum safe record used and electronic reconciliation task created.
P9-AT-015	Patient returns next day.	New encounter created; prior discharge remains intact.

23. Mandatory Negative Tests

- Hardcoded title grants discharge authority
- Patient movement treated as handover acceptance
- Network delivery treated as responsibility acceptance
- Discharge closes pending-result tasks
- Inpatient eMAR copied as discharge medication list
- Preliminary result represented as final
- AI auto-signs or completes discharge
- Receiver gains unrestricted chart access from package receipt
- Correction overwrites signed summary
- Readmission reopens prior encounter automatically
- Portal delivery treated as patient understanding
- Shift handover omits critical unresolved alerts
- Transfer duplicates monitoring and medication tasks
- Cancelled transfer changes patient location
- No-owner continuity task disappears
- Sensitive content leaks in transmission notification
- Downtime transition never reconciled

24. Traceability

Area	Prefix	Evidence
Governance	CLN-P9-GOV	Transition governance review
Types	CLN-P9-TYP	Registry tests
Handover	CLN-P9-HND	Version/content tests
Communication	CLN-P9-COM	Structured handover tests
Shift	CLN-P9-SHF	Assignment tests
Internal transfer	CLN-P9-INT	Movement/acceptance tests
Readiness	CLN-P9-RDY	Checklist/override tests
Discharge	CLN-P9-DSC	Closure tests
Summary	CLN-P9-SUM	Attestation/source tests
Patient	CLN-P9-PAT	Communication tests
External	CLN-P9-EXT	Delivery/acceptance tests
Follow-up	CLN-P9-FUP	Task continuity tests
Medication	CLN-P9-MED	Reconciliation tests
Results	CLN-P9-RES	Ownership tests
AI	CLN-P9-AI	Human-control tests
Correction	CLN-P9-COR	Version tests
Downtime	CLN-P9-DWN	Recovery tests

25. Registered Phase 9 Gaps

Gap ID	Resolution Required	Blocking
CLN-P9-GAP-001	Approve transition types, templates and acceptance points.	Yes
CLN-P9-GAP-002	Approve SBAR or alternative structured	Yes

	handover model.	
CLN-P9-GAP-003	Approve sending/receiving responsibility rules for internal transfers.	Yes
CLN-P9-GAP-004	Approve discharge readiness checklists by encounter class.	Yes
CLN-P9-GAP-005	Approve readiness override authority and escalation.	Yes
CLN-P9-GAP-006	Approve discharge summary sections and co-signature policy.	Yes
CLN-P9-GAP-007	Approve patient-instruction language/accessibility workflow.	Yes
CLN-P9-GAP-008	Approve external transfer disclosure and acceptance contracts.	Yes
CLN-P9-GAP-009	Approve self-discharge/left-before-completion process.	Yes
CLN-P9-GAP-010	Approve post-discharge task ownership and fallback.	Yes
CLN-P9-GAP-011	Approve downtime transition/discharge reconciliation.	Yes
CLN-P9-GAP-012	Confirm FHIR/CDA discharge-summary and Task profiles.	Before interoperability release

25.0 Observation and Form Continuity at Transition

CLN-P9-OBS-001 On transfer, Clinical shall recalculate prospective observation work using the destination client-space Clinical Profile while preserving prior Observations, due-event history and rendered forms.

CLN-P9-OBS-002 Requirements arising solely from the previous client space may end according to configured transition rules.

CLN-P9-OBS-003 Requirements supported by an active order, protocol, care plan, medication rule or patient-specific form shall continue when still applicable.

CLN-P9-OBS-004 Unresolved due and overdue observations shall transfer once to the eligible receiving context without duplication.

CLN-P9-OBS-005 A Labour Care Guide shall remain historically available after transfer even when active labour-monitoring schedules have ended.

25.1 Consumption Reconciliation at Transition

CLN-P9-CNS-001 Transfer, discharge readiness and encounter closure shall identify unresolved Consumption Sessions and assign reconciliation ownership without rewriting valid medication-administration records.

26. Phase 9 Completion Gate

- Phases 1 to 8 incorporated without contradiction.
- Handover types, content and acceptance rules approved.
- Internal transfer movement/responsibility coordination approved.
- Discharge readiness and override governance approved.
- Summary, patient instructions and external transfer package approved.
- Medication, pending-result and continuity-task ownership approved.
- Correction, cancellation, readmission and downtime rules approved.
- Permissions, interfaces, events, reports and tests approved.
- Every blocking gap assigned an owner and acceptance test.

Phase 10 shall consolidate specialty, reporting, interoperability, non-functional and deployment requirements without weakening the transition controls established here.

Appendix A. Minimum Transition Record

Group	Fields
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Identity	Public ID, patient, encounter, transition type/version
Context	Sending/receiving facility, client space, service and destination
Actors	Sender, receiver, responsible clinicians and recorder
Time	Prepared, sent, delivered, accepted and effective
Clinical state	Condition, trajectory, risks, treatment and devices
Outstanding work	Orders, results, observations, tasks and consultations
Plan	Priorities, goals, review times and contingency
Documents	Handover, discharge summary and patient instructions
Acceptance	Receiving decision, questions and responsibility start
Provenance	Source cut-off, references, sensitivity, correlation and audit

Appendix B. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phases 1 to 8.
- WHO communication during patient handovers and SBAR materials.
- HL7 FHIR Task and workflow materials.
- HL7 FHIR and C-CDA discharge-summary materials.

PHASE 10

SPECIALTY EXTENSIONS, INTEROPERABILITY, REPORTING AND RELEASE ASSURANCE

Phase 10 consolidates specialty-extension governance, interoperability, reporting, privacy, performance, resilience, deployment, migration and final release assurance without redefining the clinical safety rules established in Phases 1 to 9.

Document Control and Phase Authority

Field	Rule
Purpose	Complete the Clinical SRD with cross-cutting specialty, integration, reporting and production-assurance requirements.
Dependencies	Every requirement in Phases 1 to 9 remains authoritative unless formally resolved through controlled change.
Main Module contract	Clinical consumes shared identity, titles, permissions, client spaces, terminology, unit conversion, audit, events and platform services.
Extension rule	Specialties extend common Clinical capabilities through governed configuration and contracts, not parallel clinical cores.
Implementation	Functional specification only. Laravel migrations, policies, services, APIs, queues and deployment topology belong in the EDD.
Release	Controlled draft, not approved for production.

Phase 10 outcomes

- Governed specialty extension model.
- Versioned APIs, events and FHIR conformance artifacts.
- Scope-safe clinical reporting and analytics.
- Explicit availability, capacity, latency and recovery requirements.
- Secure deployment, migration and operational controls.
- Final traceability, validation and production-readiness gates.

1. Consolidated Architectural Principles

CLN-P10-ARC-001 Clinical shall remain a modular monolith or other approved Laravel-native deployment with explicit domain boundaries and no duplicated shared-platform masters.

CLN-P10-ARC-002 Specialty capabilities shall reuse the common patient, encounter, documentation, diagnosis, order, medication, observation, result and transition models.

CLN-P10-ARC-003 Every write operation shall enforce tenant, patient, encounter, client-space, care-relationship, atomic permission, privilege and record-state checks as applicable.

CLN-P10-ARC-004 Internal trust based solely on network location, service name or module ownership shall be prohibited.

CLN-P10-ARC-005 All external and cross-module writes shall be idempotent, version-aware and auditable.

CLN-P10-ARC-006 The system shall preserve authoritative ownership and provenance instead of copying external data into competing mutable records.

CLN-P10-ARC-007 Configuration shall be versioned, effective-dated, approved, testable and reversible.

Consolidation rule: Phase 10 may strengthen cross-cutting controls but shall not create alternate authorization, signature, correction, unit, order, medication, result or transition semantics.

2. Specialty Extension Framework

Extension Area	Permitted Capability	Prohibited Pattern
Documentation	Specialty note types/templates and CDE groups	Separate signature engine
Assessment	Specialty scores, examinations and pathways	Unversioned calculation logic
Orders	Specialty catalogues and order sets	Bypassing Phase 5 authorization
Medication	Specialty safeguards and monitoring	Independent eMAR/order source
Observations	Specialty CDEs, devices and ranges	Competing unit master
Results	Specialty display and review routing	Editing source reports

Transitions	Specialty readiness and summaries	Closing unresolved obligations
UI	Workspace composition and dashboards	Hardcoded title access

CLN-P10-SPC-001 Each specialty extension shall register a stable key, owner, scope, dependencies, permissions, configuration, interoperability mappings and release version.

CLN-P10-SPC-002 Activation shall require clinical-governance, information-security, interoperability and operational approval.

CLN-P10-SPC-003 Specialty extensions shall declare which common states and workflows they consume and shall not redefine canonical states locally.

CLN-P10-SPC-004 Disabling an extension shall preserve historical content and shall not break referenced records.

CLN-P10-SPC-005 Extension data shall remain discoverable through the common patient chart subject to authorization and sensitivity rules.

2.1 Candidate specialty packs

- Emergency and acute care
- Inpatient medicine
- Critical care
- Pediatrics and newborn
- Maternal health
- Mental health
- Oncology
- Renal and dialysis
- Cardiology
- Neurology
- Rehabilitation and allied health
- Community and home care

3. Interoperability Governance

CLN-P10-INT-001 Every interface shall have an owner, purpose, data contract, authoritative source, consumer, version, security classification, retry policy, SLA and support route.

CLN-P10-INT-002 FHIR, HL7 v2, DICOM and document exchange shall use approved profiles and implementation guides appropriate to the deployment jurisdiction and partner capability.

CLN-P10-INT-003 The system shall publish machine-readable capability and conformance artifacts for supported interfaces.

CLN-P10-INT-004 Unknown codes, unsupported versions, patient mismatches and incompatible units shall produce structured errors or quarantine, not silent coercion.

CLN-P10-INT-005 Interface changes shall pass contract, backward-compatibility, security and clinical-safety tests before release.

CLN-P10-INT-006 External identifiers shall retain assigning authority and shall not replace internal public IDs.

CLN-P10-INT-007 Technical acknowledgement, application acceptance, clinical acceptance and workflow completion shall remain distinct.

4. FHIR Conformance Baseline

Clinical Domain	Primary Mapping Candidates
Patient context	Patient, Encounter, EpisodeOfCare, Location, Practitioner, CareTeam
Documentation	Composition, Bundle document, DocumentReference
Problems/plans	Condition, Goal, CarePlan
Orders	ServiceRequest, MedicationRequest, Task
Medication	Medication, MedicationDispense, MedicationAdministration, MedicationStatement
Observations/results	Observation, DiagnosticReport, ImagingStudy, Specimen
Provenance/security	Provenance, AuditEvent, Consent, security labels
Definitions	StructureDefinition, ValueSet, CodeSystem, ConceptMap,

CapabilityStatement

CLN-P10-FHR-001 Profiles shall define cardinality, terminology bindings, Must Support elements, extensions and invariants for KashTre use cases.

CLN-P10-FHR-002 The deployed system shall expose an implementation CapabilityStatement matching actual configured behavior.

CLN-P10-FHR-003 Conformance resources shall be versioned and included in automated validation.

CLN-P10-FHR-004 Search, write, messaging, subscription and document capabilities shall be enabled only when explicitly supported and secured.

CLN-P10-FHR-005 FHIR resources shall not bypass Clinical domain services or Laravel policies when modifying clinical state.

5. API and Event Standards

CLN-P10-API-001 APIs shall use versioned resource contracts, explicit content types, schema validation, bounded pagination and structured errors.

CLN-P10-API-002 Object-level and function-level authorization shall be evaluated on every request.

CLN-P10-API-003 Mass assignment shall be prevented through explicit validated request objects and allowed fields.

CLN-P10-API-004 Clinical identifiers shall be opaque; sequential database IDs shall not be exposed as authorization boundaries.

CLN-P10-API-005 Rate limits and resource-consumption controls shall protect clinical availability without blocking configured emergency workflows.

CLN-P10-EVT-001 Events shall include event ID, type, schema version, tenant, aggregate/public ID, aggregate version, occurred-at, producer, correlation and causation IDs.

CLN-P10-EVT-002 Consumers shall be idempotent and shall tolerate retries and out-of-order delivery according to contract.

CLN-P10-EVT-003 Sensitive narrative shall not be placed in event headers or unrestricted telemetry.

6. Reporting and Analytics

Report Class	Examples	Control
Operational	Queues, overdue work, exceptions	Current scoped data
Clinical oversight	Critical responses, medication omissions, deterioration	Governed clinical definitions
Quality/safety	Corrections, overrides, adverse events	De-identified or limited dataset where appropriate
Regulatory	Required statutory submissions	Approved jurisdictional mapping
Management	Workload, turnaround and capacity	No unnecessary patient detail
Research/export	Approved datasets	Purpose, consent and disclosure controls

CLN-P10-RPT-001 Every report shall declare owner, purpose, inclusion/exclusion criteria, data sources, refresh frequency, terminology/unit versions and authorization scope.

CLN-P10-RPT-002 Counts, totals and denominators shall be reproducible from versioned definitions.

CLN-P10-RPT-003 Patient-level drill-through shall require separate permission and current contextual authorization.

CLN-P10-RPT-004 Entered-in-error data shall be excluded from ordinary clinical metrics unless explicitly included for safety analysis.

CLN-P10-RPT-005 Corrections and late-arriving data shall trigger controlled recalculation with reporting-period provenance.

CLN-P10-RPT-006 Exports shall be logged, encrypted, time-limited where applicable and subject to row/field minimization.

7. Privacy, Confidentiality and Consent

CLN-P10-PRV-001 Clinical shall apply data minimization, purpose limitation, sensitivity labels, disclosure restrictions and retention policies according to jurisdiction and organizational policy.

CLN-P10-PRV-002 Restricted conditions, notes, results and specialty records shall not leak through counts, search facets, caches, notifications, logs or reports.

CLN-P10-PRV-003 Break-glass access shall require reason, enhanced audit, time-limited scope and review.

CLN-P10-PRV-004 Consent shall be represented separately from authorization and shall not grant permissions the user otherwise lacks.

CLN-P10-PRV-005 Disclosure packages and redacted derivatives shall not modify the source legal record.

CLN-P10-PRV-006 Non-production environments shall not use identifiable production clinical data unless formally approved and protected.

8. Security and Zero-Trust Controls

CLN-P10-SEC-001 Authentication and authorization shall be evaluated for user and service identities before protected resource access.

CLN-P10-SEC-002 No implicit trust shall be granted solely because traffic originates inside the network or from another KashTre module.

CLN-P10-SEC-003 Service-to-service calls shall use managed identities or approved credentials, encrypted transport and least-privilege scopes.

CLN-P10-SEC-004 Secrets shall be stored in an approved secret-management service and shall not be embedded in code, images or configuration repositories.

CLN-P10-SEC-005 Session, token, device and privileged-operation policies shall support revocation and step-up authentication where configured.

CLN-P10-SEC-006 Security testing shall cover broken object authorization, authentication, property authorization, resource consumption, function authorization and unsafe third-party API consumption.

CLN-P10-SEC-007 Security incidents affecting clinical integrity or availability shall trigger clinical-safety assessment and continuity procedures.

9. Audit, Logging and Observability

Signal	Required Content
Clinical audit	Who, what, patient, action, reason, outcome, before/after reference
Application log	Service, severity, correlation, safe diagnostic detail
Security log	Authentication, authorization, privileged access and anomaly
Interface log	Message/event IDs, endpoint, acknowledgement, retry and error
Metric	Latency, error, throughput, queue depth, saturation and freshness
Trace	Cross-service correlation without unrestricted clinical payload

CLN-P10-OBS-001 Audit records shall be tamper-evident, time-synchronized, access-controlled and retained under approved policy.

CLN-P10-OBS-002 Logs shall not contain passwords, tokens, secrets or unnecessary clinical narrative.

CLN-P10-OBS-003 Monitoring shall detect failures, latency degradation, queue backlog, stale data, synchronization errors and security anomalies.

CLN-P10-OBS-004 Alert ownership, severity, response target and escalation shall be defined for every production monitor.

CLN-P10-OBS-005 Operational dashboards shall not expose patient-identifiable information unless strictly required and authorized.

10. Performance and Capacity

Area	Requirement Method
Interactive response	Named workflow percentile targets validated by load test
Write commitment	Clinical transaction and outbox commit targets
Search/chart	Scoped query targets for representative chart sizes
Queues	Backlog age and drain-rate targets

Integration	Message throughput, burst and retry capacity
Concurrency	Simultaneous users, devices and patient records
Data growth	Retention, index and archive projections

CLN-P10-PER-001 Numeric service levels shall be approved per deployment profile and recorded in the EDD and operational SLO catalogue.

CLN-P10-PER-002 Performance tests shall use clinically representative data volumes, permissions, chart complexity and concurrent workflows.

CLN-P10-PER-003 Safety checks and authorization shall not be disabled to meet performance targets.

CLN-P10-PER-004 Long-running reports, exports and integrations shall use bounded asynchronous processing and shall not exhaust interactive capacity.

CLN-P10-PER-005 Capacity planning shall cover peak shifts, medication rounds, device bursts, result batches and downtime recovery.

11. Availability, Resilience and Recovery

CLN-P10-RES-001 Availability, recovery time and recovery point objectives shall be defined for each deployment tier and critical capability.

CLN-P10-RES-002 Single points of failure shall be identified and mitigated according to approved risk and architecture.

CLN-P10-RES-003 Durable queues/outbox patterns shall protect committed clinical actions from transient integration failure.

CLN-P10-RES-004 Backup restoration shall be tested and shall validate database, attachments, configuration, audit and encryption keys as applicable.

CLN-P10-RES-005 Disaster recovery exercises shall verify patient identity, clinical integrity, ordering, eMAR, results, audit and reconciliation.

CLN-P10-RES-006 Degraded-mode and downtime workflows shall be documented, trained, testable and reconciled after restoration.

CLN-P10-RES-007 Recovery shall prioritize safety and data integrity over rapid but unverified restoration.

12. Data Integrity, Retention and Archiving

CLN-P10-DAT-001 Clinical records shall use database constraints, transactions, optimistic concurrency and immutable history appropriate to their lifecycle.

CLN-P10-DAT-002 Soft deletion shall not be used as a substitute for clinical cancellation, correction or entered-in-error states.

CLN-P10-DAT-003 Retention and legal-hold policies shall be configurable by record class and jurisdiction.

CLN-P10-DAT-004 Archive and restore shall preserve readability, provenance, terminology/unit versions, signatures and linked attachments.

CLN-P10-DAT-005 Hashing or signatures shall be used where approved to detect alteration of critical documents, payloads or exports.

CLN-P10-DAT-006 Purge operations shall require formal authority, evidence and verification and shall not break retained referential integrity.

13. Configuration and Change Governance

CLN-P10-CFG-001 Configuration changes shall progress through draft, review, approval, scheduled activation, monitoring and retirement states.

CLN-P10-CFG-002 High-risk configuration includes permissions, medication safeguards, ranges, scores, order sets, templates, routing, interfaces and unit policies.

CLN-P10-CFG-003 High-risk changes shall require segregation of duties, impact analysis, test evidence and rollback plan.

CLN-P10-CFG-004 Production configuration shall be exportable, diffable and reproducible without exposing secrets.

CLN-P10-CFG-005 Emergency changes shall be time-limited where possible and undergo retrospective review.

CLN-P10-CFG-006 A configuration rollback shall not rewrite historical records created under the retired version.

14. Deployment and Release Management

Stage	Minimum Gate
Build	Reproducible artifact, dependency lock and provenance
Static checks	Code quality, secret, dependency and policy scans
Automated tests	Unit, integration, contract, authorization and negative tests
Clinical validation	Approved scenarios and clinical-safety review
Migration rehearsal	Production-scale backup, migration and rollback test
Pre-production	Representative topology, integrations and monitoring
Approval	Named business, clinical, security, data and operations sign-off
Deployment	Controlled rollout with observability and abort criteria
Post-release	Reconciliation, smoke tests and heightened monitoring

CLN-P10-REL-001 Production deployment shall use approved immutable artifacts and environment-specific managed configuration.

CLN-P10-REL-002 Database migrations shall be backward-compatible where required, restartable, monitored and reversible or compensatable.

CLN-P10-REL-003 Feature flags shall be permission-controlled, auditable and prohibited from bypassing safety or authorization controls.

CLN-P10-REL-004 Release rollback shall include data, queue, interface and configuration reconciliation, not application binaries alone.

CLN-P10-REL-005 Known defects affecting patient safety, identity, authorization or clinical integrity shall block production release unless an approved risk decision and mitigation exist.

15. Migration and Legacy Data

CLN-P10-MIG-001 Migration shall preserve source system, patient identifiers/authorities, encounter, author, dates, statuses, units, terminology, versions and attachments where available.

CLN-P10-MIG-002 Data quality rules shall classify accepted, transformed, quarantined and rejected records with reasons.

CLN-P10-MIG-003 Migrated records shall be distinguishable from natively created records without appearing clinically inferior when validated.

CLN-P10-MIG-004 Unknown or ambiguous units, patients, authors or statuses shall not be silently guessed.

CLN-P10-MIG-005 Reconciliation shall compare source counts, identifiers, status distributions, hashes or other approved controls.

CLN-P10-MIG-006 Migration cutover shall include freeze, delta, rollback, downtime and clinical-access plans.

16. Testing and Validation Strategy

- Unit and domain-state tests
- Laravel policy and authorization matrix tests
- Patient/encounter/client-space isolation tests
- API schema and contract tests
- FHIR/profile and terminology validation
- Event idempotency and ordering tests
- Unit-conversion property and gold-case tests
- Clinical algorithm gold cases
- Concurrency and stale-version tests
- Downtime and recovery exercises
- Load, endurance and failover tests
- Accessibility and usability tests
- Security testing and threat scenarios
- Migration reconciliation tests
- Clinical user acceptance and simulation

CLN-P10-TST-001 Every SHALL requirement shall map to one or more test cases, inspections or operational controls.

CLN-P10-TST-002 Negative tests from Phases 1 to 9 shall be mandatory regression tests.

CLN-P10-TST-003 Test data shall include edge cases, incomplete data, conflicting states, corrected records and cross-tenant attack attempts.

CLN-P10-TST-004 Automated tests shall not replace clinical workflow validation by qualified reviewers.

CLN-P10-TST-005 Release evidence shall be retained and linked to the exact artifact and configuration versions.

17. Accessibility, Localization and Usability

CLN-P10-UX-001 Clinical interfaces shall support keyboard operation, visible focus, meaningful labels, screen-reader semantics and non-colour status cues.

CLN-P10-UX-002 Date, time, number, language and unit presentation shall follow approved locale and clinical-context policies without changing stored meaning.

CLN-P10-UX-003 Critical actions shall use clear confirmation and recovery patterns while avoiding alert fatigue.

CLN-P10-UX-004 Patient-facing content shall support approved translations, plain language and accessibility needs.

CLN-P10-UX-005 Usability evaluation shall include representative professional titles, care settings, devices and high-risk workflows.

18. Operations, Support and Incident Management

CLN-P10-OPS-001 Runbooks shall cover interface failures, queue backlog, identity mismatch, unit-service outage, critical alerts, eMAR downtime, result delays and data correction.

CLN-P10-OPS-002 Support access to production clinical data shall be least privilege, approved, time-limited and audited.

CLN-P10-OPS-003 Incidents shall record impact, affected patients/workflows, containment, recovery, reconciliation and corrective actions.

CLN-P10-OPS-004 Clinical-safety incidents and near misses shall feed governed improvement without altering source records.

CLN-P10-OPS-005 Operational ownership and escalation contacts shall be maintained for every critical service and interface.

CLN-P10-OPS-006 Maintenance windows shall account for clinical operations and provide approved degraded-mode communication.

19. Final Permission Additions

Permission	Purpose
<code>clinical.specialty_extension.manage</code>	Manage specialty extension definitions
<code>clinical.interoperability_config.manage</code>	Manage interface contracts/profiles
<code>clinical.report_definition.manage</code>	Manage governed report definitions
<code>clinical.report.run_sensitive</code>	Run sensitive patient-level reports
<code>clinical.export.create</code>	Create authorized export
<code>clinical.configuration.approve</code>	Approve high-risk configuration
<code>clinical.release.validate</code>	Record release validation evidence
<code>clinical.migration.reconcile</code>	Resolve migration exceptions
<code>clinical.support.break_glass</code>	Time-limited approved support access
<code>clinical.operations.audit</code>	View operational and interface audit

CLN-P10-PRM-001 All permissions shall be registered in Main and assigned through configurable title bundles and explicit privileges; Laravel policies shall enforce context without hardcoded title strings.

20. Consolidated Operational Reports

- Patient-safety alerts and response times
- Unsigned documentation and co-signatures
- Problem/diagnosis review exceptions
- Order acknowledgements and fulfilment delays
- Medication omissions, overrides and discrepancies
- Observation/score/monitoring exceptions

- Unreviewed and corrected results
- Handover and discharge exceptions
- Interface errors, quarantines and retries
- Access, break-glass and export activity
- Performance, queue age and data freshness
- Deployment, migration and reconciliation status

CLN-P10-ORP-001 The report catalogue shall identify the authoritative phase requirement and owner for each metric.

CLN-P10-ORP-002 Dashboards shall distinguish live operational state from retrospectively recalculated analytics.

21. Consolidated Acceptance Scenarios

ID	Scenario	Expected Result
P10-AT-001	Specialty pack is activated.	Common states, permissions and historical compatibility remain intact.
P10-AT-002	FHIR client queries capabilities.	Published CapabilityStatement matches enabled version and interactions.
P10-AT-003	API requests another tenant's record ID.	Object authorization blocks access without existence leakage.
P10-AT-004	Unknown source unit arrives.	Record quarantined; value is not relabelled.
P10-AT-005	Event consumer receives duplicate/out-of-order events.	Idempotency/version controls preserve correct state.
P10-AT-006	Correction changes prior-period metric.	Report recalculates with definition and refresh provenance.
P10-AT-007	Internal service calls from trusted network without authority.	Request denied.
P10-AT-008	Production restore is tested.	Clinical data, attachments, audit, queues and keys reconcile.
P10-AT-009	Migration contains ambiguous patient match.	Record quarantined for authorized reconciliation.
P10-AT-010	Feature flag attempts to bypass double check.	Configuration rejected or safety control remains enforced.
P10-AT-011	Large export starts during peak use.	Bounded asynchronous processing protects interactive workflows.
P10-AT-012	Deployment migration partially fails.	Runbook stops/rolls back or compensates with reconciliation evidence.
P10-AT-013	Support engineer needs urgent access.	Time-limited approved access with full audit.
P10-AT-014	Network segment is internal.	No implicit trust; user/service and resource authorization still required.
P10-AT-015	Phase regression suite runs.	All mandatory negative and clinical gold cases pass before release.

22. Mandatory Negative Tests

- Specialty extension bypasses common signature or correction
- FHIR endpoint writes directly around domain policy
- Cross-tenant object ID enumeration
- Mass assignment changes protected status
- Internal network call bypasses authentication
- Event retry duplicates clinical action
- Unsupported terminology silently accepted
- Report denominator changes without version
- Sensitive count leaks from dashboard
- Log contains tokens or unrestricted clinical narrative
- Backup restores database but not attachments/audit

- Feature flag disables patient identity check
- Migration guesses patient, unit or author
- Rollback leaves incompatible events/queues
- Non-production uses unapproved identifiable data
- Known clinical-safety defect released without approved decision
- Accessibility status conveyed only by colour

23. Consolidated Traceability

Phase	Primary Scope	Release Evidence
1	Authorization and workforce context	Policy and isolation tests
2	Patient, encounter and workspace	Identity/state tests
3	Documentation/legal record	Signature/version tests
4	Problems, diagnoses and plans	Clinical-status tests
5	Orders and requests	Authorization/closed-loop tests
6	Medication and eMAR	Administration/safeguard tests
7	Observations, units and scores	Conversion/algorithm tests
8	Results and follow-up	Matching/escalation tests
9	Transitions and discharge	Acceptance/continuity tests
10	Cross-cutting release assurance	Conformance, NFR and deployment tests

CLN-P10-TRC-001 The final requirements traceability matrix shall map requirement, design component, test, owner, release and evidence location.

CLN-P10-TRC-002 Unresolved contradictions shall block baseline approval.

23.1 Clinical-to-Inventory Consumption Contract

CLN-P10-CNS-001 The Clinical-to-Inventory Consumption Session contract shall be included in interface, idempotency, outage, reconciliation and traceability testing.

CLN-P10-CNS-002 Release evidence shall demonstrate Approved Pool usage, non-approved floor-stock usage, incidental-supply capture, waste, breakage, correction and retry handling.

CLN-P10-CNS-003 The companion EDD shall define the versioned payload, response states, outbox processing, retry policy, reconciliation queues and observability for Consumption Sessions.

23.2 Form Engine and Observation-Work Release Contract

CLN-P10-FRM-001 The companion EDD shall define Form Definition versioning, client-space Clinical Profiles, Effective Observation Plan derivation, task consolidation, observation matching, rendering, reconciliation and historical reproduction.

CLN-P10-FRM-002 Release evidence shall demonstrate that one Observation can satisfy and populate multiple applicable forms without duplicate patient facts or duplicate bedside work.

CLN-P10-FRM-003 Performance testing shall include clinically representative patients with multiple active forms, dense observation histories, concurrent capture and repeated rendering.

CLN-P10-FRM-004 Interoperability design shall map governed form definitions and responses appropriately while preserving authoritative Observation resources and provenance.

CLN-P10-FRM-005 Failure of form rendering, prepopulation or refresh shall be observable and recoverable without loss or rollback of committed Observations.

CLN-P10-FRM-006 The release gate shall include client-space default activation, Add Form, transfer recalculation, deduplication, correction propagation, print reproduction and authorization tests.

24. Registered Phase 10 Gaps

Gap ID	Resolution Required	Blocking
CLN-P10-GAP-001	Approve initial specialty-extension catalogue and owners.	Yes
CLN-P10-GAP-002	Approve interoperability profiles,	Yes

	versions and partner contracts.	
CLN-P10-GAP-003	Approve API/event standards and schema registry.	Yes
CLN-P10-GAP-004	Approve report catalogue, definitions and disclosure controls.	Yes
CLN-P10-GAP-005	Approve deployment-tier performance and capacity targets.	Yes
CLN-P10-GAP-006	Approve availability, RTO, RPO and disaster-recovery objectives.	Yes
CLN-P10-GAP-007	Approve retention, archive, legal-hold and purge policies.	Yes
CLN-P10-GAP-008	Approve security architecture, privileged access and secret management.	Yes
CLN-P10-GAP-009	Approve observability, alert ownership and operational SLOs.	Yes
CLN-P10-GAP-010	Approve release, rollback and emergency-change governance.	Yes
CLN-P10-GAP-011	Approve migration scope, source systems and reconciliation controls.	Before migration
CLN-P10-GAP-012	Complete full SRD-to-EDD-to-test traceability and resolve all prior phase gaps.	Yes

25. Final SRD Baseline Gate

- All ten phase drafts reviewed and reconciled.
- No duplicate or conflicting canonical state models.
- Main Module integration contracts confirmed.
- Specialty extension framework approved.
- FHIR, API, event and document conformance approved.
- Security, privacy, audit and reporting controls approved.
- Performance, availability, resilience and recovery targets approved.
- Deployment, migration, operations and rollback processes approved.
- All blocking gaps from Phases 1 to 10 closed or formally dispositioned.
- Traceability matrix and clinical-safety case accepted.
- Production release remains prohibited until EDD, implementation and validation evidence satisfy this gate.

Phase 10 closes the functional drafting sequence. The next controlled activity is consolidation into the Clinical SRD v6.1 baseline and creation or completion of the Laravel-native Clinical EDD against the approved requirements traceability matrix.

26. Specialty Extension Registration Contract

Contract Area	Mandatory Detail
Identity	Stable extension key, semantic version, owner, clinical sponsor and support team
Scope	Specialty use cases, patient populations, encounter classes and client spaces
Dependencies	Required common capabilities, catalogues, terminology, devices and external services
Data	CDEs, templates, value sets, ranges, scores and retention classification
Authority	Atomic permissions, required privileges, co-signature and override rules
Workflow	Entry points, states, transitions, tasks, alerts and closure evidence
Integration	Inbound/outbound APIs, events, documents and authoritative

	ownership
Safety	Hazards, mitigations, mandatory negative tests and downtime behavior
Operations	SLOs, monitors, runbooks, release strategy and rollback
Governance	Approval records, effective dates, change log and retirement plan

CLN-P10-SPC-010 An extension package shall be rejected when it cannot declare authoritative ownership, required permissions, state transitions or historical-retention behavior.

CLN-P10-SPC-011 Specialty configuration shall use public IDs for shared concepts and shall not bind to environment-specific database identifiers.

CLN-P10-SPC-012 An extension shall provide deterministic seed/migration behavior and shall be safe to install repeatedly without duplicate definitions.

CLN-P10-SPC-013 Extension upgrades shall declare compatible source versions, transformed configuration, data migrations, validation checks and rollback limitations.

CLN-P10-SPC-014 A specialty pack shall include a minimum validation bundle covering permissions, patient isolation, lifecycle, correction, reporting and downtime.

CLN-P10-SPC-015 Retirement shall block new use prospectively while preserving historical display, source resolution, audit and export behavior.

27. Detailed Security Verification Baseline

Control Domain	Required Verification Evidence
Authentication	Identity-provider configuration, MFA/step-up rules, session expiry, revocation and service identity tests
Authorization	Object, property, function, tenant, patient, encounter, client-space and privilege matrix tests
Input handling	Schema validation, encoding, sanitization, file validation, injection and fuzz tests
Tokens/sessions	Audience, issuer, expiry, replay, rotation, logout and compromised-token handling
Cryptography	Approved algorithms, key ownership, rotation, backup and certificate-expiry monitoring
Data protection	Encryption, minimization, masking, cache controls, export restrictions and secure deletion
API security	Rate limits, pagination limits, mass-assignment prevention and unsafe-consumer tests
Logging	Security events, safe error messages, alerting and absence of secrets/clinical payload leakage
Dependencies	SBOM, provenance, vulnerability scanning, licensing and remediation evidence
Operations	Privileged access, break-glass, incident response and forensic preservation

CLN-P10-SEC-010 The EDD shall select and record an approved application-security verification target and map each applicable control to implementation and test evidence.

CLN-P10-SEC-011 Security acceptance shall include authenticated and unauthenticated attack paths, horizontal and vertical privilege escalation, stale assignments and cross-tenant identifiers.

CLN-P10-SEC-012 Uploaded documents and attachments shall be validated for type, size, malicious content, active content and authorized retrieval.

CLN-P10-SEC-013 Error responses shall not disclose patient existence, internal identifiers, stack traces, SQL, secrets or authorization policy detail useful for attack.

CLN-P10-SEC-014 Automated dependency findings shall have severity, exploitability assessment, owner, due date, disposition and release impact.

CLN-P10-SEC-015 Software build provenance and component inventory shall be retained for every production artifact.

CLN-P10-SEC-016 Security controls inherited from Main or infrastructure shall have named owners and evidence, rather than being assumed.

28. Service-Level Objective Catalogue

SLO Field	Required Definition
Service/capability	Exact workflow, API, queue or integration covered
User population	Care setting, facility tier and expected concurrency
Indicator	Latency, availability, freshness, durability, error rate or recovery measure
Target	Approved numeric threshold and percentile/window
Exclusions	Declared maintenance or dependency exclusions with governance
Measurement	Telemetry source, calculation, sampling and retention
Alerting	Warning/critical thresholds, evaluation window and owner
Response	Runbook, escalation, communication and clinical contingency
Review	Reporting cadence and target-change governance

CLN-P10-SLO-001 SLOs shall be defined separately for patient lookup, chart open, note save/sign, order commit, eMAR commit, observation commit, result posting and transition completion.

CLN-P10-SLO-002 Integration SLOs shall distinguish producer delay, transport delay, Clinical processing delay, quarantine time and consumer acknowledgement.

CLN-P10-SLO-003 Data-freshness SLOs shall be defined for assignments, medication orders, device observations, diagnostic results and critical alerts.

CLN-P10-SLO-004 Error budgets shall not authorize disabling patient-safety, authorization, audit or data-integrity controls.

CLN-P10-SLO-005 Breaches shall trigger operational review and, where patient care may be affected, clinical-risk assessment and continuity action.

29. Recovery Tier and Reconciliation Model

Tier	Illustrative Scope	Required Recovery Evidence
Tier A: Immediate clinical safety	Identity, active chart, orders, eMAR, critical alerts	Priority recovery, degraded workflow and urgent reconciliation
Tier B: Time-critical clinical	Observations, results, handover, continuity tasks	Bounded recovery and backlog processing
Tier C: Supporting clinical	Templates, non-urgent reports, specialty administration	Scheduled recovery with dependency validation
Tier D: Analytical/administrative	Historical analytics and non-operational extracts	Deferred recovery with integrity checks

CLN-P10-DR-001 Final tier assignment and numeric RTO/RPO shall be approved per deployment and shall not be inferred from this illustrative catalogue.

CLN-P10-DR-002 Recovery sequencing shall account for dependencies so that Clinical does not resume before identity, permissions, time, terminology and unit services are trustworthy.

CLN-P10-DR-003 After recovery, automated and manual reconciliation shall cover database commits, outbox/inbox records, queues, external acknowledgements and offline activity.

CLN-P10-DR-004 Reconciliation shall identify missing, duplicate, conflicting and out-of-order clinical actions and assign each exception to an accountable owner.

CLN-P10-DR-005 Disaster-recovery exercises shall include at least one patient-safety scenario spanning more than one module and one external interface.

CLN-P10-DR-006 A recovery exercise shall produce timing evidence, discrepancies, patient-impact assessment, corrective actions and retest decision.

30. Migration Workstreams and Cutover Controls

Workstream	Minimum Controls
Discovery	Source systems, owners, data classes, volumes, quality and legal constraints
Mapping	Source-to-target fields, code systems, units, statuses and provenance
Cleansing	Approved transformations with no invented clinical meaning
Dry run	Repeatable extraction, transformation, load and reconciliation
Clinical validation	Representative charts, edge cases and historical interpretation
Cutover	Freeze, delta capture, downtime, communication and go/no-go
Reconciliation	Counts, identifiers, hashes, status/unit distributions and sampled chart review
Remediation	Quarantine ownership, correction and residual-risk acceptance
Decommission	Read-only access, archive, retention and support transition

CLN-P10-MIG-010 Migration mappings shall explicitly handle provisional, final, corrected, cancelled and entered-in-error states.

CLN-P10-MIG-011 Historical free text shall not be converted into coded diagnosis, allergy, medication or result assertions without governed human validation.

CLN-P10-MIG-012 Unknown authors shall retain source-attribution text and migration provenance rather than being assigned to a convenient user.

CLN-P10-MIG-013 Historical quantities shall preserve source unit strings and be normalized only where the Shared Unit Engine has an approved deterministic mapping.

CLN-P10-MIG-014 Migration shall preserve cross-record relationships where trustworthy and shall identify broken or ambiguous links.

CLN-P10-MIG-015 Cutover approval shall require reconciliation thresholds, open-exception inventory, rollback feasibility and clinical-operational acceptance.

31. Requirements Traceability Matrix Specification

RTM Column	Purpose
Requirement ID/version	Stable source and revision
Requirement text	Approved normative statement
Phase/domain	Ownership and context
Hazard/risk	Related safety or security concern
Design component	EDD service, policy, table, API, event or UI
Configuration	Relevant versioned registry/policy
Test IDs	Positive, negative, integration and clinical tests
Evidence	Report, log, screenshot or signed validation record
Owner	Business, clinical and technical accountability
Release status	Planned, implemented, verified, deferred or rejected
Residual risk	Approved limitation and mitigation
Change link	Decision/change request and impact assessment

CLN-P10-RTM-001 Every normative SHALL statement shall have a unique requirement ID and a traceability row.

CLN-P10-RTM-002 A requirement shall not be considered verified merely because a design component exists; test evidence and acceptance are required.

CLN-P10-RTM-003 Deferred requirements shall identify reason, impact, mitigation, owner, target release and approving authority.

CLN-P10-RTM-004 Changes to shared states, units, permissions or interoperability contracts shall trigger impact analysis across all affected phase requirements.

CLN-P10-RTM-005 The final baseline shall report requirements with missing design, missing test, failed test, unresolved risk or unapproved evidence.

32. Production Readiness Evidence Pack

Evidence Group	Required Artifacts
Requirements	Approved SRD, gaps, decisions and RTM
Design	EDD, architecture, data model, API/event contracts and threat model
Build	Artifact provenance, dependency lock, SBOM and configuration manifest
Testing	Automated results, clinical scenarios, security, accessibility, load and recovery evidence
Data	Migration reports, reconciliation and retained exceptions
Operations	Runbooks, monitors, SLOs, on-call ownership and downtime packs
Governance	Clinical safety, privacy, security, interoperability and business approvals
Deployment	Change plan, rollback, smoke tests, communications and post-release review

CLN-P10-PRD-001 Every evidence artifact shall identify version, environment, date, owner, approval state and applicable release.

CLN-P10-PRD-002 Go-live approval shall document all open defects and gaps with severity, patient impact, workaround, owner and explicit disposition.

CLN-P10-PRD-003 The release command shall remain unavailable until mandatory gates are satisfied by machine-verifiable and human approval checks as applicable.

CLN-P10-PRD-004 Post-deployment validation shall verify authentication, patient lookup, chart access, clinical write, outbox delivery, eMAR, result ingestion, audit and monitoring.

CLN-P10-PRD-005 A failed safety-critical smoke test shall trigger the approved abort or rollback procedure.

CLN-P10-PRD-006 The heightened-monitoring period shall end only after reconciliation is complete and named operational and clinical owners accept stability.

33. Detailed Final Baseline Review Checklist

- All requirement IDs are unique and version controlled.
- Every state model has one canonical owner and transition definition.
- All permissions exist in Main and have policy tests.
- Every quantitative CDE references a Main Module canonical unit.
- Every cross-module object has authoritative ownership and stable identifiers.
- Every external interface has contract, conformance, security and operational evidence.
- Every correction path preserves history and dependent review.
- Every critical workflow has acknowledgement, escalation and downtime behavior.
- Every report defines numerator, denominator, exclusions, versions and permissions.
- Every specialty pack reuses common services and has retirement behavior.
- Every SLO has telemetry, owner, threshold and runbook.
- Every migration exception has status and owner.
- Every production gate has evidence and approval.
- Every open risk is explicitly accepted, mitigated, deferred or blocking.

CLN-P10-BAS-001 Baseline approval shall produce a signed decision record listing included documents, versions, unresolved items and effective date.

CLN-P10-BAS-002 After baseline approval, changes shall use controlled impact assessment and shall not edit the approved baseline silently.

CLN-P10-BAS-003 The Clinical EDD shall reference this baseline by immutable version and shall identify every intentional deviation.

34. Confirmed Production Runtime Baseline

Component	Confirmed Production State	Phase 10 Treatment
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PHP runtime	PHP 8.3	Required runtime baseline for current production deployment
Composer PHP constraint	^8.1	Compatibility constraint; not a statement that production runs PHP 8.1
Laravel framework	Laravel 10.48.x	Current production framework branch and patch family
Database	MySQL 8+	Existing target pending exact production patch confirmation
Frontend	Vue 3	Existing target pending exact production package-lock confirmation

CLN-P10-PLT-001 The EDD, build pipeline, deployment manifests, container or server configuration and production evidence shall identify PHP 8.3 as the deployed runtime baseline.

CLN-P10-PLT-002 The composer.json PHP constraint ^8.1 shall be preserved as a package-resolution constraint unless deliberately changed; deployment validation shall separately enforce PHP 8.3.

CLN-P10-PLT-003 The application shall be tested against the exact production Laravel 10.48.x dependency lock and PHP 8.3 patch version before release.

CLN-P10-PLT-004 Composer install shall use the committed lock file, production-only dependencies, optimized autoloading and a reproducible artifact build.

CLN-P10-PLT-005 Configuration and health checks shall detect runtime drift between approved PHP/Laravel versions and deployed instances.

CLN-P10-PLT-006 The deployment evidence pack shall record php --version, enabled extensions, composer.lock hash, installed Laravel version, application commit and build artifact digest.

CLN-P10-PLT-007 Required PHP extensions shall be explicitly listed and verified in CI and at deployment rather than assumed from the server image.

CLN-P10-PLT-008 OPcache, queue workers, scheduler, cache, session, filesystem and process-supervision settings shall be versioned in the EDD and validated per environment.

CLN-P10-PLT-009 Queue workers and long-running processes shall be restarted or safely recycled after deployment so that they execute the approved artifact and configuration.

CLN-P10-PLT-010 Application debug mode and detailed exception output shall be disabled in production, with safe correlation identifiers available for support.

CLN-P10-PLT-011 Database, cache, queue and external-service compatibility shall be validated with PHP 8.3 and the locked Laravel 10.48.x dependency set.

CLN-P10-PLT-012 Because the Laravel 10 support window has ended, production use shall have a formally owned framework-upgrade or extended-support risk treatment with target release, compensating controls and regression scope.

CLN-P10-PLT-013 A future Laravel major-version upgrade shall be handled as a controlled platform migration and shall not silently alter Clinical domain semantics, permissions, queues, serialization or database behavior.

CLN-P10-PLT-014 Platform upgrade testing shall rerun all Phase 1 to 10 negative tests, authorization matrices, queue/idempotency tests, clinical gold cases and migration rehearsals.

Platform Evidence	Minimum Verification
Runtime	Exact PHP version and extensions on every application/worker node
Framework	Installed Laravel version and composer.lock hash
Build	Source commit, dependency provenance and artifact digest
Configuration	Environment manifest excluding secrets, cache state and feature flags
Database	Server version, connection mode, charset/collation and migration level
Workers	Queue worker/scheduler versions, restart time and health
Security	Debug disabled, TLS, secret source and vulnerability disposition
Operations	Health checks, logs, metrics, alerting and rollback record

Appendix A. Minimum Interface Contract

Group	Fields
Identity	Interface ID, owner, producer, consumer and environment
Purpose	Use case, authoritative source and permitted purposes
Contract	Protocol, version, schema/profile and terminology
Security	Identity, scopes, encryption, sensitivity and consent
Reliability	Acknowledgement, idempotency, ordering, retry and timeout
Operations	SLA/SLO, monitoring, alert owner and support route
Change	Compatibility, test evidence, activation and rollback
Audit	Correlation, payload hash, outcome and retention

Appendix B. Sources Reviewed

- KashTre Clinical Module SRD v6.0 and addendum.
- KashTre Clinical Module SRD v6.1 Phases 1 to 9.
- HL7 FHIR R5 conformance and CapabilityStatement materials.
- OWASP API Security Project materials.
- NIST SP 800-207 Zero Trust Architecture.